



## openFinance API Framework

### Data Dictionary

Version 2.2.3

24 January 2025

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## 1 Introduction

### 1.1 From Core XS2A Interface to openFinance API

With [PSD2] the European Union has published a directive on payment services in the internal market. Among others [PSD2] contains regulations on services to be operated by so called Third Party Payment Service Providers (TPP) on behalf of a Payment Service User (PSU). These services are

- Payment Initiation Service (PIS) to be operated by a Payment Initiation Service Provider (PISP) TPP as defined by article 66 of [PSD2],
- Account Information Service (AIS) to be operated by an Account Information Service Provider (AISP) TPP as defined by article 67 of [PSD2], and
- Confirmation on the Availability of Funds Service (FCS) to be used by a Payment Instrument Issuing Service Provider (PIISP) TPP as defined by article 65 of [PSD2].

To implement these services (subject to PSU consent) a TPP needs to access the account of the PSU. The account is managed by another PSP called the Account Servicing Payment Service Provider (ASPSP). To support the TPP in accessing the accounts managed by an ASPSP, each ASPSP has to provide an "access to account interface" (XS2A interface). Such an interface has been defined in the Berlin Group NextGenPSD2 XS2A Framework.

This XS2A Framework is now broadened to extended services and developed into a Version 2 API family. This interface is addressed in the following as **openFinance API**. This openFinance API differs from the XS2A interface in several dimensions:

- The extended services might not rely anymore solely on PSD2.
- Other important regulatory frameworks which apply are e.g. GDPR.
- The openFinance API can address different types of **API Clients** as access clients, e.g. TPPs regulated by an NCA according to PSD2, or corporates not regulated by an NCA.
- The extended services might require contracts between the access client and the ASPSP.
- While the client identification at the openFinance API can still be based on eIDAS certificates, they do not need to be necessarily PSD2 compliant eIDAS certificates.
- The extended services might require e.g. the direct involvement of the access client's bank for KYC processes.

**Note:** The notions of API Client and ASPSP are used because of the technical standardisation perspective of the openFinance API. These terms are analogous to "asset broker" and "asset holder" resp. in the work of the ERPB on a SEPA API access scheme.

**Note:** In implementations, the API services of several ASPSPs might be provided on an aggregation platform. Such platforms will be addressed in the openFinance API Framework as "API provider".

The following account access methods are covered by this framework:

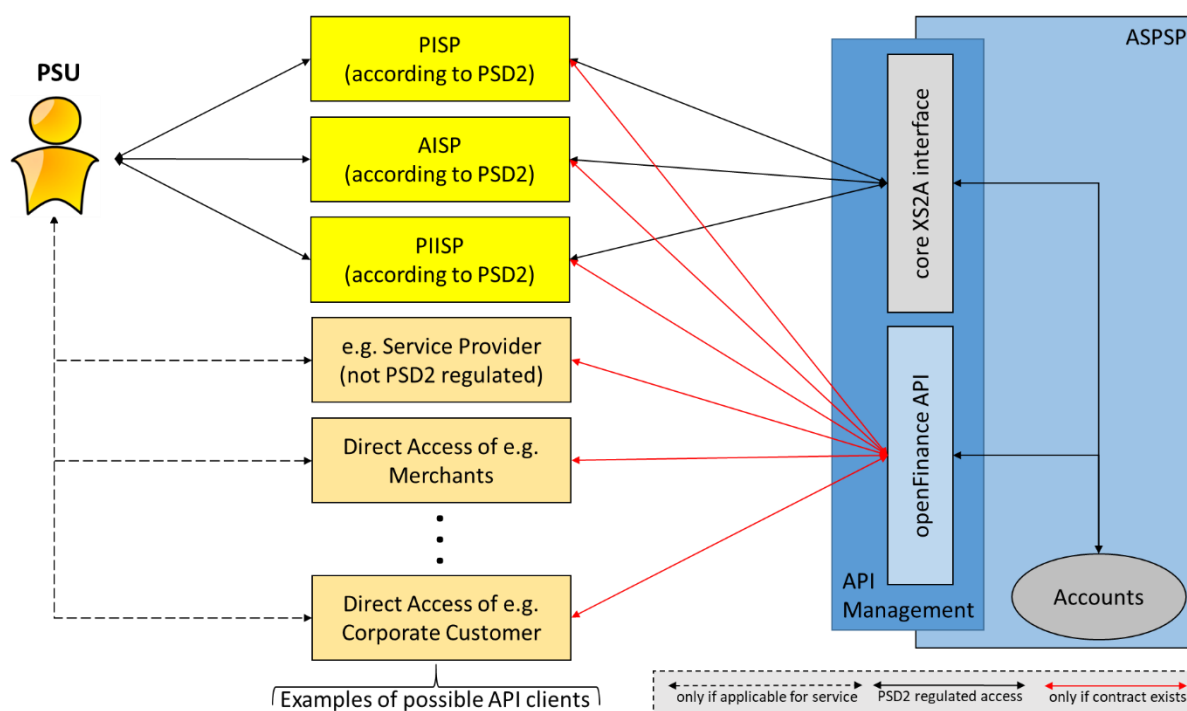


Figure 1: Core XS2A API and openFinance API

## 1.2 Data Dictionary for the openFinance API Framework

This document specifies the complex data structure to support the core XS2A API Services as well as the consent API specified for the openFinance API Framework. Also first V2 Premium Services are already covered.

Future V2 Premium Services will rely on this data dictionary as well, but might also define their own set of complex data structures which are not supported yet in the current version of the data dictionary.

## 1.3 Document Structure

This document covers complex data types in Section 2 and code lists in Section 3.

## 1.4 Document History

| Version | Change/Note                                                                                                                                                                                                                                                                                                                                                                                                                              | Approved                           |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 2.0     | Initial version of data dictionary for the openFinance API Framework Version 2                                                                                                                                                                                                                                                                                                                                                           | 05 October 2023<br>openFinance TF  |
| 2.1     | Updating the dictionary for attributes from Extended Payment Services, Extended Account Information Services, RTP Services, Push AIS Services, Admin Services as well as Discovery Services.                                                                                                                                                                                                                                             | 08 February 2024<br>openFinance TF |
| 2.2     | Updates related to openFinance API Framework Release 2.1<br><br>Details of the changes are covered in section 5.1                                                                                                                                                                                                                                                                                                                        | 31 July 2024 openFinance TF        |
| 2.2.1   | Introducing new data attributes and codes for the Mandate API.<br><br>Updated Message Codes and some minor Errata<br><br>Details of the changes are covered in Section 5.2                                                                                                                                                                                                                                                               | 21 October 2024 oFA TF             |
| 2.2.2   | Introducing new attributes for extensions of the Discovery Services for new services<br><br>Introducing new data attributes for the new VOP Services and the Document Services.<br><br>Updated several code sets, some minor errata.<br><br>Adds some attributes in several Data Types used in Discovery Services to support new features from the openFinance API Release 2.1.<br><br>Details on the changes are covered in Section 5.3 | 16 December 2024 oFA TF            |

| Version | Change/Note                                                                                                                                                                                                                                                                                                                                                                                                                  | Approved               |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 2.2.3   | <p>Added new data types and code lists for the new VOP Service and VOP integration into payment services.</p> <p>Added related attributes in the data types needed for the Discovery API</p> <p>Changed methodology of dash based attributes to lowerCamelCase for header parameters which are transported in the payload of the Discovery API.</p> <p>Errata</p> <p>Details of the changes are contained in Section 5.4</p> | 24 January 2025 oFA TF |

**RFU:** Please note that the Berlin Group openFinance API Framework is still under constant development. Technical issues, which are already in discussion within the Berlin Group openFinance working structure are mentioned in this document by "RFU" indicating "Remark for Future" to make the reader aware of upcoming potential changes.

## 2 Complex Data Types

In the following constructed data types are defined as used within the openFinance API Framework.

**NOTE:** Some of the data models have been changed from Version 1.3.x to Version 2.x of the openFinance API Framework. All attributes, where changes in subattributes applied, are marked by **yellow colour**.

**NOTE:** It is indicated below each data type, in which openFinance API Framework Implementation Guidelines the related data type is used in attributes of API requests and API responses on **first** level.

**NOTE:** The NISP project is a project platform to support NextGenPSD2 APIs and in future also openFinance APIs. This project platform had a.o. worked on an ASPSP testcase catalogue. This testcase catalogue is generic due to the potential variants of NextGenPSD2 APIs and so-called "Implementer Options" were defined for an ASPSP configuration spreadsheet to support derivation of test cases from the generic test case catalogue. During this derivation, the implementer options supported filtering the relevant test cases applicable to an implementation following the defined configuration. Since these Implementer Options are also used by ASPSP to partially document the PSD2 API implementations and since the NISP project has submitted some of the test case related work related to discovery services to the Berlin Group, the data types defined within the context of discovery services [oFA-Disc] below use "(IOx1, ... ,xn)" in the attribute description column to indicate that the addressed attribute is related to NISP Implementer Option 1 to NISP Implementer Option n, where  $n \geq 1$ . In future versions, these references will be deleted.

### 2.1 Account Access

| Attribute    | Type                           | Condition    | Description                                                                                           |
|--------------|--------------------------------|--------------|-------------------------------------------------------------------------------------------------------|
| payments     | Array of Account Access Rights | {Or Optional | Addressing required access rights to current accounts.<br><br>The array may not be empty.             |
| cards        | Array of Account Access Rights | Or Optional  | Addressing required access rights to single cards accounts. The array may not be empty.               |
| cardAccounts | Array of Account Access Rights | Or Optional  | Addressing required access rights to card reconciliation accounts.<br><br>The array may not be empty. |

| Attribute  | Type                           | Condition    | Description                                                                                |
|------------|--------------------------------|--------------|--------------------------------------------------------------------------------------------|
| savings    | Array of Account Access Rights | Or Optional  | Addressing required access rights to saving accounts.<br><br>The array may not be empty.   |
| loans      | Array of Account Access Rights | Or Optional  | Addressing required access rights to loan accounts.<br><br>The array may not be empty.     |
| securities | Array of Account Access Rights | Or Optional} | Addressing required access rights to security accounts.<br><br>The array may not be empty. |

This data type is used in [oFA-Con].

## 2.2 Account Access Rights

The account access rights as posted by the API Client.

| Attribute | Type                        | Condition   | Description                                                               |
|-----------|-----------------------------|-------------|---------------------------------------------------------------------------|
| account   | Account Reference           | Conditional | Is mandated if the consentType equals "detailed", not supported otherwise |
| rights    | List of Access Rights Codes | Mandatory   |                                                                           |

This data type is used in [oFA-Con] and by this document.

## 2.3 Account Access Rights Extended

The account access rights as provided by the ASPSP.

| Attribute | Type              | Condition   | Description                                                                               |
|-----------|-------------------|-------------|-------------------------------------------------------------------------------------------|
| account   | Account Reference | Conditional | Provided after customer authentication if not already submitted by the API Client itself. |

| Attribute | Type                        | Condition | Description                                                  |
|-----------|-----------------------------|-----------|--------------------------------------------------------------|
| rights    | List of Access Rights Codes | Mandatory |                                                              |
| _links    | Links                       | Optional  | A hyperlink to the endpoint of the related account category. |

This data type is used in [oFA-Con] and by this document.

## 2.4 Account Access and Authorisation Rights

| Attribute             | Type                        | Condition | Description                                                                                                                                                 |
|-----------------------|-----------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| account               | Account Reference           | Mandatory |                                                                                                                                                             |
| rights                | Array of Access Right Codes | Mandatory | Defines the reading rights on the related account identified above.                                                                                         |
| limits                | Array of Limit              | Optional  | Defines the limits applicable only to this account.                                                                                                         |
| paymentAuthorisations | Array of Product Rights     | Mandatory | Defines authorisation related rights on level of dedicated payment products including specific limits.<br><br>Empty array, if no payment right is provided. |
| paymentCancellations  | Array of Product Rights     | Mandatory | Defines payment cancellation rights on level of dedicated payment products.<br><br>Empty array, if no payment cancellation right is provided.               |

This data type is used in [oFA-Disc].

## 2.5 Account Access and Authorisation Rights1

| Attribute | Type              | Condition | Description |
|-----------|-------------------|-----------|-------------|
| account   | Account Reference | Mandatory |             |



| Attribute             | Type                        | Condition | Description                                                                                            |
|-----------------------|-----------------------------|-----------|--------------------------------------------------------------------------------------------------------|
| rights                | Array of Access Right Codes | Mandatory |                                                                                                        |
| paymentAuthorisations | Array of Product Rights     | Mandatory | Defines authorisation related rights on level of dedicated payment products including specific limits. |

This data type is used in [oFA-Disc].

## 2.6 Account Access Extended

| Attribute  | Type                                    | Condition    | Description                                                                                                                                                                                                      |
|------------|-----------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| payments   | Array of Account Access Rights Extended | {Or Optional | Addressing required access rights to current accounts.<br><br>The array may not be empty.                                                                                                                        |
| cards      | Array of Account Access Rights Extended | Or Optional  | Addressing required access rights to card accounts. The accounts can be card reconciliation accounts as well as single credit cards, dependent of the AIS offer of the ASPSP.<br><br>The array may not be empty. |
| savings    | Array of Account Access Rights Extended | Or Optional  | Addressing required access rights to saving accounts.<br><br>The array may not be empty.                                                                                                                         |
| loans      | Array of Account Access Rights Extended | Or Optional  | Addressing required access rights to loan accounts.<br><br>The array may not be empty.                                                                                                                           |
| securities | Array of Account                        | Or           | Addressing required access rights to security                                                                                                                                                                    |

| Attribute | Type                   | Condition | Description                                  |
|-----------|------------------------|-----------|----------------------------------------------|
|           | Access Rights Extended | Optional} | accounts.<br><br>The array may not be empty. |

This data type is used in [oFA-Con].

## 2.7 Account Details

**Remark:** The ASPSP shall give at least one of the account reference identifiers listed as optional below.

| Attribute  | Type                           | Condition   | Description                                                                                                                                                                                                                                            |
|------------|--------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| resourceId | Max70Text                      | Mandatory   | This is the data element to be used in the path when retrieving data from a dedicated account.                                                                                                                                                         |
| iban       | IBAN                           | Optional    | IBAN of the account.                                                                                                                                                                                                                                   |
| bban       | BBAN                           | Optional    | BBAN of the account. This data elements is used for payment accounts which have no IBAN.                                                                                                                                                               |
| msisdn     | Max35Text                      | optional    | An alias to access a payment account via a registered mobile phone number. This alias might be needed e.g. in the payment initiation services. . The support of this alias must be explicitly documented by the ASPSP for the corresponding API Calls. |
| other      | Generic Account Identification | Conditional | An alias with a proprietary coding                                                                                                                                                                                                                     |
| currency   | Currency Code                  | Mandatory   | Account currency                                                                                                                                                                                                                                       |
| ownerName  | Max140Text                     | Optional    | Name of the legal account owner. If there is more than one owner, then e.g. two names might be noted here.<br><br>For a corporate account, the corporate name is used for this attribute.                                                              |

| Attribute       | Type                   | Condition | Description                                                                                                                                                                                 |
|-----------------|------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |                        |           | Even if supported by the ASPSP, the provision of this field might depend on the fact whether an explicit consent to this specific additional account information has been given by the PSU. |
| ownerNames      | Array of Account Owner | Optional  | List of owner names together with a role, where applicable.                                                                                                                                 |
| name            | Max70Text              | Optional  | Name of the account, as assigned by the ASPSP, in agreement with the account owner in order to provide an additional means of identification of the account.                                |
| displayName     | Max70Text              | Optional  | Name of the account as defined by the PSU within online channels.                                                                                                                           |
| product         | Max35Text              | Optional  | Product Name of the Bank for this account, proprietary definition                                                                                                                           |
| cashAccountType | Cash Account Type      | Optional  | ExternalCashAccountType1Code from ISO 20022                                                                                                                                                 |
| status          | Account Status Code    | Optional  | Account status. If this field is not used, than the account is available in the sense of this specification.                                                                                |
| bic             | BICFI                  | Optional  | The BIC associated to the account.                                                                                                                                                          |
| linkedAccounts  | Max70 Text             | Optional  | This data attribute is a field, where an ASPSP can name a cash account associated to the described account, e.g. in case of a securities account or a card account.                         |
| usage           | Max4 Text              | Optional  | Specifies the usage of the account <ul style="list-style-type: none"> <li>- PRIV: private personal account</li> <li>- ORGA: professional account</li> </ul>                                 |

| Attribute           | Type                                  | Condition   | Description                                                                                                                                                                                                                                                                                                                                 |
|---------------------|---------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| details             | Max500<br>Text                        | Optional    | Specifications that might be provided by the ASPSP<br><br>- characteristics of the account<br><br>- characteristics of the relevant card                                                                                                                                                                                                    |
| balances            | Array of<br>Balances                  | Conditional |                                                                                                                                                                                                                                                                                                                                             |
| tariffs             | Max1000Text                           | Optional    | Currently only supported for securities accounts:<br><br>Text information on tariffs related to this securities account.                                                                                                                                                                                                                    |
| applicableFees      | Array of<br>Securities<br>Account Fee | Optional    | Currently only supported for securities accounts:<br><br>Array of elements, each representing one fee (rule) that is individually applicable for this Securities account. If, for example, an ASPSP applies a standard "managementFee" X, but the consumer negotiated a discount D, an element should represent (X-D), but neither X nor D. |
| interest            | Array of<br>Interest                  | Optional    | Currently only supported for savings accounts and loan accounts:<br><br>A set of interest rates associated to this account, e.g. interests for savings or credits.                                                                                                                                                                          |
| relatedDates        | Account<br>related Dates              | Optional    | Currently only supported for savings accounts and loan accounts:<br><br>Dates related to this account, e.g. start Date of a loan in case of a loan account.                                                                                                                                                                                 |
| collateralsInvolved | Boolean                               | Optional    | Currently only supported for loan accounts:<br><br>Is set to true if a collateral is involved securing the loan (e.g. security deposit, physical collaterals like houses, cars). More information might be provided under the "details" attribute.                                                                                          |

| Attribute         | Type    | Condition | Description                                                                                                                                                                                                                                                       |
|-------------------|---------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| guaranteeInvolved | Boolean | Optional  | Currently only supported for loan accounts:<br><br>Is set to true in case of specific collaterals where a guarantee of payment of the loan is given by a different person.                                                                                        |
| _links            | Links   | Optional  | Links to the account, which can be directly used for retrieving account information from this dedicated account.<br><br>Links to "balances" and/or "transactions"<br><br>These links are only supported, when the corresponding consent has been already granted. |

This data type is used in [oFA-IG-Com], [oFA-IG-XAIS].

## 2.8 Account Owner

| Attribute | Type      | Condition | Description                                                                                       |
|-----------|-----------|-----------|---------------------------------------------------------------------------------------------------|
| name      | Max70Text | Mandatory | Account owner name                                                                                |
| role      | Max35Text | Optional  | The following proprietary codes are used:<br><br>"owner", "legalRepresentative", "authorisedUser" |

This data type is used in [oFA-IG-Com].

## 2.9 Account Reference

This type is containing any account identification which can be used on payload-level to address specific accounts. The ASPSP will document which account reference type it will support. Exactly one of the attributes defined as "conditional" shall be used.

**Remark:** The currency of the account is needed, where the currency is an account characteristic identifying certain sub-accounts under one external identifier like an

IBAN. These sub-accounts are separated accounts from a legal point of view and have separated balances, transactions etc.

| Attribute       | Type                           | Condition       | Description                                                                                                       |
|-----------------|--------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------|
| iban            | IBAN                           | Conditional     |                                                                                                                   |
| bban            | BBAN                           | Conditional     | This data elements is used for payment accounts which have no IBAN.                                               |
| pan             | Max35Text                      | Conditional     | Primary Account Number (PAN) of a card, can be tokenised by the ASPSP due to PCI DSS requirements.                |
| maskedPan       | Max35Text                      | Conditional     | Primary Account Number (PAN) of a card in a masked form.                                                          |
| msisdn          | Max35Text                      | Conditional     | An alias to access a payment account via a registered mobile phone number.                                        |
| other           | Generic Account Identification | Conditional     | An alias with a proprietary coding                                                                                |
| typeCode        | Cash Account Type              | {Or - Optional} | Remark: Shall not be used in the consent model, since account category provides enough information for the ASPSP. |
| typeProprietary | Max35Text                      | Or – Optional}  | Remark: Shall not be used in the consent model, since account category provides enough information for the ASPSP. |
| currency        | Currency Code                  | Optional        | ISO 4217 Alpha 3 currency code                                                                                    |
| proxy           | Proxy Account Identification   | Conditional     |                                                                                                                   |

This data type is used in [oFA-IG-Com], [oFA-PFDom]. [oFA -IG-PAIS], [oFA-IG-VOP].

**2.10 Account Reference1**

| Attribute | Type          | Condition   | Description                                                                                        |
|-----------|---------------|-------------|----------------------------------------------------------------------------------------------------|
| iban      | IBAN          | Conditional |                                                                                                    |
| bban      | BBAN          | Conditional | This data element is used for payment accounts which have no IBAN.                                 |
| pan       | Max35Text     | Conditional | Primary Account Number (PAN) of a card, can be tokenised by the ASPSP due to PCI DSS requirements. |
| maskedPan | Max35Text     | Conditional | Primary Account Number (PAN) of a card in a masked form.                                           |
| currency  | Currency Code | Optional    | ISO 4217 Alpha 3 currency code                                                                     |

This data type is used in [oFA-PFDom].

**2.11 Account Reference2**

| Attribute | Type          | Condition | Description                    |
|-----------|---------------|-----------|--------------------------------|
| iban      | IBAN          | Mandatory |                                |
| currency  | Currency Code | Optional  | ISO 4217 Alpha 3 currency code |

**2.12 Account Reference3**

| Attribute | Type                           | Condition   | Description                                                        |
|-----------|--------------------------------|-------------|--------------------------------------------------------------------|
| iban      | IBAN                           | Conditional |                                                                    |
| bban      | BBAN                           | Conditional | This data element is used for payment accounts which have no IBAN. |
| other     | Generic Account Identification | Conditional | An alias with a proprietary coding                                 |

| Attribute | Type          | Condition | Description                    |
|-----------|---------------|-----------|--------------------------------|
| currency  | Currency Code | optional  | ISO 4217 Alpha 3 currency code |

This data type is used in [oFA-PFDom].

### 2.13 Account Reference4

| Attribute | Type                           | Condition | Description |
|-----------|--------------------------------|-----------|-------------|
| iban      | IBAN                           | Optional  |             |
| other     | Generic Account Identification | Optional  |             |

### 2.14 Account related Dates

| Attribute                | Type    | Condition | Description                                                                                                                                                                                                       |
|--------------------------|---------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| contractStartDate        | ISODate | Optional  | Start of the account contract.                                                                                                                                                                                    |
| contractEndDate          | ISODate | Optional  | End of the account contract.                                                                                                                                                                                      |
| contractAvailabilityDate | ISODate | Optional  | <p>Next Date where money can be taken off a saving account without loss of interest, if initiated now (in case of a saving account).</p> <p>Next Date where a loan can be fully repaid without any extra fee.</p> |

### 2.15 Account Report

| Attribute | Type                  | Condition   | Description                                                                        |
|-----------|-----------------------|-------------|------------------------------------------------------------------------------------|
| booked    | Array of transactions | Conditional | Shall be contained if bookingStatus parameter is set to "booked", "both" or "all". |



| Attribute   | Type                  | Condition | Description                                                                                                                                                                                                                                     |
|-------------|-----------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pending     | Array of transactions | Optional  | Not contained if the bookingStatus parameter is set to "booked" or "information".                                                                                                                                                               |
| information | Array of transactions | Optional  | Only contained if the bookingStatus is set to "information" or "all" and if supported by ASPSP.                                                                                                                                                 |
| _links      | Links                 | Mandatory | The following links might be used within this context: <ul style="list-style-type: none"> <li>• account (mandatory)</li> <li>• first (optional)</li> <li>• next (optional)</li> <li>• previous (optional)</li> <li>• last (optional)</li> </ul> |

This data type is used in [oFA-IG-Com].

## 2.16 Account Resource Reference

| Attribute  | Type | Condition   | Description                                                                    |
|------------|------|-------------|--------------------------------------------------------------------------------|
| resourceId | UUID | Conditional | To be used in authorisation processes, when updating a resource by an account. |

This data type is used in [oFA-PFSM]

## 2.17 Accrued Interest

Represents the accrued interest by a securities position:

| Attribute   | Type            | Condition | Description                                                                    |
|-------------|-----------------|-----------|--------------------------------------------------------------------------------|
| daysAccrued | Number          | Optional  | Specifies the number of days used for calculating the accrued interest amount. |
| amounts     | Array of Amount | Optional  | Amount of the accrued interest. Each item represents                           |

| Attribute | Type | Condition | Description                                                                                                      |
|-----------|------|-----------|------------------------------------------------------------------------------------------------------------------|
|           |      |           | the same monetary value in different currencies, e.g. account currency, currency of the security's denomination. |

## 2.18 Additional Party Information

| Attribute            | Type                   | Condition | Description                                                   |
|----------------------|------------------------|-----------|---------------------------------------------------------------|
| tradeName            | Max70Text              | Optional  | Trade name of the related party. For display to the PSU only. |
| merchantCategoryCode | Merchant Category Code | Optional  | Merchant Category Code as assigned by ISO:                    |
| logoUrl              | Max2048 Text           | Optional  | A hyperlink to the logo of the party.                         |
| geoLocation          | GEO Location           | Optional  |                                                               |
| merchant Logo        | String                 | n.a.      | Base 64 coded graphic. Reserved for future extensions.        |

## 2.19 Additional Request Information

| Attribute                | Type                                 | Condition | Description                                                                     |
|--------------------------|--------------------------------------|-----------|---------------------------------------------------------------------------------|
| paymentConditions        | Payment Conditions                   | optional  | Conditions for the execution of the payment.                                    |
| creationDateTime         | ISODateTime                          | optional  | creationDateTime of the originator                                              |
| requestedExpiry DateTime | ISODateTime                          | mandatory | Date and Time at which this request expires.                                    |
| originatorPspId          | Financial Institution Identification | Optional  | Originator payment service provider in case of indirect participation regarding |

| Attribute     | Type                                 | Condition   | Description                                                                                                                                                                                                                                      |
|---------------|--------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                                      |             | a related API Access Scheme or bilateral contract                                                                                                                                                                                                |
| receiverPspId | Financial Institution Identification | Conditional | Receiving payment service provider as defined by an API Access Scheme or bilateral contracts in case of not addressing the receiving institution directly. It is mandatory in case where the technical API Provider is not the receiver as such. |

This data type is used in [oFA-IG-VOP].

## 2.20 Additional Request Information1

| Attribute       | Type                                 | Condition | Description                                                                                                                       |
|-----------------|--------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
| originatorPspId | Financial Institution Identification | Optional  | Originator payment service provider in case of indirect participation regarding a related API Access Scheme or bilateral contract |

This data type is used in [oFA-Con], [oFA-PFDom].

## 2.21 Address

| Attribute      | Type         | Condition | Description |
|----------------|--------------|-----------|-------------|
| streetName     | Max70Text    | Optional  |             |
| buildingNumber | String       | Optional  |             |
| townName       | String       | Optional  |             |
| postCode       | String       | Optional  |             |
| country        | Country Code | Mandatory |             |

## 2.22 Agent Description

| Attribute                | Type                                 | Condition | Description                                                                       |
|--------------------------|--------------------------------------|-----------|-----------------------------------------------------------------------------------|
| financialInstitutionId   | Financial Institution Identification | mandatory |                                                                                   |
| branchIdentificationId   | Max35Text                            | Optional  | Unique and unambiguous identification of a branch of a financial institution.     |
| branchIdentificationName | Max140Text                           | Optional  | Name by which an agent is known and which is usually used to identify that agent. |

## 2.23 Agent Description1

| Attribute              | Type                                 | Condition | Description |
|------------------------|--------------------------------------|-----------|-------------|
| financialInstitutionId | Financial Institution Identification | mandatory |             |

This data type is used in [oFA-PFDom].

## 2.24 Agent Description2

| Attribute              | Type                                  | Condition | Description |
|------------------------|---------------------------------------|-----------|-------------|
| financialInstitutionId | Financial Institution Identification1 | mandatory |             |

This data type is used in [oFA-PFDom].

## 2.25 Amendment Information Details

| Attribute         | Type      | Condition | Description                                                                     |
|-------------------|-----------|-----------|---------------------------------------------------------------------------------|
| originalMandateId | Max35Text | Optional  | Mandatory of change occur in "Mandate Identification" otherwise not to be used. |

| Attribute                | Type               | Condition | Description                                                                             |
|--------------------------|--------------------|-----------|-----------------------------------------------------------------------------------------|
| originalCreditorSchemeld | Party Description4 | Optional  | Mandatory of change occur in "Creditor Scheme Identification" otherwise not to be used. |
| originalDebtorAccount    | Account Reference4 | Optional  | Mandatory of change occur in "Debtor Account" otherwise not to be used.                 |
| originalDebtorAgent      | Agent Description2 | Optional  | If changes occur in debtor agent and SMNDA is not used in "originalDebtorAccount"       |

## 2.26 Amount

| Attribute | Type          | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|---------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| currency  | Currency Code | Mandatory | ISO 4217 Alpha 3 currency code                                                                                                                                                                                                                                                                                                                                                                                            |
| amount    | String        | Mandatory | <p>The amount given with fractional digits, where fractions must be compliant to the currency definition. Up to 14 significant figures. Negative amounts are signed by minus.</p> <p>The decimal separator is a dot.</p> <p><b>Example:</b> Valid representations for EUR with up to two decimals are:</p> <ul style="list-style-type: none"> <li>• 1056</li> <li>• 5768.2</li> <li>• -1.50</li> <li>• 5877.78</li> </ul> |

This data type is used in [oFA-IG-Com], [oFA-PFSM], [oFA-IG-XPIS]., [oFA-IG-RTP].

## 2.27 Amount dependent Rate

| Attribute  | Type   | Condition | Description                                                                                                                                       |
|------------|--------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| percentage | String | Mandatory | The (signed) rate of the interest.<br><br>Up to 20 significant figures. The decimal separator is a dot.<br><br>Trailing zeroes must be truncated. |
| fromAmount | Amount | Optional  | Amount from which the interest rate applies.                                                                                                      |
| toAmount   | Amount | Optional  | Amount up to which the interest rate applies.                                                                                                     |

**Remark:** ISO 20022 is defining Percentage as a decimal type, where 5.0 expresses 5% interest rate. This specification is using a signed percentage to be able to express also negative percentage codes, i.e. -1.2 will express -1,2% of interest.

## 2.28 Amount Details

| Attribute         | Type                        | Condition | Description |
|-------------------|-----------------------------|-----------|-------------|
| instructedAmount  | Amount and Exchange Details | Optional  |             |
| transactionAmount | Amount and Exchange Details | Optional  |             |

### Example:

```
"amountDetails":  
  {  
    "instructedAmount" :  
      {  
        "currency": "DKK",  
        "amount": "123.50",
```

```

        "sourceCurrency": "DKK",
        "targetCurrency": "EUR",
        "exchangeRate": "7.47",
      },
      "transactionAmount" :
      {
        "currency": "EUR",
        "amount": "16.53",
        "sourceCurrency": "DKK",
        "targetCurrency": "EUR",
        "exchangeRate": "7.60",
      }
    }
  }

```

## 2.29 Amount and Exchange Details

| Attribute      | Type          | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------|---------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| currency       | Currency Code | Mandatory | ISO 4217 Alpha 3 currency code                                                                                                                                                                                                                                                                                                                                                                                            |
| amount         | String        | Mandatory | <p>The amount given with fractional digits, where fractions must be compliant to the currency definition. Up to 14 significant figures. Negative amounts are signed by minus.</p> <p>The decimal separator is a dot.</p> <p><b>Example:</b> Valid representations for EUR with up to two decimals are:</p> <ul style="list-style-type: none"> <li>• 1056</li> <li>• 5768.2</li> <li>• -1.50</li> <li>• 5877.78</li> </ul> |
| sourceCurrency | Currency Code | Optional  | Currency from which an amount is to be converted in a currency conversion.                                                                                                                                                                                                                                                                                                                                                |
| exchangeRate   | String        | Optional  | Factor used to convert an amount from one currency into another. This reflects the price at which one currency was bought with another currency.                                                                                                                                                                                                                                                                          |

| Attribute      | Type          | Condition | Description                                                                                                                         |
|----------------|---------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------|
|                |               |           | <b>Note:</b> If this field is present, then also the field sourceCurrency must be provided.                                         |
| unitCurrency   | Currency Code | Optional  | Currency in which the rate of exchange is expressed in a currency exchange. In the example 1EUR = xxxCUR, the unit currency is EUR. |
| targetCurrency | Currency Code | Optional  | Currency into which an amount is to be converted in a currency conversion.                                                          |
| quotationDate  | ISODate       | Optional  | Date at which an exchange rate is quoted.                                                                                           |
| contractId     | String        | Optional  | Unique identification to unambiguously identify the foreign exchange contract.                                                      |

### 2.30 API Instance Parameters

| Attribute      | Type        | Condition | Description                                                                                                                                                   |
|----------------|-------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiName        | Max35Text   | Mandatory | "Berlin Group openFinance", "giroAPI", "EPC SPAA", "Berlin Group XS2A", "Berlin Group Direct Access".<br><br>More entries might be added by the API Provider. |
| releaseVersion | Max35Text   | Mandatory | Release version, not necessary connected to the standard version.                                                                                             |
| validFrom      | ISODate     | Optional  |                                                                                                                                                               |
| validTo        | ISODate     | Optional  |                                                                                                                                                               |
| releaseNote    | Max2000Text | Optional  | Offers a summary on the new release.                                                                                                                          |
| _links         | Links       | Mandatory | A list of hyperlinks to be recognised by the TPP. The actual hyperlinks used in the response depend on the dynamical                                          |



| Attribute | Type | Condition | Description                                                                                                                                                                                                                                                                                                                      |
|-----------|------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |      |           | <p>decisions of the ASPSP when processing the request.</p> <p>Remark: All links can be relative or full links, to be decided by the ASPSP.</p> <p>The following valid values for links are supported:</p> <ul style="list-style-type: none"> <li>- aspspParameters</li> <li>- aspspContacts</li> <li>- aspspDowntimes</li> </ul> |

This data type is used in [oFA-Disc].

### 2.31 ASPSP API Access

| Attribute              | Type                                  | Condition | Description                                                                                            |
|------------------------|---------------------------------------|-----------|--------------------------------------------------------------------------------------------------------|
| lastUpdate             | ISODateTime                           | Mandatory | Last Update in one of the attributes of entities referenced by links provided in apiInstances          |
| aspspName              | Max70Text                             | Mandatory | Name ASPSP:                                                                                            |
| aspspBranch            | Max70Text                             | Optional  | Branch or Division in case of multiple APIs per ASPSP which relate to different branches/divisions     |
| aspspChannel           | Max70Text                             | Optional  | Online Channel in case of multiple APIs per ASPSP Online Channel, e.g. "small corporates" or "retail". |
| aspspIdentifications   | Array of BICFI                        | Mandatory | All bank identifications which are covered by the addressed API.                                       |
| clearingSystemMemberId | Clearing System Member Identification | Optional  |                                                                                                        |
| accessScheme           | Max35Text                             | Optional  | Scheme-identification                                                                                  |

| Attribute        | Type             | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------|------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| aspspDescription | Max140Text       | Mandatory | Description of API provider party.                                                                                                                                                                                                                                                                                                                                                                                              |
| validFrom        | ISODate          | Optional  |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| validTo          | ISODate          | Optional  |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| contacts         | Array of Contact | Optional  |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| _links           | Links            | Optional  | <p>A list of hyperlinks to be recognised by the API Client. The actual hyperlinks used in the response depend on the dynamical decisions of the ASPSP when processing the request.</p> <p>Remark: All links can be relative or full links, to be decided by the ASPSP.</p> <p>The following valid values for links are supported:</p> <ul style="list-style-type: none"> <li>- aspspContacts</li> <li>- apiInstances</li> </ul> |

This data type is used in [oFA-Disc].

## 2.32 Authentication Object

| Attribute             | Type                | Condition   | Description                                                                                                                                                                                                                                  |
|-----------------------|---------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| authenticationType    | Authentication Type | Mandatory   | Type of the authentication method.                                                                                                                                                                                                           |
| authenticationVersion | String              | Conditional | Depending on the "authenticationType". This version can be used by differentiating authentication tools used within performing OTP generation in the same authentication type. This version can be referred to in the ASPSP's documentation. |

| Attribute              | Type      | Condition | Description                                                                                                                                                                                                                                                                                                                                        |
|------------------------|-----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| authenticationMethodId | Max35Text | Mandatory | An identification provided by the ASPSP for the later identification of the authentication method selection.                                                                                                                                                                                                                                       |
| name                   | String    | Mandatory | <p>This is the name of the authentication method defined by the PSU in the Online Banking frontend of the ASPSP. Alternatively this could be a description provided by the ASPSP like "SMS OTP on phone +49160 xxxxx 28".</p> <p>This name shall be used by the TPP when presenting a list of authentication methods to the PSU, if available.</p> |
| explanation            | String    | Optional  | detailed information about the SCA method for the PSU                                                                                                                                                                                                                                                                                              |

This data type is used in [oFA-Con], [oFA-IG-Com], [oFA-PFSM], [oFA-IG-XPIS] .

### 2.33 Authentication Type

More authentication types might be added during implementation projects and documented in the ASPSP documentation.

| Name     | Description                                                                                                                                                                                                                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SMS_OTP  | An SCA method, where an OTP linked to the transaction to be authorised is sent to the PSU through a SMS channel.                                                                                                                                                                                                                               |
| CHIP_OTP | An SCA method, where an OTP is generated by a chip card, e.g. an TOP derived from an EMV cryptogram. To contact the card, the PSU normally needs a (handheld) device. With this device, the PSU either reads the challenging data through a visual interface like flickering or the PSU types in the challenge through the device key pad. The |

| Name      | Description                                                                                                                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | device then derives an OTP from the challenge data and displays the OTP to the PSU.                                                                                                                                                                                           |
| PHOTO_OTP | An SCA method, where the challenge is a QR code or similar encoded visual data which can be read in by a consumer device or specific mobile app.<br><br>The device resp. the specific app then derives an OTP from the visual challenge data and displays the OTP to the PSU. |
| PUSH_OTP  | An OTP is pushed to a dedicated authentication APP and displayed to the PSU.                                                                                                                                                                                                  |
| SMTP_OTP  | An OTP is sent via email to the PSU.                                                                                                                                                                                                                                          |

## 2.34 Balance

| Attribute                 | Type         | Condition | Description                                                                                                                                           |
|---------------------------|--------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| balanceAmount             | Amount       | Mandatory |                                                                                                                                                       |
| balanceType               | Balance Type | Mandatory |                                                                                                                                                       |
| creditLimitIncluded       | Boolean      | Optional  | A flag indicating if the credit limit of the corresponding account is included in the calculation of the balance, where applicable.                   |
| lastChangeDateTime        | ISODateTime  | Optional  | This data element might be used to indicate e.g. with the expected or booked balance that no action is known on the account, which is not yet booked. |
| referenceDate             | ISODate      | Optional  | indicates the date of the balance                                                                                                                     |
| lastCommitted Transaction | Max35Text    | Optional  | entryReference of the last committed transaction to support the TPP in identifying whether all PSU transactions are already known.                    |

This data type is used in [oFA-IG-Com].

### 2.35 Binary Document

Base64 encoding of the binary file representing the enclosed document or template, such as a PDF file, image file, XML file, MT message. The maximum size of the binary document is 10MB.

### 2.36 Card Account Details

| Attribute   | Type          | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|---------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| resourceId  | String        | Mandatory | This is the data element to be used in the path when retrieving data from a dedicated account.                                                                                                                                                                                                                                                                                                      |
| maskedPan   | Max35Text     | Mandatory | Primary Account Number (PAN) of the main card in masked form..                                                                                                                                                                                                                                                                                                                                      |
| currency    | Currency Code | Mandatory | Account currency                                                                                                                                                                                                                                                                                                                                                                                    |
| ownerName   | Max140Text    | Optional  | <p>Name of the legal account owner. If there is more than one owner, then e.g. two names might be noted here.</p> <p>For a corporate account, the corporate name is used for this attribute.</p> <p>Even if supported by the ASPSP, the provision of this field might depend on the fact whether an explicit consent to this specific additional account information has been given by the PSU.</p> |
| name        | Max70Text     | Optional  | Name of the account, as assigned by the ASPSP, in agreement with the account owner in order to provide an additional means of identification of the account.                                                                                                                                                                                                                                        |
| displayName | Max70Text     | Optional  | Name of the account as defined by the PSU within online channels.                                                                                                                                                                                                                                                                                                                                   |

| Attribute       | Type              | Condition | Description                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|-------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| product         | Max35Text         | Optional  | Product Name of the Bank for this account, proprietary definition                                                                                                                                                                                                                                                                                             |
| debitAccounting | Boolean           | Optional  | <p>If true, the amounts of debits on the reports are quoted positive with the related consequence for balances.</p> <p>If false, the amount of debits on the reports are quoted negative.</p>                                                                                                                                                                 |
| status          | String            | Optional  | <p>Account status. The value is one of the following:</p> <ul style="list-style-type: none"> <li>• "enabled": account is available</li> <li>• "deleted": account is terminated</li> <li>• "blocked": account is blocked e.g. for legal reasons</li> </ul> <p>If this field is not used, than the account is available in the sense of this specification.</p> |
| usage           | Max140Text        | Optional  | <p>Specifies the usage of the account</p> <ul style="list-style-type: none"> <li>- PRIV: private personal account</li> <li>- ORGA: professional account</li> </ul>                                                                                                                                                                                            |
| details         | Max1000Text       | Optional  | <p>Specifications that might be provided by the ASPSP</p> <ul style="list-style-type: none"> <li>- characteristics of the account</li> <li>- characteristics of the relevant card</li> </ul>                                                                                                                                                                  |
| creditLimit     | Amount            | Optional  | Defines the credit limit of the PSU aggregated for all cards related to this card account in total.                                                                                                                                                                                                                                                           |
| balances        | Array of Balances | Optional  | The specific card account balances associated to this card accounts.                                                                                                                                                                                                                                                                                          |
| _links          | Links             | Optional  | Links to the cardAccount, which can be directly used for retrieving account information from this dedicated account.                                                                                                                                                                                                                                          |

| Attribute | Type | Condition | Description                                                                                                                                          |
|-----------|------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |      |           | <p>Links to "balances" and/or "cardTransactions"</p> <p>These links are only supported, when the corresponding consent has been already granted.</p> |

This data type is used in [oFA-IG-Com].

### 2.37 Card Account Report

| Attribute | Type                       | Condition   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------|----------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| booked    | Array of Card Transactions | Conditional | Shall be contained if bookingStatus parameter is set to "booked" or "both".                                                                                                                                                                                                                                                                                                                                                                   |
| pending   | Array of Card Transactions | Optional    | Not contained if the bookingStatus parameter is set to "booked".                                                                                                                                                                                                                                                                                                                                                                              |
| _links    | Links                      | Mandatory   | <p>The following links might be used within this context:</p> <ul style="list-style-type: none"> <li>• cardAccount (mandatory when providing transaction reports on card reconciliation accounts under /card-accounts)</li> <li>• card (mandatory when providing transaction reports on single card entry level under /cards)first (optional)</li> <li>• next (optional)</li> <li>• previous (optional)</li> <li>• last (optional)</li> </ul> |

This data type is used in [oFA-IG-Com].

## 2.38 Card Transactions

| Attribute                       | Type                                   | Condition | Description                                                                                                                                                                                                                                                              |
|---------------------------------|----------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cardTransactionId               | Max35Text                              | Optional  | Unique end to end identity.                                                                                                                                                                                                                                              |
| terminalId                      | Max35Text                              | Optional  | Identification of the Terminal, where the card has been used.                                                                                                                                                                                                            |
| transactionDate                 | ISODate                                | Optional  | date of the actual card transaction                                                                                                                                                                                                                                      |
| acceptorTransaction<br>DateTime | ISODate<br>Time                        | Optional  | Timestamp of the actual card transaction within the acceptance system                                                                                                                                                                                                    |
| bookingDate                     | ISODate                                | Optional  | booking date of the related booking on the card account                                                                                                                                                                                                                  |
| valueDate                       | ISODate                                | Optional  | <p>Date at which assets become available to the account owner in case of a credit entry, or cease to be available to the account owner in case of a debit entry.</p> <p>For card transactions this is the payment due date of related booked transactions of a card.</p> |
| transactionAmount               | Amount                                 | Mandatory | The amount of the transaction as billed to the card account.                                                                                                                                                                                                             |
| grandTotalAmount                | Amount                                 | Optional  | Total amount of the instalment including charges, insurance and taxes in addition to the funded amount.                                                                                                                                                                  |
| currencyExchange                | Array of<br>Report<br>Exchange<br>Rate | Optional  | For card accounts, this often is restricted by the ASPSP to use only one exchange rate.                                                                                                                                                                                  |
| originalAmount                  | Amount                                 | Optional  | Original amount of the transaction at the Point of Interaction in original currency                                                                                                                                                                                      |



| Attribute                      | Type                   | Condition | Description                                                                                                              |
|--------------------------------|------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------|
| markupFee                      | Amount                 | Optional  | Any fee related to the transaction in billing currency.                                                                  |
| markupFeePercentage            | String                 | Optional  | Percentage of the involved transaction fee in relation to the billing amount, e.g. "0.3" for 0,3%                        |
| cardAcceptorId                 | Max35Text              | Optional  | Identification of the Card Acceptor (e.g. merchant) as given in the related card transaction.                            |
| cardAcceptorName               | Max70Text              | Optional  | Name of the Card Acceptor (e.g. merchant) as provided in the related card transaction                                    |
| cardAcceptorAddress            | Address                | Optional  | Address of the Card Acceptor as given in the related card transaction.                                                   |
| cardAcceptorPhone              | Phone Number           | Optional  | Merchant phone number                                                                                                    |
| merchantCategoryCode           | Merchant Category Code | Optional  | Merchant Category Code of the Card Acceptor as given in the related card transaction.                                    |
| maskedPAN                      | Max35Text              | Optional  | The masked PAN of the card used in the transaction.                                                                      |
| transactionDetails             | Max1000Text            | Optional  | Additional details given for the related card transactions.                                                              |
| invoiced                       | Boolean                | Optional  | Flag indicating whether the underlying card transaction is already invoiced.                                             |
| proprietaryBankTransactionCode | Max35Text              | Optional  | proprietary bank transaction code as used within a community or within an ASPSP e.g. for MT94x based transaction reports |

### 2.39 Card Transaction

| Attribute   | Type                     | Condition | Description |
|-------------|--------------------------|-----------|-------------|
| card        | Card Type                | Optional  |             |
| transaction | Card Transaction Details | Optional  |             |

### 2.40 Card Transaction Details

| Attribute           | Type                 | Condition | Description |
|---------------------|----------------------|-----------|-------------|
| paymentContext      | Payment Context Type | Optional  |             |
| sequenceNumber      | Max35Text            | Optional  |             |
| transactionDateTime | ISODateTime          | Optional  |             |

### 2.41 Card Type

| Attribute      | Type      | Condition | Description |
|----------------|-----------|-----------|-------------|
| cardBrand      | Max35Text | Optional  |             |
| maskedPAN      | Max30Text | Optional  |             |
| cardHolderName | Max45Text | Optional  |             |

### 2.42 Challenge

| Attribute | Type   | Condition | Description                                                                                                                                                                                |
|-----------|--------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| image     | String | Optional  | <p>PNG data (max. 512 kilobyte) to be displayed to the PSU, Base64 encoding, cp. [RFC4648].</p> <p>This attribute is used only, when PHOTO_OTP or CHIP_OTP is the selected SCA method.</p> |

| Attribute              | Type             | Condition | Description                                                                                                                                  |
|------------------------|------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| data                   | Array of Strings | Optional  | A collection of challenge data                                                                                                               |
| imageLink              | String           | Optional  | A link where the ASPSP will provides the challenge image for the TPP.                                                                        |
| otpMaxLength           | Integer          | Optional  | The maximal length for the OTP to be typed in by the PSU.                                                                                    |
| otpFormat              | String           | Optional  | The format type of the OTP to be typed in. The admitted values are "characters" or "integer".                                                |
| additional Information | String           | Optional  | Additional explanation for the PSU to explain e.g. fallback mechanism for the chosen SCA method. The TPP is obliged to show this to the PSU. |

This data type is used in [oFA-Con] ,[oFA-IG-Com], [oFA-PFSM], [oFA-IG-XPIS].

## 2.43 Clearing System Member Identification

| Attribute                               | Type                                | Condition      | Description                                                                                                  |
|-----------------------------------------|-------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------|
| memberId                                | Max35Text                           | Optional       | Identification of a member of a clearing system.                                                             |
| clearingSystemIdentificationCode        | Clearing System Identification Code | {Or - Optional | Identification of a clearing system, in a coded form as published in an external list.                       |
| clearingSystemIdentificationProprietary | Max35Text                           | Or – Optional} | Identification code for a clearing system, that has not yet been identified in the list of clearing systems. |

This data type is used in [oFA-Disc].

## 2.44 Client Message Information

| Attribute | Type         | Condition   | Description                                                                                |
|-----------|--------------|-------------|--------------------------------------------------------------------------------------------|
| category  | String       | Mandatory   | Only "ERROR" or "WARNING" permitted                                                        |
| code      | Message Code | Mandatory   |                                                                                            |
| path      | String       | Conditional | Path of the element of the request message which provoked the error message, if applicable |
| text      | Max500Text   | Optional    | Additional explaining text.                                                                |

This data type is used in [oFA-IG-Com], [oFA-PFSM], [oFA-IG-XPIS], [oFA-IG-RTP]

## 2.45 Contact

| Attribute      | Type            | Condition | Description |
|----------------|-----------------|-----------|-------------|
| contactType    | Contact Type    | Mandatory |             |
| name           | Max70Text       | Optional  |             |
| postalAddress  | Postal Address  | Optional  |             |
| contactDetails | Contact Details | Optional  |             |

This data type is used in [oFA-Disc].

## 2.46 Contact Details

| Attribute    | Type             | Condition | Description                                                                      |
|--------------|------------------|-----------|----------------------------------------------------------------------------------|
| namePrefix   | Name Prefix Code | Optional  | Specifies the terms used to formally address a person.                           |
| name         | Max70Text        | Optional  | Name by which a party is known and which is usually used to identify that party. |
| phoneNumber  | Max35Text        | Optional  | pattern \+[0-9]{1,3}-[0-9()+\-]{1,30}                                            |
| mobileNumber | Max35Text        | Optional  | pattern \+[0-9]{1,3}-[0-9()+\-]{1,30}                                            |

| Attribute       | Type             | Condition | Description                                                       |
|-----------------|------------------|-----------|-------------------------------------------------------------------|
| faxNumber       | Max35Text        | Optional  | pattern \+[0-9]{1,3}-[0-9()+\-]{1,30}                             |
| emailAddress    | Max70Text        | Optional  | Address for electronic mail (e-mail).                             |
| emailPurpose    | Max35Text        | Optional  | Purpose for which an email address may be used.                   |
| jobTitle        | Max35Text        | Optional  | Title of the function.                                            |
| responsibility  | Max35Text        | Optional  | Role of a person in an organisation.                              |
| department      | Max35Text        | Optional  | Identification of a division of a large organisation or building. |
| preferredMethod | Preferred Method | Optional  | Preferred method used to reach the contact.                       |

## 2.47 Creditor Details and Account

| Attribute       | Type              | Condition | Description |
|-----------------|-------------------|-----------|-------------|
| creditor        | Party Description | Mandatory |             |
| creditorAccount | Account Reference | Mandatory |             |

This data type is used in [oFA-Con].

## 2.48 Creditor Reference Information

| Attribute                | Type                         | Condition    | Description                                              |
|--------------------------|------------------------------|--------------|----------------------------------------------------------|
| reference                | Max35Text                    | Mandatory    | The actual reference.                                    |
| referenceTypeCode        | Creditor Reference Type Code | {Or Optional | Type of the reference, provided as code.                 |
| referenceTypeProprietary | Max35Text                    | Or Optional} | Type of the reference, provided in proprietary encoding. |

| Attribute       | Type      | Condition | Description              |
|-----------------|-----------|-----------|--------------------------|
| referenceIssuer | Max35Text | Optional  | Issuer of the reference. |

## 2.49 Direct Debit Transaction (Mandate Information)

| Attribute                 | Type                        | Condition | Description                                                                                      |
|---------------------------|-----------------------------|-----------|--------------------------------------------------------------------------------------------------|
| mandateRelatedInformation | Mandate Related Information | Mandatory |                                                                                                  |
| creditorSchemeld          | Party Description3          | Optional  | <u>SDD</u> :<br>Identification = Creditor-Identifier (CI)<br>"schemeNameProprietary" :<br>"SEPA" |

This data type is used in [oFA-PFDom].

## 2.50 Duration

| Attribute | Type    | Condition | Description |
|-----------|---------|-----------|-------------|
| fromDate  | ISODate | Mandatory |             |
| toDate    | ISODate | Optional  |             |

## 2.51 Enclosed File

| Attribute       | Type               | Condition       | Description                                 |
|-----------------|--------------------|-----------------|---------------------------------------------|
| typeCode        | Document Type Code | {Or<br>Optional | Type of document or template.               |
| typeProprietary | Max35Text          | Optional<br>Or} | Proprietary document type definition.       |
| identification  | Max35Text          | Mandatory       | Identification of the document or template. |
| issueDate       | ISODate            | {Or             | Issue date of the document.                 |

| Attribute                    | Type                 | Condition | Description                                                                                                                              |
|------------------------------|----------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------|
| issueDateTime                | ISODateTime          | Or}       | Issue date and time of the document.                                                                                                     |
| name                         | Max140Text           | Optional  | Name of document or transaction, for example, tax invoice.                                                                               |
| languageCode                 | Language Code        | Optional  | Unique identifier for a language used in the document.                                                                                   |
| formatCode                   | Document Format Code | {Or       | Format of the document or template, such as PDF, XML, XSLT.                                                                              |
| formatProprietary            | Max35Text            | Or}       | Proprietary format encoding.                                                                                                             |
| formatProprietary SchemeName | Max35Text            | Optional  | May be used only if also formatProprietary is used.                                                                                      |
| formatProprietary Issuer     | Max35Text            | Optional  | May be used only if also formatProprietary is used.                                                                                      |
| fileName                     | Max140Text           | Optional  | Technical name of the file.                                                                                                              |
| digitalSignature             | Party And Signature  | Optional  | Digital signature of the enclosed binary file.<br><b>Note:</b> Not supported for now. Might be added later in case of market needs.      |
| enclosure                    | Binary Document      | Mandatory | Base64 encoding of the binary file representing the enclosed document or template, such as a PDF file, image file, XML file, MT message. |

This data type is used in [oFA-PFDom].

## 2.52 Entry Details

This data type describe entry details of a batch booking entry. The details are restricted to data attributes relevant in this case.

| Attribute  | Type              | Condition | Description                                                                       |
|------------|-------------------|-----------|-----------------------------------------------------------------------------------|
| references | References        | Optional  | might contain the endToEndId as well as the new UETR field                        |
| creditor   | Party Description | Optional  | Name and potentially an identification of the creditor if a "Debited" transaction |

| Attribute                         | Type                          | Condition   | Description                                                                      |
|-----------------------------------|-------------------------------|-------------|----------------------------------------------------------------------------------|
| transactionAmount                 | Amount                        | Mandatory   | The amount of the transaction as billed to the account.                          |
| currencyExchange                  | Array of Report Exchange Rate | Optional    |                                                                                  |
| creditorAccount                   | Account Reference             | Conditional |                                                                                  |
| creditorAgent                     | Agent Description1            | Optional    | Proprietary identifications might be provided under financialInstitutionId/other |
| ultimateCreditor                  | Party Description4            | Optional    |                                                                                  |
| debtor                            | Party Description             | Optional    | Name and potentially identification of the debtor if a "Credited" transaction    |
| debtorAccount                     | Account Reference             | Conditional |                                                                                  |
| debtorAgent                       | Agent Description1            | Optional    | Proprietary identifications might be provided under financialInstitutionId/other |
| ultimateDebtor                    | Party Description4            | Optional    |                                                                                  |
| remittanceInformationUnstructured | Array of Max140Text           | Optional    |                                                                                  |
| remittanceInformationStructured   | Array of Re-mittance          | Optional    |                                                                                  |
| purposeCode                       | Purpose Code                  | Optional    |                                                                                  |



## 2.53 Error Information

This is a data element to support the declaration of additional errors in the context of [RFC7807].

| Attribute | Type         | Condition | Description                                                                                                                                                                                                                         |
|-----------|--------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| title     | Max70Text    | Optional  | Short human readable description of error type. Could be in local language. To be provided by ASPSPs.                                                                                                                               |
| detail    | Max500Text   | Optional  | Detailed human readable text specific to this instance of the error. XPath might be used to point to the issue generating the error in addition.<br><br><b>RFU:</b> In future, a dedicated field might be introduced for the XPath. |
| code      | Message Code | Mandatory | Message code to explain the nature of the underlying error.                                                                                                                                                                         |

This data type is used in [oFA-PFSM].

## 2.54 Evaluated Amount

This data structure represents an evaluated amount (without indicating the method of evaluation) when an unambiguous balance cannot be provided, but an evaluation, e.g. the total value of a securities deposit:

| Attribute          | Type        | Condition | Description                                                 |
|--------------------|-------------|-----------|-------------------------------------------------------------|
| amount             | Amount      | Mandatory | Amount which is the result of the corresponding evaluation. |
| evaluationDateTime | ISODateTime | {Or       | Date / date and time of the evaluation.                     |
| evaluationDate     | ISODate     | Or}       |                                                             |

## 2.55 Evaluated Price

This data structure represents an evaluated price (e.g. of a security) when an unambiguous balance cannot be provided, but an evaluation, e.g. the total value of a securities deposit

| Attribute     | Type                          | Condition | Description                                                                                                                                                                                                     |
|---------------|-------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| amount        | Amount                        | {Or       | If the price is evaluated as an amount, amount of the price in its respective currency.                                                                                                                         |
| percentage    | String                        | Or}       | Percentage of the price with (optional) fractional digits. Up to 20 significant figures. Negative amounts are signed by minus.<br><br>The decimal separator is a dot.<br><br>Trailing zeroes must be truncated. |
| priceDateTime | ISODateTime                   | {Or       | Date / date and time of the evaluation of this price.                                                                                                                                                           |
| priceDate     | ISODate                       | Or}       |                                                                                                                                                                                                                 |
| priceType     | Type of Price Code            | Optional  | Type of the price.                                                                                                                                                                                              |
| sourceOfPrice | Source of Price               | Optional  | Indicates the source of the respective price.                                                                                                                                                                   |
| description   | Max500Text                    | Optional  | Additional description of the source of price. E.g., if the source of price indicates a local market, the ASPSP can include the local market's name in this field.                                              |
| exchangeRates | Array of Report Exchange Rate | Optional  | Relevant exchange rates for the determined price.                                                                                                                                                               |

## 2.56 Extended Party Description

| Attribute | Type       | Condition | Description        |
|-----------|------------|-----------|--------------------|
| name      | Max140Text | Optional  | Name of the party. |

| Attribute           | Type                 | Condition | Description                                                                                                                                                        |
|---------------------|----------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| postal address      | Postal Address       | Optional  | Postal Address of the party.                                                                                                                                       |
| identification      | Party Identification | Optional  | Identification of the party.                                                                                                                                       |
| countryOf Residence | Country Code         | Optional  | Country in which a person resides (the place of a person's home). In the case of a company, it is the country from which the affairs of that company are directed. |
| contactDetails      | Contact Details      | Optional  | Set of elements used to indicate how to contact the party.                                                                                                         |

## 2.57 Financial Institution Identification

| Attribute              | Type                                         | Condition | Description                                                                                              |
|------------------------|----------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------|
| bicfi                  | BICFI                                        | optional  | Code allocated to a financial institution.                                                               |
| clearingSystemMemberId | Clearing System Member Identification        | optional  | Information used to identify a member within a clearing system.                                          |
| name                   | Max140Text                                   | optional  | Name of the financial institution                                                                        |
| postalAddress          | Postal Address                               | Optional  | Postal Address of the financial institution.                                                             |
| other                  | Generic Financial Institution Identification | Optional  | Unique identification of an organisation, as assigned by an institution, using an identification scheme. |

## 2.58 Financial Institution Identification1

| Attribute | Type  | Condition | Description                                |
|-----------|-------|-----------|--------------------------------------------|
| bicfi     | BICFI | Mandatory | Code allocated to a financial institution. |

## 2.59 Financial Instrument

This data structure represents financial instrument.

| Attribute       | Type                                        | Condition | Description                                                                                                                                                                                                                     |
|-----------------|---------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| isin            | ISIN                                        | {Or       | ISIN of the financial Instrument.<br><br>The primitive ISIN type is defined as in ISO20022.                                                                                                                                     |
| other           | Generic Financial Instrument Identification | Or}       | If an ISIN is not available, the ASPSP may instead include an identification by other means.                                                                                                                                    |
| name            | Max70Text                                   | Optional  | Name of the financial instrument.                                                                                                                                                                                               |
| normalisedPrice | Evaluated Price                             | Optional  | Price per unit of the financial instrument in case of a price amount. Percentage price otherwise.<br><br>If used in the context of a transaction, the price here shall indicate the applied price (without fees) for the trade. |

**Remark:** Publicly available information on the financial instrument like interests (e.g. in the case of bonds), various dates (maturity date, issue date,...) or details on options are intentionally not supported by the data model. Individual markets might extend the definition, for example to meet certain regulatory requirements.

## 2.60 FromToDate

This data structure represents a timeperiod.

| Attribute    | Type        | Condition | Description                               |
|--------------|-------------|-----------|-------------------------------------------|
| fromDateTime | ISODatetime | Mandatory | Date and time at which the period starts. |
| toDateTime   | ISODatetime | Mandatory | Date and time at which the period ends.   |

This data type is used in [oFA-Disc]

## 2.61 Generic Account Identification

| Attribute                 | Type                              | Condition      | Description                                          |
|---------------------------|-----------------------------------|----------------|------------------------------------------------------|
| identification            | Max35Text                         | Mandatory      | Unique and unambiguous identification of an account. |
| schemeName<br>Code        | Account<br>Identification<br>Code | {Or - Optional | An entry provided by an external ISO code list       |
| schemeName<br>Proprietary | Max35Text                         | Or – Optional} | A scheme name defined in a proprietary way.          |
| issuer                    | Max35Text                         | Optional       | Issuer of the identification                         |

## 2.62 Generic Financial Instrument Identification

This data structure represents the identification of a financial instrument by means other than ISIN.

| Attribute      | Type                                | Condition | Description                                                                               |
|----------------|-------------------------------------|-----------|-------------------------------------------------------------------------------------------|
| identification | Max35Text                           | Mandatory | Identification of a security.                                                             |
| suffix         | Max16Text                           | Optional  | Identifies the suffix of the security identification.                                     |
| typeCode       | External<br>Financial<br>Instrument | {Or       | Unique and unambiguous identification source, as assigned via a pre-determined code list. |

| Attribute       | Type                     | Condition | Description                                                                           |
|-----------------|--------------------------|-----------|---------------------------------------------------------------------------------------|
|                 | Identification Type Code |           |                                                                                       |
| typeProprietary | Max35Text                | Or}       | Unique and unambiguous identification source using a proprietary identification type. |

### 2.63 Generic Financial Institution Identification

| Attribute              | Type                                      | Condition      | Description                                                       |
|------------------------|-------------------------------------------|----------------|-------------------------------------------------------------------|
| identification         | Max35Text                                 | Mandatory      | Unique and unambiguous identification of a financial institution. |
| schemeName Code        | Financial Institution Identification Code | {Or - Optional | An entry provided by an external ISO code list                    |
| schemeName Proprietary | Max35Text                                 | Or – Optional} | A scheme name defined in a proprietary way.                       |
| issuer                 | Max35Text                                 | Optional       | Issuer of the identification                                      |

### 2.64 Generic Organisation Identification

| Attribute              | Type                             | Condition      | Description                                               |
|------------------------|----------------------------------|----------------|-----------------------------------------------------------|
| identification         | Max35Text                        | Mandatory      | Unique and unambiguous identification of an organisation. |
| schemeName Code        | Organisation Identification Code | {Or - Optional | An entry provided by an external ISO code list            |
| schemeName Proprietary | Max35Text                        | Or – Optional} | A scheme name defined in a proprietary way.               |
| issuer                 | Max35Text                        | Optional       | Issuer of the identification                              |

## 2.65 Geo Location

Format using [RFC2426], i.e. GEO:<latitude>;< longitude >.

This data type is used in [oFA-PFSM].

## 2.66 href Type

| Attribute | Type   | Condition | Description |
|-----------|--------|-----------|-------------|
| href      | String | Mandatory |             |

## 2.67 Index

| Attribute              | Type                  | Condition | Description                                                                                                                                                                                                                             |
|------------------------|-----------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| index                  | Max35Text             | Optional  | Though arbitrary text is technically supported, it is recommended to use the following keywords:<br><br>"CPI" – customer Price Index<br><br>"FC" – foreign currency<br><br>"other" – for others, not clearly specified index relations. |
| rootIndexValue         | Max35Text             | Optional  | A number (represented as a string) giving the basic index value which is the basis for the loan contract.                                                                                                                               |
| exchangeRate           | Amount dependent Rate | Optional  | Intended for usage in case of index = "FC".                                                                                                                                                                                             |
| additional Information | Max500Text            | Optional  | Explanations e.g. to specify index = "other".                                                                                                                                                                                           |

## 2.68 Initiation

Initiation provides informations about entries in an /initiations sub resource of a payment resource.

| Attribute         | Type               | Condition | Description                                                  |
|-------------------|--------------------|-----------|--------------------------------------------------------------|
| initiationId      | Max35Text          | Mandatory | resource identification of the addressed payment initiation. |
| instructedAmount  | Amount             | Mandatory | Instructed Amount of the addressed initiation.               |
| transactionStatus | Transaction Status | Mandatory | Status of the addressed initiation.                          |
| _links            | Links              | Mandatory | links of href type "paymentInitiation"                       |

This data type is contained in [oFA-IG-XPIS].

## 2.69 Interest

| Attribute      | Type                           | Condition | Description                                                                                                                                                            |
|----------------|--------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type           | Interest Type Code             | Optional  | Type of the interest. Currently, interests are distinguished between those with fixed rates ("FIXD") and those with rates derived from an index ("INDE").              |
| relatedIndices | Array of Index                 | Optional  | List of indices related to the interest rate.                                                                                                                          |
| rate           | Array of Amount dependent Rate | Mandatory | An array of interest rates, which might apply to different amount ranges.                                                                                              |
| fromDateTime   | ISODateTime                    | Optional  | The date from which this interest rate is applicable. If no exact time is applicable, 00:00:00 is chosen as time definition.                                           |
| toDateTime     | ISODateTime                    | Optional  | The date until this interest rate is applicable. If no exact time is applicable, 23:59:59 is chosen as time definition.                                                |
| reason         | Max35Text                      | Optional  | Specifies the reason of the interest.<br><br>Though arbitrary text is technically supported, it is recommended to use the following keywords with pre-defined meaning: |



| Attribute        | Type        | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|-------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |             |           | <ul style="list-style-type: none"> <li>- <b>arrears</b>: The addressed interest rate applies only in case of arrears.</li> <li>- <b>alternateRateMaximumOf</b>: The addressed alternate interest rate applies in case where this rate implies a higher interest in a dedicated interest period than the regular interest rate provided without reason element.</li> <li>- <b>alternateRateMinimumOf</b>: The addressed alternate interest rate applies in cases where this rate implies a lower interest in a dedicated interest period than the regular interest rate provided without reason element.</li> </ul> <p>ASPSPs may define further keywords and shall publish them in their documentation.</p> |
| change Mechanism | Max1024Text | Optional  | An explanation of a mandated change mechanism.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## 2.70 Lean Account Report

| Attribute   | Type                  | Condition   | Description                                           |
|-------------|-----------------------|-------------|-------------------------------------------------------|
| booked      | Array of transactions | Conditional | booked entries on the related account                 |
| pending     | Array of transactions | Optional    | pending entries on the related account                |
| information | Array of transactions | Optional    | information entries related to the addressed account. |

This data type is used in [oFA -IG-PAIS].

## 2.71 Limit

| Attribute | Type       | Condition | Description |
|-----------|------------|-----------|-------------|
| limitType | Limit Type | Mandatory |             |

| Attribute   | Type   | Condition | Description |
|-------------|--------|-----------|-------------|
| limitAmount | Amount | Mandatory |             |

## 2.72 Links

The structure of Links is conform to [HAL].

| Attribute                               | Type      | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------|-----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| scaRedirect                             | href Type | Optional  | A link to an ASPSP site where SCA is performed within the Redirect SCA approach.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| scaOAuth                                | href Type | Optional  | The link refers to a JSON document specifying the OAuth details of the ASPSP's authorisation server. JSON document 49 follows the definition given in [RFC 8414].                                                                                                                                                                                                                                                                                                                                                                                                                 |
| confirmation                            | href Type | Optional  | <p>"confirmation": Might be added by the ASPSP if either the "scaRedirect" or "scaOAuth" hyperlink is returned in the same response message. This hyperlink defines the URL to the resource which needs to be updated with</p> <ul style="list-style-type: none"> <li>a confirmation code as retrieved after the plain redirect authentication process with the ASPSP authentication server or</li> <li>an access token as retrieved by submitting an authorization code after the integrated OAuth based authentication process with the ASPSP authentication server.</li> </ul> |
| startAuthorisation                      | href Type | Optional  | A link to an endpoint, where the authorisation of a transaction or the authorisation of a transaction cancellation shall be started with a POST command. No specific data is needed for this process start.                                                                                                                                                                                                                                                                                                                                                                       |
| startAuthorisationWithPsuIdentification | href Type | Optional  | The link to an endpoint where the authorisation of a transaction or of a transaction cancellation shall be started, where PSU identification shall be uploaded with the corresponding call.                                                                                                                                                                                                                                                                                                                                                                                       |

| Attribute                                        | Type      | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------|-----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| updatePsuIdentification                          | href Type | Optional  | The link to the payment initiation or account information resource, which needs to be updated by the PSU identification if not delivered yet.                                                                                                                                                                                                                                                                                                         |
| startAuthorisationWithProprietaryData            | hrefType  | Optional  | <p>A link to the endpoint, where the authorisation of a transaction or of a transaction cancellation shall be started, and where proprietary data needs to be updated with this call. The TPP can find the scope of missing proprietary data in the ASPSP documentation.</p> <p>The usage of this hyperlink is not further specified in the specification but is used analogously to e.g. the startAuthorisation withPsuIdentification hyperlink.</p> |
| updateProprietaryData                            | href Type | Optional  | The link to the payment initiation or account information resource, which needs to be updated by the proprietary data.                                                                                                                                                                                                                                                                                                                                |
| startAuthorisationWithPsuAuthentication          | href Type | Optional  | The link to an endpoint where the authorisation of a transaction or of a transaction cancellation shall be started, where PSU authentication data shall be uploaded with the corresponding call.                                                                                                                                                                                                                                                      |
| updatePsuAuthentication                          | href Type | Optional  | The link to the payment initiation or account information resource, which needs to be updated by a PSU password and eventually the PSU identification if not delivered yet.                                                                                                                                                                                                                                                                           |
| startAuthorisationWithEncryptedPsuAuthentication | href Type | Optional  | The link to an endpoint where the authorisation of a transaction or of a transaction cancellation shall be started, where encrypted PSU authentication data shall be uploaded with the corresponding call.                                                                                                                                                                                                                                            |
| updateEncryptedPsuAuthentication                 | href Type | Optional  | The link to the payment initiation or account information resource, which needs to be updated by an encrypted PSU password and eventually the PSU identification if not delivered yet.                                                                                                                                                                                                                                                                |

| Attribute                                           | Type      | Condition | Description                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------|-----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| startAuthorisationWithAuthenticationMethodSelection | href Type | Optional  | This is a link to an endpoint where the authorisation of a transaction or of a transaction cancellation shall be started, where the selected SCA method shall be uploaded with the corresponding call.                                                                                   |
| selectAuthenticationMethod                          | href Type | Optional  | This is a link to a resource, where the TPP can select the applicable second factor authentication methods for the PSU, if there were several available authentication methods.                                                                                                          |
| startAuthorisationWithTransactionAuthorisation      | href Type | Optional  | A link to an endpoint, where an authorisation of a transaction or a cancellation can be started, and where the response data for the challenge is uploaded in the same call for the transaction authorisation or transaction cancellation at the same time in the Embedded SCA Approach. |
| authoriseTransaction                                | href Type | Optional  | The link to the payment initiation or consent resource, where the "Transaction Authorisation"Request" is sent to. This is the link to the resource which will authorise the payment or the consent by checking the SCA authentication data within the Embedded SCA approach.             |
| updateResourceByDebtorAccountResource               | href Type | Optional  | A link to a payment related resource which needs to be updated by an account, referenced by a resource identification.                                                                                                                                                                   |
| self                                                | href Type | Optional  | The link to the payment initiation resource created by the request itself.<br><br>This link can be used later to retrieve the transaction status of the payment initiation.                                                                                                              |
| status                                              | href Type | Optional  | A link to retrieve the status of the transaction resource.                                                                                                                                                                                                                               |
| transactionfees                                     | href Type | Optional  | The link is to the status resource.                                                                                                                                                                                                                                                      |

| Attribute          | Type      | Condition | Description                                                                                                            |
|--------------------|-----------|-----------|------------------------------------------------------------------------------------------------------------------------|
|                    |           |           | This link is only added within the authorisation process in case fee information is available via the status resource. |
| scaStatus          | href Type | Optional  | A link to retrieve the status of the authorisation or cancellation-authorisation sub-resource.                         |
| account            | href Type | Optional  | A link to the resource providing the details of one account                                                            |
| savingsAccount     | href Type | Optional  | A link to the resource providing the details of a savings account.                                                     |
| loanAccount        | href Type | Optional  | A link to the resource providing the details of a loan account.                                                        |
| balances           | href Type | Optional  | A link to the resource providing the balance of a dedicated account.                                                   |
| transactions       | href Type | Optional  | A link to the resource providing the transaction history of a dedicated account.                                       |
| cardAccount        | href Type | Optional  | A link to the resource providing the details of one card account.                                                      |
| cardTransactions   | href Type | Optional  | A link to the resource providing the transaction history of a dedicated card account.                                  |
| transactionDetails | href Type | Optional  | A link to the resource providing details of a dedicated transaction.                                                   |
| ibanCheck          | href Type | Optional  | A link to the endpoint offering the addressed iban check result.                                                       |
| paymentInitiation  | href Type | Optional  | A link to an initiation related to a payment resource.                                                                 |
| securitiesAccount  | href Type | Optional  | A link to the resource providing the details of one securitiesAccount.                                                 |
| positions          | href Type | Optional  | A link to the resource providing the list of positions of one securitiesAccount.                                       |

| Attribute           | Type               | Condition | Description                                                                                                                                                                                                                      |
|---------------------|--------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| orders              | href Type          | Optional  | A link to the resource providing the list of orders of one securitiesAccount.                                                                                                                                                    |
| orderDetails        | href Type          | Optional  | A link to the resource providing details of one specific order.                                                                                                                                                                  |
| relatedOrders       | Array of href Type | Optional  | An array of links to request order details of related orders related to the order that is represented by the data structure containing this element (e.g. other orders that originated from the same order split as this order). |
| relatedTransactions | Array of href Type | Optional  | An array of links to request securities transaction details of transactions that resulted from this order.                                                                                                                       |
| subscription        | href Type          | Optional  | A link to the resource providing the details of a subscription for Push AIS Services.                                                                                                                                            |
| entryStatusRevoked  | Array of href Type | Optional  | Links to entry endpoints where the entry status is revoked.                                                                                                                                                                      |
| first               | href Type          | Optional  | Navigation link for paginated account reports.                                                                                                                                                                                   |
| next                | href Type          | Optional  | Navigation link for paginated account reports.                                                                                                                                                                                   |
| previous            | href Type          | Optional  | Navigation link for paginated account reports.                                                                                                                                                                                   |
| last                | href Type          | Optional  | Navigation link for paginated account reports.                                                                                                                                                                                   |
| download            | href Type          | Optional  | Download link for huge AIS data packages.                                                                                                                                                                                        |
| paymentInitiation   | href Type          | Optional  | link to an initiation related to a payment resource.                                                                                                                                                                             |
| confirmInitiation   | href Type          | Optional  | link to a confirmation endpoint, where a payment initiation needs to be confirmed explicitly by the API Client for execution.                                                                                                    |

This data type is part of the API Framework definition, introduced in [oFA-PFSM], but used in all Implementation Guidelines (not further specified here).

## 2.73 Mandate Related Information

| Attribute                   | Type                          | Condition | Description              |
|-----------------------------|-------------------------------|-----------|--------------------------|
| mandateId                   | Max35Text                     | Mandatory |                          |
| dateOfSignature             | ISO Date                      | Mandatory |                          |
| amendmentInformationDetails | Amendment Information Details | Optional  |                          |
| electronicSignature         | Max1025Text                   | Optional  | Only used for e-mandates |

## 2.74 Mandate Type

| Attribute                  | Type                  | Condition       | Description                                                    |
|----------------------------|-----------------------|-----------------|----------------------------------------------------------------|
| localInstrumentCode        | Local Instrument Code | {Or             | External List<br>e.g. Core, B2B, DDMC (for electronic mandate) |
| localInstrumentProprietary | Max35Text             | Or}             |                                                                |
| serviceLevelCode           | Service Level Code    | {Or<br>Optional | External code                                                  |
| serviceLevelProprietary    | Max35Text             | Or<br>Optional} |                                                                |
| categoryPurposeCode        | Category Purpose      | {Or<br>Optional | External List                                                  |
| categoryPurposeProprietary | Max35Text             | Or<br>Optional} |                                                                |
| classificationCode         | Classification Code   | {Or<br>Optional | FIXE, USGB, VARI (internal code)                               |
| classificationProprietary  | Max35Text             | Or Optional}    |                                                                |

## 2.75 Mandate Status

| Code                | Description                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------|
| received            | The mandate data have been received and are technically correct. The data is not authorised yet.                     |
| rejected            | The mandate data have been rejected e.g. since no successful authorisation has taken place.                          |
| partiallyAuthorised | The mandate is due to a multi-level authorisation, some but not all mandated authorisations have been performed yet. |
| valid               | The mandate is accepted and valid for direct debit transactions specified in the mandate object.                     |
| revokedByPsu        | The mandate has been revoked by the PSU towards the ASPSP.                                                           |
| expired             | The mandate expired.                                                                                                 |
| terminatedByTpp     | The corresponding TPP has terminated the mandate by applying the DELETE method to the mandate resource.              |

The ASPSP might add further codes. These codes then shall be contained in the ASPSP's documentation of the openFinance API.

## 2.76 Market Identification

| Attribute                        | Type      | Condition | Description                                 |
|----------------------------------|-----------|-----------|---------------------------------------------|
| mic                              | Max4Text  | {Or       | ISO 10383 code of the market place.         |
| market Identifier<br>Proprietary | Max35Text | Or}       | Proprietary identifier of the market place. |



## 2.77 Occurrences

| Attribute           | Type           | Condition | Description                                                        |
|---------------------|----------------|-----------|--------------------------------------------------------------------|
| sequenceType        | Sequence Type  | Mandatory | RCUR, OOFF                                                         |
| frequencyType       | Frequency Code | Optional  |                                                                    |
| duration            | Duration       | Optional  | Length of time for which the mandate remains valid.                |
| firstCollectionDate | ISODate        | Optional  | Date of the first collection of a direct debit as per the mandate. |
| finalCollectionDate | ISODate        | Optional  | Date of the final collection of a direct debit as per the mandate. |

## 2.78 Organisation Identification

| Attribute | Type                                         | Condition | Description                                                                                                                                   |
|-----------|----------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| anyBIC    | BIC                                          | {Or       | A code allocated to a business entity or to a financial institution by a Registration Authority under an international identification scheme. |
| lei       | LEI                                          | Or        | Legal Entity Identifier.                                                                                                                      |
| others    | Array of Generic Organisation Identification | Or}       | Unique identification of an organisation, as assigned by an institution, using an identification scheme.                                      |

## 2.79 Original Transaction Information and Status

| Attribute                          | Type               | Condition     | Description                                                                                                                                                                                                                                |
|------------------------------------|--------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originalInstructionId              | Max35Text          | Conditional   | Unique identification, as assigned by the original instructing party for the original instructed party, to unambiguously identify the original instruction.                                                                                |
| originalUetr                       | UUID               | Conditional   | Universally unique identifier to provide the original end-to-end reference of a payment transaction.                                                                                                                                       |
| transactionStatus                  | Transaction Status | Mandatory     | Specifies the status of a transaction, as published in an external payment transaction status code list.                                                                                                                                   |
| statusReasonInformationCode        | Status Reason Code | {Or-Optional} | Provides detailed information on the status reason.                                                                                                                                                                                        |
| statusReasonInformationProprietary | Max35Text          | Or Optional}  | Proprietary additional information on the reason for e.g. rejecting the request.<br><br>Remark: If an ISO Code is available it should be used instead of Proprietary Reasons. Further restrictions could be defined by API Access Schemes. |
| statusReasonAdditional Information | Max105Text         | Optional      | Further details on the status reason.                                                                                                                                                                                                      |

This data type is used in [oFA-IG-Com].

## 2.80 Original Transaction Information and VOP Status

The data type defined below is an extension of the data type Original Transaction Information and Status defined above. The attributes with **yellow background** are added.

| Attribute                          | Type                  | Condition     | Description                                                                                                                                                                                                                                |
|------------------------------------|-----------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originalInstructionId              | Max35Text             | Conditional   | Unique identification, as assigned by the original instructing party for the original instructed party, to unambiguously identify the original instruction.                                                                                |
| originalUetr                       | UUID                  | Conditional   | Universally unique identifier to provide the original end-to-end reference of a payment transaction.                                                                                                                                       |
| transactionStatus                  | Transaction Status    | Mandatory     | Specifies the status of a transaction, as published in an external payment transaction status code list.                                                                                                                                   |
| statusReasonInformationCode        | Status Reason Code    | {Or-Optional} | Provides detailed information on the status reason.                                                                                                                                                                                        |
| statusReasonInformationProprietary | Max35Text             | Or Optional}  | Proprietary additional information on the reason for e.g. rejecting the request.<br><br>Remark: If an ISO Code is available it should be used instead of Proprietary Reasons. Further restrictions could be defined by API Access Schemes. |
| statusReasonAdditionalInformation  | Max105Text            | Optional      | Further details on the status reason.                                                                                                                                                                                                      |
| creditorNameMatch                  | Party Name Match Code | Conditional   | Will be provided, if an account/name match was requested, if the related code in the interbank VOP service was "CMTC", "NMTC" or "NOAP" ,i.e. if the singleTransactionStatus also equals "RVCM"                                            |

| Attribute                   | Type                            | Condition   | Description                                                                                                                                                                                           |
|-----------------------------|---------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| creditorOrganisationIdMatch | Party Identification Match Code | Conditional | Will be provided, if an account/identification match was requested and if the related code in the interbank VOP service was "NMTC" or "NOAP", i.e. if the singleTransactionStatus also equals "RVCM". |
| creditorName                | Max140Text                      | Optional    | Provided only in case of "CMTC" as code in creditorNameMatch                                                                                                                                          |

This data type is used in [oFA-IG-VOP].

## 2.81 Parameters API Security

| Attribute                          | Type    | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------|---------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| messageSigningRequired             | Boolean | Mandatory | <p>Mandate the API Client to sign the requests on application level.</p> <ul style="list-style-type: none"> <li>• True:<br/>The fields x-jws-signature and Digest have to be included in the message header. In addition, the TLS Certificate is used on transport layer.</li> <li>• False:<br/>Only the TLS Certificate on the transport layer is used. The ASPSP has to ensure that a check of access rights can be performed. This check has to be done previous to changing the transport layer and access right information which are contained in the certificate being potentially removed, e.g. by firewalls. (IO1)</li> </ul> |
| messageSigningCertificate Mandated | Boolean | Mandatory | Indicates whether the ASPSP mandates the API Client to include the certificate used for message signing into the message (as part of the JWS Protected Header).                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Attribute                           | Type                              | Condition   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------|-----------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| supportedMessageSignatureAlgorithms | Array of Signature Algorithm Code | Conditional | Array [] containing one or more identifier indicating signature algorithms according to RFC7518, i.e. RS256, PS512, etc. applicable to http message signing.                                                                                                                                                                                                                                                                                                                                                                                             |
| scaMethods                          | Array of Authentication Type      | Mandatory   | Supported SCA Methods<br>(IO11)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| scaApproaches                       | Array of SCA Approach Type        | Mandatory   | One or several of the following values:<br><ul style="list-style-type: none"><li>• Decoupled</li><li>• Embedded</li><li>• OAuth2</li><li>• Redirect</li></ul><br>(IO5)                                                                                                                                                                                                                                                                                                                                                                                   |
| scaApproachesMethodsMapping         | SCA Approach Methods Mapping      | Optional    | Configuration of supported SCA methods per applicable SCA Approaches<br><ul style="list-style-type: none"><li>• Decoupled</li><li>• Embedded</li><li>• OAuth2</li><li>• Redirect</li><li>• ASPSP-Channel</li></ul><br>(IO12)                                                                                                                                                                                                                                                                                                                             |
| oauth2PrestepRequired               | Boolean                           | Mandatory   | OAuth2 required as a pre-step for PSU authentication (of the first factor)<br><br>The ASPSP faces the following options:<br><ul style="list-style-type: none"><li>• True:<br/>The ASPSP requires a OAuth2 prestep before sending the first request to the openFinance or XS2A API. The API Client has to perform an OAuth2 based authorization prior to the openFinance service request. This authorisation covers only the first factor of an SCA of the full (simple) authorisation if no second factor is used, e.g. in case of exemptions.</li></ul> |

| Attribute                                      | Type                              | Condition | Description                                                                                                                                                                                                                         |
|------------------------------------------------|-----------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                |                                   |           | <p>If an OAuth2 prestep is used, the field Authorization in the header has to carry the bearer token OAuth2 authorization.</p> <ul style="list-style-type: none"> <li>False:<br/>No OAuth2 prestep is used.</li> </ul> <p>(IO6)</p> |
| oauth2PrestepMetadataUri                       | Max140Text                        | Optional  | The URI of the OAuth implementation Metadata information.                                                                                                                                                                           |
| oauth2OfflineScopeRequired                     | Boolean                           | Optional  | Defines, whether the OAuth scope "offline_access" needs to be supported when addressing the API where the PSU is not online.                                                                                                        |
| scaProcessFlowRequirement                      | SCA Process Flow Requirement Code | Optional  | <p>There are some options regarding the SCA process flow: there are various points in the process flow where the selection of the SCA method could be performed.</p> <p>(IO9)</p>                                                   |
| clientExplicitAuthorisationPreferenceSupported | Boolean                           | Mandatory | Indicates whether the ASPSP will not start the authorisation implicitly, if the header parameter Client-Explicit-Authorisation-Preferred equals true.                                                                               |
| supportedStartAuthorisationHyperlinks          | Array of Max35Text                | Mandatory | <p>All "startAuthorisationXXX" hyperlinks supported in this API.</p> <p>If the Array is empty, start authorisation is not supported in this API instance.</p> <p>(IO39)</p>                                                         |
| supportedUpdateAuthorisationHyperlinks         | Array of Max35Text                | Mandatory | <p>All "updateAuthorisationXXX" hyperlinks supported in this API.</p> <p>If the Array is empty, update authorisation is not supported in this API instance.</p>                                                                     |

| Attribute                                | Type                                   | Condition   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------|----------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| firstFactorViaApiForDecoupledScaRequired | Boolean                                | Mandatory   | <p>In case of a decoupled SCA approach, parts of the PSU authentication might still be required via the API.</p> <p>There are essentially two variants for the implementation:</p> <p>False i.e. PSSU Identification sufficient:</p> <p>Only the PSU-Id is transmitted via the API. Information beyond that are not transferred via the API.</p> <p>True: (OF-)Authentication required:</p> <p>The first factor of the SCA is handled via the API. The remaining part of the SCA is not processed via the API.</p> <p>(IO14)</p> |
| supportedBodySignatureAlgorithms         | Array of Signature Algorithm Code      | Conditional | <p>Array [] containing one or more identifier indicating signature algorithms according to RFC7518, i.e. RS256, PS512, etc. applicable to body signing algorithms.</p>                                                                                                                                                                                                                                                                                                                                                           |
| passwordEncryptionRequired               | Boolean                                | Mandatory   | <p>PSU Authentication data delivered via the XS2A-Interface (Embedded Approach) shall be encrypted at application level.</p> <p>(IO40)</p>                                                                                                                                                                                                                                                                                                                                                                                       |
| supportedKeyEncryptionAlgorithms         | Array of Key Encryption Algorithm Code | Conditional | <p>Array [] containing one or more identifier indicating encryption algorithms for key encryption according to RFC7518, i.e. RSAES-OAEP, RSA-OAEP-256, etc.</p>                                                                                                                                                                                                                                                                                                                                                                  |
| supportedContentEncryptionAlgorithms     | Array of Content Encryption            | Conditional | <p>Array [] containing one or more identifier indicating encryption algorithms for encryption of the content according to RFC7518, i.e. A256GCM, etc.</p>                                                                                                                                                                                                                                                                                                                                                                        |

| Attribute                                 | Type           | Condition | Description                                                                                                |
|-------------------------------------------|----------------|-----------|------------------------------------------------------------------------------------------------------------|
|                                           | Algorithm Code |           |                                                                                                            |
| implicitTransactionAuthorisationSupported | Boolean        | Mandatory | Implicit start of transaction authorisation supported.<br><br>(IO38)                                       |
| psuNameInAuthorisationSupported           | Boolean        | Mandatory | Indicates, whether providing of the PSU Name in the authorisation sub resource is supported.<br><br>(IO68) |
| clientScaPreferenceSupported              | Boolean        | Mandatory | Indicates whether the header Client-SCA-Approach-Preference is recognised by the ASPSP.<br><br>(IO71)      |

## 2.82 Parameters Payment Service

| Attribute                   | Type            | Condition | Description                                                                                                           |
|-----------------------------|-----------------|-----------|-----------------------------------------------------------------------------------------------------------------------|
| serviceType                 | Service Type    | Mandatory | The addressed payment initiation service, e.g. "XMDPIS", where the configuration data below relates to.               |
| serviceVersion              | Max10Text       | Mandatory | Version of Implementation Guidelines of the addressed service, e.g. "2.1".                                            |
| statusNotificationSupported | Boolean         | Mandatory | Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources. |
| psuHeaderData               | PSU Header Data | Mandatory | Defines whether PSU ID, Type etc are mandated by the ASPSP in the first transaction request.                          |



| Attribute                    | Type               | Condition   | Description                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------|--------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| confirmationRequired         | Boolean            | Mandatory   | The confirmation step in authorization Redirect SCA or OAuth2 SCA Approach required for this service.<br><br>(IO57)                                                                                                                                                                                                                                                                                                                 |
| psuAccountSelectionSupported | Boolean            | Mandatory   | The ASPSP offers to select the PSU account in a second step after a password has been presented.                                                                                                                                                                                                                                                                                                                                    |
| multiLevelScaSupported       | Boolean            | Mandatory   | Indicates, whether the service is offered as a service mandating a multi-level authorisation for some PSU.<br><br>(IO27)                                                                                                                                                                                                                                                                                                            |
| scaForCancellationRequired   | Boolean            | Conditional | It is optional to require an SCA to cancel the related payment service. <ul style="list-style-type: none"> <li>• True:<br/>SCA can be required for payment cancellations where applicable. The ASPSP is free to drop the SCA in some cases.</li> <li>• False:<br/>SCA is not required to cancel payments of the related service.</li> </ul> Attribute not provided: Cancellation of the related service is not supported.<br>(IO26) |
| supportedStatusFormats       | Array of Max35Text | Mandatory   | Supported codes:<br><br>"JSON",<br><br>"JSON-Extended",<br><br>"pain.002.0x.0x", where x is instantiated with the correct values.<br><br>(IO44)                                                                                                                                                                                                                                                                                     |

| Attribute                   | Type                          | Condition   | Description                                                                                                                                                                                                                                                       |
|-----------------------------|-------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| psuNameInStatusSupported    | Boolean                       | Mandatory   | Indicates whether the psuName is provided in resource status.<br><br>(IO67)                                                                                                                                                                                       |
| clientVopRequestedSupported | Boolean                       | Mandatory   | Indicates, whether the steering of integration of VOP into payment initiation is supported.                                                                                                                                                                       |
| clientVopReuestIdSupported  | Boolean                       | Conditional | Indicates whether the ASPSP will supporte the header Client-VOP-Request-ID.<br><br>This is only provided, if the clientVopRequestedSupported attribute equals true                                                                                                |
| clientVopRequestIdMandated  | Boolean                       | Conditional | Indicates whether the ASPSP mandates the API Client to use the Client-VOP-Request-Id if VOP is applicable to the related product <b>and</b> if VOP has not been opted out.<br><br>This is only provided, if the clientVopRequestedSupported attribute equals true |
| jsonPaymentProducts         | Array of Payment Product JSON | Optional    | List of supported JSON based payment products. with related product specific additional information.<br><br>(IO2, IO3, IO4)                                                                                                                                       |
| xmlPaymentProducts          | Array of Payment Product XML  | Optional    | List of supported XML based payment products.                                                                                                                                                                                                                     |
| textPaymentProducts         | Array of Payment Product Text | Optional    | List of supported text based payment products.                                                                                                                                                                                                                    |

| Attribute                | Type                        | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------|-----------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| debtorNameReturned       | Boolean                     | Mandatory | Indicates whether the ASPSP will return the debtor name in JSON/XML based payment initiations after successful SCA.<br><br>(IO58)                                                                                                                                                                                                                                                                                                       |
| fundsAvailableSupported  | Boolean                     | Optional  | This attribute indicates whether the attribute "fundsAvailable" is supported in payment status.                                                                                                                                                                                                                                                                                                                                         |
| transactionFeesSupported | Boolean                     | Mandatory | Transaction fees transported via the openFinance API in Payment Initiation Services.<br><br>(IO10)                                                                                                                                                                                                                                                                                                                                      |
| supportedScaExemptions   | Array of SCA Exemption Code | [0..*]    | Exemptions from SCA are possible in general. An ASPSP has to decide whether and if so under which condition to allow for an exemption from SCA for payment initiation. The applied condition is arbitrary but it has to be exactly one condition that determines the scenario for an SCA exemption.<br><br>Empty array implies that SCA exemption is not supported.<br><br>No entry of this field implies no information.<br><br>(IO22) |
| minLead                  | Integer                     | Optional  | The minimum of bank working days from the current date which is needed for the requested execution date.                                                                                                                                                                                                                                                                                                                                |
| maxLead                  | Integer                     | Optional  | The maximum of calendar days from the current date which is supported for the requested execution date.                                                                                                                                                                                                                                                                                                                                 |

| Attribute                                       | Type                                     | Condition | Description                                                                                                                                                                                                                                                 |
|-------------------------------------------------|------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| maximumExpiryDays<br>Reservation                | Integer                                  | Optional  | The maximum expiry interval for a reservation of funds.                                                                                                                                                                                                     |
| supportedExecutionPeriods                       | Array of<br>Frequency<br>Code            | Optional  | Supported frequency codes for recurring payments.                                                                                                                                                                                                           |
| supportedMaximum<br>ExecutionDay                | Integer                                  | Optional  | Only relevant for recurring payments.<br><br>Either 30 or 31 as possible values. If 30 is the maximum execution day in a month, then "ultimo" is not supported as value.                                                                                    |
| supportedExecutionRules                         | Array of<br>Max35Text                    | Optional  | The supported values are "preceding" and "following".                                                                                                                                                                                                       |
| supportedAccountReference                       | Array of<br>Account<br>Reference<br>Code | Mandatory | List of account reference codes which are supported for the debtor account.<br><br>(IO18)                                                                                                                                                                   |
| accountCurrencyCode<br>Required                 | Boolean                                  | Mandatory | Defines whether the PSU account identification mandates the currency code for identification.                                                                                                                                                               |
| multiCurrencyAccounts<br>Supported              | Boolean                                  | Mandatory | The ASPSP supports multi-currency accounts for payment initiation services.<br><br>(IO17)                                                                                                                                                                   |
| supportedAdditionalCharacter<br>SetTextfields   | Array of<br>Max1Text                     | Mandatory | Characters in addition to the standard character set as defined in [OFA-PFSM], which are accepted in the payment API for text fields (names, address and remittance information). If the array is empty, no additional characters are permitted in the API. |
| transformedAdditionalCharact<br>erSetTextfields | Array of<br>Max1Text                     | Optional  | Additional Characters accepted for text fields within payments (see above), which are transformed into a                                                                                                                                                    |

| Attribute                 | Type                      | Condition | Description                                                                                                                                     |
|---------------------------|---------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |                           |           | different character/characters in addition.                                                                                                     |
| supportedCountryCodes     | Array of Country Codes    | Optional  | List of country codes for cross-border payment products.                                                                                        |
| supportedCurrencyCodes    | Array of Currency Code    | Optional  | List of currencies which are accepted for cross-border payments.<br><br>(IO52)                                                                  |
| supportedInstructionCodes | Array of Instruction Code | Optional  | Specifies further instructions concerning the processing of a payment instruction, provided by the sending clearing agent to the next agent(s). |
| batchBookingFlagSupported | Boolean                   | Optional  | if equals true, then the ASPSP will book the bulk transactions following this flag.                                                             |
| maximumNumberTransactions | Integer                   | Optional  | maximum number of transactions in a bulk payment                                                                                                |
| totalAmountRequired       | Boolean                   | Optional  | Applicable only for bulk payments                                                                                                               |
| _links                    | Links                     | Mandatory | The endpoint of the addressed payment service.                                                                                                  |

This data type is used in [oFA-Disc].

### 2.83 Parameters Account Information Service

| Attribute   | Type         | Condition | Description                                                                                                         |
|-------------|--------------|-----------|---------------------------------------------------------------------------------------------------------------------|
| serviceType | Service Type | Mandatory | The addressed account information service, e.g. "AIS-PAY" "AIS-CAC", where the configuration data below relates to. |

| Attribute                             | Type                       | Condition | Description                                                                                                                                                                 |
|---------------------------------------|----------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| supportedAccessRight Codes            | Array of Access Right Code | Mandatory | Array of access right codes which might be used when submitting a consent to an account information service related to the addressed AIS service.<br><br>(IO30,54,56)       |
| supportedReportFormats                | Array of Max35Text         | Optional  | supported scheme names for XML based payment initiation, e.g. "camt.05x.0x.0x" as well as "JSON" and "MT94x".<br><br>(IO31)                                                 |
| globalStatementsSupported             | Boolean                    | Mandatory | Defines whether statements on a dedicated endpoint /statements as introduced in [oFA-TDM2] are supported within this service type or not.                                   |
| supportedEndpoints                    | Array of Max35Text         | Mandatory | "accountDetails", "balances", "transactions", "statements", "orders" as permitted values for the array entries.<br><br>(IO59, 61)                                           |
| supportedQueryParameters Balances     | Array of Max35Text         | Mandatory | Query parameters supported for balances.<br><br>"dateFrom" as potential value.<br><br>(IO66)                                                                                |
| supportedQueryParameters Transactions | Array of Max35Text         | Mandatory | Query parameters supported for transactions.<br><br>"dateFrom", "dateTo", "entryReferenceFrom", "bookingStatus", "deltaList", "withBalance", "pageSize" as potential values |

| Attribute                            | Type               | Condition   | Description                                                                                                                                                                                                                                        |
|--------------------------------------|--------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      |                    |             | (IO32,33)                                                                                                                                                                                                                                          |
| maxPageSize                          | Integer            | Conditional | <p>Delivers the maximum page size in numbers of transactions to be reported.</p> <p>To be provided if the pageSize query parameter is supported.</p>                                                                                               |
| supportedBookingStatusCode           | Array of Max35Text | Conditional | <p>IF the bookingStatus is supported for the addressed service, this attribute is mandated. It provides all booking status codes supported as query parameters.</p> <p>""booked", "pending", "both", "information", "all" as potential values.</p> |
| supportedQueryParameters<br>Accounts | Array of Max35Text | Mandatory   | <p>Query parameters supported for account list/details.</p> <p>Potential values are "withBalance" and "evaluationCurrency".</p> <p>(IO32)</p>                                                                                                      |
| multiCurrencyAccountDetails          | Max35Text          | Optional    | <p>"aggregation", "sub-account" "aggregationAndSub-account" for potential solutions offered for account details in case of multi-currency accounts (IO41)</p>                                                                                      |
| supportedInterestReasons             | Array of Max35Text | Conditional | <p>Shows all supported values for interest reasons defined for the data type Interest for account types with an interest definition.</p> <p>Shall be used in case the reason attribute is used at all.</p>                                         |
| countingOfAisRequests                | Max35Text          | Mandatory   | <p>This attribute is defining the way of counting a 4 times a day counting for PSU not present (cp. [EBA-RTS])</p>                                                                                                                                 |

| Attribute                          | Type                  | Condition   | Description                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------|-----------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    |                       |             | <p>"none": no dedicated AIS request counting is performed for this service.</p> <p>"timeslot": ,Counting is done by aggregating all related account information endpoint accesses in a certain time interval as one access.</p> <p>"byEndpointCall": Counting is done by allowing for each endpoint 4 times access per day, counting them explicitly by endpoint.</p> <p>(IO47)</p> |
| maximumInfomationInterval          | Integer               | Mandatory   | <p>Maximum addressable transaction information of the past, interval in days.</p> <p>Might be further restricted on account level.</p>                                                                                                                                                                                                                                              |
| supportedBalanceTypes              | Array of Balance Type | Mandatory   | <p>Balance Types supported.</p> <p>(IO34,64,65)</p>                                                                                                                                                                                                                                                                                                                                 |
| creditLineIncludedSupported        | Boolean               | Mandatory   |                                                                                                                                                                                                                                                                                                                                                                                     |
| standingOrdersSupported            | Boolean               | Mandatory   |                                                                                                                                                                                                                                                                                                                                                                                     |
| supportedPageSize                  | Integer               | Optional    |                                                                                                                                                                                                                                                                                                                                                                                     |
| fileDownloadRequired               | Boolean               | Mandatory   | <p>Transaction list offered only as download link in case of too high volumes.</p> <p>(IO36,35)</p>                                                                                                                                                                                                                                                                                 |
| signedAccountInformation Supported | Boolean               | Mandatory   | Signature on body level as defined in [oFA-PFSM].                                                                                                                                                                                                                                                                                                                                   |
| supportedSignature Algorithms      | Array of Signature    | Conditional | Array [] containing one or more identifier indicating signature                                                                                                                                                                                                                                                                                                                     |



| Attribute                            | Type           | Condition | Description                                                                                                                                       |
|--------------------------------------|----------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | Algorithm Code |           | algorithms according to RFC7518, i.e. RS256, PS512, etc.                                                                                          |
| encryptedAccountInformationSupported | Boolean        | Mandatory | Encryption on application level supported by the ASPSP as defined in [oFA-PFSM] and if a related certificate has been provided by the API Client. |
| _links                               | Links          | Mandatory | The endpoint of the addressed account information service, e.g. of the related "/accounts" endpoint.                                              |

This data type is used in [oFA-Disc].

## 2.84 Parameters Consent Service

| Attribute                   | Type            | Condition | Description                                                                                                              |
|-----------------------------|-----------------|-----------|--------------------------------------------------------------------------------------------------------------------------|
| serviceType                 | Service Type    | Mandatory | The addressed account information service, e.g. "CONS-AIS"                                                               |
| statusNotificationSupported | Boolean         | Mandatory | Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources.    |
| psuHeaderData               | PSU Header Data | Mandatory | Defines whether PSU ID, Type etc are mandated by the ASPSP in the first transaction request.                             |
| confirmationRequired        | Boolean         | Mandatory | The confirmation step in authorization Redirect SCA or OAuth2 SCA Approach required for this service.<br><br>(IO57)      |
| multiLevelScaSupported      | Boolean         | Mandatory | Indicates, whether the service is offered as a service mandating a multi-level authorisation for some PSU.<br><br>(IO27) |
| inconsistentConsentRejected | Boolean         | Optional  | Indicates, whether an inconsistent consent is rejected before PSU                                                        |

| Attribute                       | Type                            | Condition   | Description                                                                                                                                          |
|---------------------------------|---------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |                                 |             | authentication, e.g. where the PSU ID does not match the required IBAN.<br>(IO53)                                                                    |
| supportedAccountReference       | Array of Account Reference Code | Mandatory   | List of account reference codes which are supported for the account in the related account category in consents of type "detailed".<br><br>(IO60,62) |
| psuNameInResourceSupported      | Boolean                         | Optional    | Indicates whether the psuName is provided in resource status. (Only relevant for Version 1.3.x).<br><br>(IO67)                                       |
| psuNameInStatusSupported        | Boolean                         | Mandatory   | Indicates whether the psuName is provided in authorisation status.                                                                                   |
| frequencyPerDay                 | Integer                         | Conditional | The maximum frequency offered for access without PSU being online. Only relevant for account information consent.                                    |
| consentValidityDays             | Integer                         | Optional    | Maximum consent validity days without the validity being restricted by the ASPSP, if applicable.                                                     |
| supportedConsentTypes           | Array of Consent Type Code      | Mandatory   |                                                                                                                                                      |
| multiCurrencyAccounts Supported | Boolean                         | Mandatory   | The ASPSP supports multi-currency accounts in the consent API.<br><br>(IO17)                                                                         |
| revocationViaAspspSupported     | Boolean                         | Mandatory   | Indicates whether the ASPSP supports the revocation of a consent by the PSU via the ASPSP.<br><br>(IO51)                                             |
| supportedAccessRight Codes      | Array of Access Right Code      | Optional    | Array of access right code admitted for the related consent, if it is not a consent on AIS. In the latter case, the supported access right codes are |

| Attribute | Type  | Condition | Description                                                                            |
|-----------|-------|-----------|----------------------------------------------------------------------------------------|
|           |       |           | specific to the account type, which is why it is defined within the AIS information.   |
| _links    | Links | Mandatory | The endpoint of the addressed consent service e.g. for "/consents/account-information" |

This data type is used in [oFA-Disc].

## 2.85 Parameters Funds Confirmation Service

| Attribute           | Type         | Condition | Description                                                                                                                                                                                                                                                            |
|---------------------|--------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| serviceType         | Service Type | Mandatory | The addressed funds confirmation service "XFCON"                                                                                                                                                                                                                       |
| cardNumberSupported | Boolean      | Mandatory | Indicates whether the card number can be used for addressing the account in addition. If true and if the addressed account is a multicurrency account, then the card will address the related sub-account of the consent process, where the card number has been used. |

This data type is used in [oFA-Disc].

## 2.86 Parameters Mandate Services

| Attribute                   | Type         | Condition | Description                                                                                                           |
|-----------------------------|--------------|-----------|-----------------------------------------------------------------------------------------------------------------------|
| serviceType                 | Service Type | Mandatory | The addressed mandate service, e.g. "MAN", where the configuration data below relates to.                             |
| serviceVersion              | Max10Text    | Mandatory | Version of Implementation Guidelines of the addressed service, e.g. "2.1".                                            |
| statusNotificationSupported | Boolean      | Mandatory | Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources. |

| Attribute                     | Type            | Condition   | Description                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|-----------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| psuHeaderData                 | PSU Header Data | Mandatory   | Defines whether PSU ID, Type etc are mandated by the ASPSP in the first transaction request.                                                                                                                                                                                                                                                                                                                                        |
| confirmationRequired          | Boolean         | Mandatory   | The confirmation step in authorization Redirect SCA or OAuth2 SCA Approach required for this service.<br><br>(IO57)                                                                                                                                                                                                                                                                                                                 |
| psuAccountSelection Supported | Boolean         | Mandatory   | The ASPSP offers to select the PSU account in a second step after a password has been presented.                                                                                                                                                                                                                                                                                                                                    |
| multiLevelScaSupported        | Boolean         | Mandatory   | Indicates, whether the service is offered as a service mandating a multi-level authorisation for some PSU.<br><br>(IO27)                                                                                                                                                                                                                                                                                                            |
| scaForCancellationRequired    | Boolean         | Conditional | It is optional to require an SCA to cancel the related payment service. <ul style="list-style-type: none"> <li>• True:<br/>SCA can be required for payment cancellations where applicable. The ASPSP is free to drop the SCA in some cases.</li> <li>• False:<br/>SCA is not required to cancel payments of the related service.</li> </ul> Attribute not provided: Cancellation of the related service is not supported.<br>(IO26) |
| psuNameInStatusSupported      | Boolean         | Mandatory   | Indicates whether the psuName is provided in resource status.<br><br>(IO67)                                                                                                                                                                                                                                                                                                                                                         |
| debtorNameMandated            | Boolean         | Mandatory   | Default is "true".                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Attribute                                   | Type                            | Condition | Description                                                                                                                                                                                                                                                |
|---------------------------------------------|---------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| debtorNameReturned                          | Boolean                         | Mandatory | Indicates whether the ASPSP will return the debtor name in submitted mandates after successful SCA.<br><br>(IO58)                                                                                                                                          |
| minLead                                     | Integer                         | Optional  | The minimum of bank working days from the current date which is needed for the first collection date.                                                                                                                                                      |
| maxLead                                     | Integer                         | Optional  | The maximum of calendar days from the current date which is supported for the first collection date.                                                                                                                                                       |
| supportedExecutionPeriods                   | Array of Frequency Code         | Optional  | Supported frequency codes for mandates.                                                                                                                                                                                                                    |
| supportedAccountReference                   | Array of Account Reference Code | Mandatory | List of account reference codes which are supported for the debtor account.<br><br>(IO18)                                                                                                                                                                  |
| supportedAdditionalCharacterSetTextfields   | Array of Max1Text               | Mandatory | Characters in addition to the standard character set as defined in[oFA-PFSM], which are accepted in the payment API for text fields (names, address and remittance information). If the array is empty, no additional characters are permitted in the API. |
| transformedAdditionalCharacterSetTextfields | Array of Max1Text               | Optional  | Additional Characters accepted for text fields within payments (see above), which are transformed into a different character/characters in addition.                                                                                                       |
| _links                                      | Links                           | Mandatory | The endpoint of the addressed mandate service.                                                                                                                                                                                                             |

This data type is used in [oFA-Disc].

## 2.87 Parameters Document Services

| Attribute                                   | Type              | Condition | Description                                                                                                                                                                                                                                                 |
|---------------------------------------------|-------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| serviceType                                 | Service Type      | Mandatory | The addressed document service, e.g. "XDOC"                                                                                                                                                                                                                 |
| statusNotificationSupported                 | Boolean           | Mandatory | Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources.                                                                                                                                       |
| supportedAdditionalCharacterSetTextfields   | Array of Max1Text | Mandatory | Characters in addition to the standard character set as defined in [oFA-PFSM], which are accepted in the payment API for text fields (names, address and remittance information). If the array is empty, no additional characters are permitted in the API. |
| transformedAdditionalCharacterSetTextfields | Array of Max1Text | Optional  | Additional Characters accepted for text fields within payments (see above), which are transformed into a different character/characters in addition.                                                                                                        |
| _links                                      | Links             | Mandatory | The endpoint of the addressed rtp service.                                                                                                                                                                                                                  |

This data type is used in [oFA-Disc].

## 2.88 Parameters Push AIS Service

| Attribute                           | Type            | Condition | Description                                                                                                              |
|-------------------------------------|-----------------|-----------|--------------------------------------------------------------------------------------------------------------------------|
| serviceType                         | Service Type    | Mandatory | The addressed Push AIS sub service, "XAIPS-XX[X]", e.g. "XAIPS-RTP".                                                     |
| statusNotificationSupported         | Boolean         | Mandatory | Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources.    |
| psuHeaderData                       | PSU Header Data | Mandatory | Defines whether PSU ID, Type etc are mandated by the ASPSP in the first transaction request.                             |
| confirmationRequired                | Boolean         | Mandatory | The confirmation step in authorization Redirect SCA or OAuth2 SCA Approach required for this service.<br><br>(IO57)      |
| multiLevelScaSupported              | Boolean         | Mandatory | Indicates, whether the service is offered as a service mandating a multi-level authorisation for some PSU.<br><br>(IO27) |
| psuNameInStatusSupported            | Boolean         | Mandatory | Indicates whether the psuName is provided in authorisation status.<br><br>(IO67)                                         |
| apiClientSecondaryPush URISupported | Boolean         | Mandatory | Support of secondary Push URI registration                                                                               |
| staticCallbackTextSupported         | Boolean         | Mandatory | Pushing callback text supported.                                                                                         |
| callbackLinksSupported              | Boolean         | Mandatory | Pushing links to related account information entries supported.                                                          |
| pushEntriesSupported                | Boolean         | Mandatory | Pushing actual account information supported.                                                                            |

| Attribute                         | Type               | Condition | Description                                                                  |
|-----------------------------------|--------------------|-----------|------------------------------------------------------------------------------|
| attributeFilteringSupported       | Boolean            | Mandatory | Support of attribute filters when pushing                                    |
| documentForwardingSupported       | Boolean            | Mandatory | Support of pushing also documents related to the account information pushed. |
| supportedAccountEntryCriteria     | Array of Max35Text | Optional  | The list of supported Account Entry Trigger Criteria                         |
| supportedAccountStatementCriteria | Array of Max35Text | Optional  | The list of supported Account Statement Trigger Criteria                     |
| supportedAccountBalanceCriteria   | Array of Max35Text | Optional  | The list of supported Account Balance Trigger Criteria                       |
| supportedRtpCriteria              | Array of Max35Text | Optional  | The list of supported RTP Trigger Criteria                                   |
| _links                            | Links              | Mandatory | The endpoint of the addressed subscription service.                          |

This data type is used in [oFA-Disc].

## 2.89 Parameters Signing Basket Service

| Attribute                   | Type            | Condition | Description                                                                                                           |
|-----------------------------|-----------------|-----------|-----------------------------------------------------------------------------------------------------------------------|
| serviceType                 | Service Type    | Mandatory | The addressed signing basket function "SGNB"                                                                          |
| statusNotificationSupported | Boolean         | Mandatory | Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources. |
| psuHeaderData               | PSU Header Data | Mandatory | Defines whether PSU ID, Type etc are mandated by the ASPSP in the first transaction request.                          |
| confirmationRequired        | Boolean         | Mandatory | The confirmation step in authorization Redirect SCA or OAuth2 SCA Approach required for this service.                 |



| Attribute              | Type                      | Condition | Description                                                                                                                                            |
|------------------------|---------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |                           |           | (IO57)                                                                                                                                                 |
| multiLevelSCASupported | Boolean                   | Mandatory | Indicates, whether the service is offered as a service mandating a multi-level authorisation for some PSU.<br>(IO27)                                   |
| transactionGroupings   | Array of Transaction Type | Mandatory | The maximum groups of transaction which might be grouped for authorisation. The constant "all" is indicating that there are no restrictions.<br>(IO25) |

This data type is used in [oFA-Disc].

## 2.90 Parameters RTP Service

| Attribute                                 | Type                          | Condition | Description                                                                                                           |
|-------------------------------------------|-------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------|
| serviceType                               | Service Type                  | Mandatory | The addressed request to pay service, e.g. "X RTP"                                                                    |
| statusNotificationSupported               | Boolean                       | Mandatory | Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources. |
| documentSupported                         | Boolean                       | Mandatory | Expresses, whether the transport of a document within the submitted RTP is supported.                                 |
| jsonPaymentProducts                       | Array of Payment Product JSON | Mandatory | List of supported JSON based payment products. with related product specific additional information.                  |
| supportedAdditionalCharacterSetTextfields | Array of Max1Text             | Mandatory | Characters in addition to the standard character set as defined in [oFA-                                              |

| Attribute                                   | Type              | Condition | Description                                                                                                                                                                         |
|---------------------------------------------|-------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             |                   |           | PFSM], which are accepted in the payment API for text fields (names, address and remittance information). If the array is empty, no additional characters are permitted in the API. |
| transformedAdditionalCharacterSetTextfields | Array of Max1Text | Optional  | Additional Characters accepted for text fields within payments (see above), which are transformed into a different character/characters in addition.                                |
| _links                                      | Links             | Mandatory | The endpoint of the addressed rtp service.                                                                                                                                          |

This data type is used in [oFA-Disc].

## 2.91 Parameters VOP Service

| Attribute                                 | Type              | Condition | Description                                                                                                                                  |
|-------------------------------------------|-------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| serviceType                               | Service Type      | Mandatory | The addressed Verification of Party service, e.g. "VOP-PAYEE"                                                                                |
| statusNotificationSupported               | Boolean           | Mandatory | Indicates whether the status notification function as specified in [oFA-PFSM] is supported for the related resources.                        |
| supportedAdditionalCharacterSetTextfields | Array of Max1Text | Mandatory | Characters in addition to the standard character set as defined in [oFA-PFSM], which are accepted in the payment API for text fields (names, |

| Attribute                                   | Type              | Condition | Description                                                                                                                                          |
|---------------------------------------------|-------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             |                   |           | address and remittance information). If the array is empty, no additional characters are permitted in the API.                                       |
| transformedAdditionalCharacterSetTextfields | Array of Max1Text | Optional  | Additional Characters accepted for text fields within payments (see above), which are transformed into a different character/characters in addition. |
| _links                                      | Links             | Mandatory | The endpoint of the addressed rtp service.                                                                                                           |

This data type is used in [oFA-Disc].

## 2.92 .Party Description

| Attribute                  | Type                         | Condition | Description                                                                                                                                                                                                                                                       |
|----------------------------|------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name                       | Max140Text                   | Optional  | Name of the party.                                                                                                                                                                                                                                                |
| postal address             | Postal Address               | Optional  | Postal Address of the party.                                                                                                                                                                                                                                      |
| identification             | Party Identification         | Optional  | Identification of the party.                                                                                                                                                                                                                                      |
| additionalPartyInformation | Additional Party Information | Optional  | Only supported in extended services for creditor related parties, if at all.<br><br><b>NOTE:</b> This information will be neither put to the related interbank payment nor to account reports/statements and might only be used to display related information to |

| Attribute | Type | Condition | Description                                       |
|-----------|------|-----------|---------------------------------------------------|
|           |      |           | the PSU during potential authorisation processes. |

This data type is used in many services.

### 2.93 Party Description1

| Attribute | Type       | Condition | Description        |
|-----------|------------|-----------|--------------------|
| name      | Max140Text | Optional  | Name of the party. |

This data type is used in [oFA-PFDom].

### 2.94 Party Description2

| Attribute      | Type           | Condition | Description                  |
|----------------|----------------|-----------|------------------------------|
| name           | Max140Text     | Optional  | Name of the party.           |
| postal address | Postal Address | Optional  | Postal Address of the party. |

This data type is used in [oFA-PFDom].

### 2.95 Party Description3

| Attribute      | Type                  | Condition | Description                  |
|----------------|-----------------------|-----------|------------------------------|
| identification | Party Identification1 | Optional  | Identification of the party. |

This data type is used in [oFA-PFDom], [oFA-IG-Adm].

## 2.96 Party Description4

| Attribute      | Type                 | Condition | Description                  |
|----------------|----------------------|-----------|------------------------------|
| name           | Max140Text           | Optional  | Name of the party.           |
| identification | Party Identification | Optional  | Identification of the party. |

This data type is used in [oFA-IG-VOP].

## 2.97 Party Identification

| Attribute      | Type                        | Condition | Description                                    |
|----------------|-----------------------------|-----------|------------------------------------------------|
| organisationId | Organisation Identification | {Or       | An entry provided by an external ISO code list |
| privateId      | Private Identification      | Or }      | A scheme name defined in a proprietary way.    |

**Note:** Only one of the entries is allowed.

This data type is used in [oFA -IG-PAIS].

## 2.98 Party Identification1

| Attribute | Type                   | Condition | Description                                 |
|-----------|------------------------|-----------|---------------------------------------------|
| privateId | Private Identification | Mandatory | A scheme name defined in a proprietary way. |

## 2.99 Party Verification Request

| Tag          | Type              | Usage     | Description                                                                                                             |
|--------------|-------------------|-----------|-------------------------------------------------------------------------------------------------------------------------|
| uetr         | UUID              | Mandatory | Identifies this party identification request. Shall be a unique identifier.                                             |
| partyAccount | Account Reference | Mandatory | The data type might be restricted by the ASPSP to e.g. "iban" use only, which would be Account Reference2 as data type. |

| Tag        | Type               | Usage     | Description                                                           |
|------------|--------------------|-----------|-----------------------------------------------------------------------|
| partyAgent | Agent Description  | Optional  |                                                                       |
| party      | Party Description4 | Mandatory | This structure shall contain the party name and/or a party identifier |

This data type is used in [oFA-IG-VOP].

## 2.100 Party Verification and Response

| Tag            | Type                            | Usage       | Description                                                                                                                                       |
|----------------|---------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| uetr           | UUID                            | Mandatory   | Identifies this party identification request. Shall be a unique identifier.                                                                       |
| partyAccount   | Account Reference               | Mandatory   | The data type might be restricted by the ASPSP to e.g. "iban" use only, which would be Account Reference2 as data type.                           |
| partyAgent     | Agent Description               | Optional    |                                                                                                                                                   |
| party          | Party Description4              | Mandatory   | This structure shall contain the party name and/or a party identifier,                                                                            |
| partyNameMatch | Party Name Match Code           | Conditional | Only provided, if a party name has been provided and if the processing of the related interbank VOP request has been finalized already.           |
| partyIdMatch   | Party Identification Match Code | Conditional | Only provided, if a party identification has been provided and if the processing of the related interbank VOP request has been finalized already. |
| matchedName    | Max140Text                      | Conditional | This attribute might be provided in case of a near match and contains the name as provided by the party agent.                                    |

| Tag                   | Type       | Usage    | Description                    |
|-----------------------|------------|----------|--------------------------------|
| additionalInformation | Max500Text | Optional | Should be human readable text. |

This data type is used in [oFA-IG-VOP].

### 2.101 Party Verification Response

| Attribute             | Type                            | Condition   | Description                                                                                                    |
|-----------------------|---------------------------------|-------------|----------------------------------------------------------------------------------------------------------------|
| uetr                  | UUID                            | Mandatory   | Replies the uetr entry of the related request.                                                                 |
| partyNameMatch        | Party Name Match Code           | Conditional | Only provided, if a party name has been provided.                                                              |
| partyIdMatch          | Party Identification Match Code | Conditional | Only provided, if a party identification has been provided.                                                    |
| matchedName           | Max140Text                      | Conditional | This attribute might be provided in case of a near match and contains the name as provided by the party agent. |
| additionalInformation | Max500Text                      | Optional    | Should be human readable text.                                                                                 |

This data type is used in [oFA-IG-VOP].

### 2.102 Payment Conditions

| Attribute                 | Type    | Condition | Description                                                                             |
|---------------------------|---------|-----------|-----------------------------------------------------------------------------------------|
| amountModificationAllowed | Boolean | optional  | Indicates if the debtor is allowed to pay a different amount then the requested amount. |
| earlyPaymentAllowed       | Boolean | optional  | Indicates if the debtor is allowed to pay before the requested execution date.          |

### 2.103 Payment Context Type

| Attribute   | Type    | Condition | Description |
|-------------|---------|-----------|-------------|
| cardPresent | Boolean | Optional  |             |

| Attribute         | Type                 | Condition | Description |
|-------------------|----------------------|-----------|-------------|
| cardDataEntryMode | Card Data Entry Mode | Optional  |             |

## 2.104 Payment Exchange Rate

| Attribute              | Type                    | Condition | Description                                                                                                                                      |
|------------------------|-------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| unitCurrency           | Currency Code           | Optional  | Currency in which the rate of exchange is expressed in a currency exchange. In the example 1EUR = xxxCUR, the unit currency is EUR.              |
| exchangeRate           | String                  | Optional  | Factor used to convert an amount from one currency into another. This reflects the price at which one currency was bought with another currency. |
| contractIdentification | Max35Text               | Optional  | Unique identification to unambiguously identify the foreign exchange contract.                                                                   |
| rateType               | Exchange Rate Type Code | Optional  | Specifies the type used to complete the currency exchange.<br><br>Only SPOT, SALE and AGRD is allowed.                                           |

This data type is used in [oFA-PFDom].

## 2.105 Payment Identification

| Attribute     | Type      | Condition | Description                                                                                                                  |
|---------------|-----------|-----------|------------------------------------------------------------------------------------------------------------------------------|
| instructionId | Max35Text | Optional  | Unique identification as assigned by an instructing party for an instructed party to unambiguously identify the instruction. |
| endToEndId    | Max35Text | Optional  |                                                                                                                              |



| Attribute | Type | Condition | Description                                                                                                                                                                                                                                               |
|-----------|------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uetr      | UUID | Optional  | <p>Universally unique identifier to provide an end-to-end reference of a payment transaction.</p> <p>It is recommended to use time-based version (version 1) and variant 8,9,a or b of the UUID as defined in [RFC4122] for UUID fields such as uetr.</p> |

This data type is used in [oFA-PFDom], [oFA-IG-XPIS].

## 2.106 Payment Identification1

| Attribute  | Type      | Condition | Description                                                                                                                                                                           |
|------------|-----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| endToEndId | Max35Text | Optional  | Unique identification assigned by the initiating party to unambiguously identify the transaction. This identification is passed on, unchanged, throughout the entire end-to-end chain |

This data type is used in [oFA-PFDom].

## 2.107 Payment Product JSON

| Attribute                       | Type               | Condition | Description                                                                                                                                                                                                                                                                     |
|---------------------------------|--------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pathProductName                 | Max35Text          | Mandatory | path definition of the related JSON formatted product, e.g. "sepa-credit-transfers"                                                                                                                                                                                             |
| supportedJSONPaymentsAttributes | Array of Max35Text | Mandatory | <p>Supported payment attributes (first level attributes) for the addressed product.</p> <p>In [oFA-PFDom] first level attributes for single payments e.g. "paymentIdentification", "creditorAccount", "instructedAmount" are defined. These attributes may come with a full</p> |

| Attribute             | Type                  | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------|-----------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                       |           | <p>data type, e.g. "Payment Identification" or with a restricted data type, e.g. "Payment Identification 1".</p> <p>This array defines, which of these attributes are accepted within this service. It further supports that the related attributes are presented with the potential restriction, e.g.</p> <p>by "paymentIdentification" for the data type "Payment Identification" or</p> <p>by "paymentIdentification_1" for the data Type "Payment Identification 1".</p> <p>Remark: By this definition, the discovery services define implicitly level one and level two of the data structure. It is the choice of the API Clients to ignore the implicit definition of the level 2 definition, if this is not needed for the supported use cases. For bulk or multi-bulk payments data structures defined in the related attributes might be defined on a lower level. However, this field still defines, whether the related attribute is supported (with the related potential restriction in the data type), potentially on a lower level.</p> <p>(IO21)</p> |
| debtorAccountMandated | Boolean               | Mandatory | <p>If true, the debtor account shall be used within this service.</p> <p>if false, the usage of the attribute is optional for the API client.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| supportedPurposeCodes | Array of Purpose Code | Mandatory | A list of the supported purpose codes for the related product.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Attribute        | Type                        | Condition | Description                                                    |
|------------------|-----------------------------|-----------|----------------------------------------------------------------|
| finalStatusCodes | Array of Transaction Status | Mandatory | Final transaction stati for the related product.<br><br>(IO45) |

## 2.108 Payment Product XML

| Attribute             | Type                        | Condition | Description                                                                                           |
|-----------------------|-----------------------------|-----------|-------------------------------------------------------------------------------------------------------|
| pathProductName       | Max35Text                   | Mandatory | path definition of the related XML formatted product, e.g. "pain.001-sepa-credit-transfers"           |
| supportedVersions     | Array of Max35Text          | Mandatory | Additional information on format versions supported, e.g. different XML schemas like "pain.001.01.09" |
| supportedPurposeCodes | Array of Purpose Code       | Mandatory | A list of the supported purpose codes for the related product.                                        |
| finalStatusCodes      | Array of Transaction Status | Mandatory | Final transaction stati for the related product.<br><br>(IO45)                                        |

## 2.109 Payment Products Text

| Attribute       | Type      | Condition | Description                                                                                                                                                                                                                              |
|-----------------|-----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pathProductName | Max35Text | Mandatory | Path definition of the related text formatted product, e.g. "dtazv" which stands for a legacy German text based payment format for cross-border payments.<br><br>The related values are defined on domestic level only., cp. [oFA-PFDom] |

| Attribute         | Type                        | Condition | Description                                                              |
|-------------------|-----------------------------|-----------|--------------------------------------------------------------------------|
| supportedVersions | Max35Text                   | Optional  | The related values are defined on domestic level only., cp. [oFA-PFDom]. |
| finalStatusCodes  | Array of Transaction Status | Mandatory | Final transaction stati for the related product.<br><br>(IO45)           |

### 2.110 Person Identification

| Attribute              | Type                       | Condition      | Description                                        |
|------------------------|----------------------------|----------------|----------------------------------------------------|
| identification         | Max35Text                  | Mandatory      | Unique and unambiguous identification of a oersib. |
| schemeName Code        | Person Identification Code | {Or - Optional | An entry provided by an external ISO code list     |
| schemeName Proprietary | Max35Text                  | Or – Optional} | A scheme name defined in a proprietary way.        |
| issuer                 | Max35Text                  | Optional       | Issuer of the identification                       |

### 2.111 Postal Address

| Attribute    | Type                | Condition      | Description                                                                                                                                                         |
|--------------|---------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| addressLines | Array of Max140Text | {Or - Optional | At most seven entries are permitted. May only be used, if none of the structured address elements "streetName", "buildingNumber", "postcode" or "townName" is used. |

| Attribute        | Type      | Condition      | Description                                                                                                                                              |
|------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |           |                | <b>Remark:</b> For SEPA transactions this is further restricted to a maximum of 1 entry.                                                                 |
| department       | Max70Text | Or – Optional} | Identification of a division of a large organisation or building.                                                                                        |
| subDepartment    | Max70Text |                | Identification of a sub-division of a large organisation or building.                                                                                    |
| streetName       | Max70Text |                | Name of a street or thoroughfare.                                                                                                                        |
| buildingNumber   | Max16Text |                | Number that identifies the position of a building on a street.                                                                                           |
| buildingName     | Max35Text |                | Name of the building or house.                                                                                                                           |
| floor            | Max70Text |                | Floor or storey within a building.                                                                                                                       |
| postBox          | Max16Text |                | Numbered box in a post office, assigned to a person or organisation, where letters are kept until called for.                                            |
| room             | Max70Text |                | Building room number.                                                                                                                                    |
| postCode         | Max16Text |                | Identifier consisting of a group of letters and/or numbers that is added to a postal address to assist the sorting of mail.                              |
| townName         | Max35Text |                | Name of a built-up area, with defined boundaries, and a local government.<br><br>Usage Rule: If address lines are not used, this attribute is mandatory. |
| townLocationName | Max35Text |                | Specific location name within the town.                                                                                                                  |
| districtName     | Max35Text |                | Identifies a subdivision within a country sub-division.                                                                                                  |

| Attribute          | Type         | Condition | Description                                                                                                    |
|--------------------|--------------|-----------|----------------------------------------------------------------------------------------------------------------|
| countrySubDivision | Max35Text    |           | Identifies a subdivision of a country such as state, region, county.                                           |
| country            | Country Code | Optional  | Nation with its own government.<br><br>Usage Rule: If address lines are not used, this attribute is mandatory. |

**Remark:** The usage of addressLines will be phased out in November 2025 at least for SEPA payments.

**Remark:** The elements of the structured address information marked by "Optional Or}" may only be used, if the element "addressLines" is not used.

### 2.112 Private Identification

| Attribute       | Type                           | Condition   | Description                                                                                                                                                     |
|-----------------|--------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| birthDate       | ISODate                        | Optional    |                                                                                                                                                                 |
| provinceOfBirth | Max35Text                      | Optional    |                                                                                                                                                                 |
| cityOfBirth     | Max35Text                      | Optional    |                                                                                                                                                                 |
| countryOfBirth  | Country Code                   | Optional    |                                                                                                                                                                 |
| others          | Array of Person Identification | Conditional | Unique identification of a person, as assigned by an institution, using an identification scheme.<br><br>Shall be used if none of the above attributes is used. |

### 2.113 Product Rights

| Attribute | Type            | Condition | Description                                                           |
|-----------|-----------------|-----------|-----------------------------------------------------------------------|
| product   | Payment Product | Mandatory | payment product definitions as used in the URI for payment initiation |

| Attribute    | Type           | Condition | Description                                                       |
|--------------|----------------|-----------|-------------------------------------------------------------------|
| signingClass | Singing Class  | Mandatory |                                                                   |
| limits       | Array of Limit | Optional  | Limits applying to the authorisation of related payment products. |

### 2.114 Proxy Account Identification

| Attribute       | Type                    | Condition      | Description                      |
|-----------------|-------------------------|----------------|----------------------------------|
| typeCode        | Proxy Account Type Code | {Or optional - | Type of the proxy identification |
| typeProprietary | Max35Text               | Or optional} – |                                  |
| identification  | Max2048Text             | mandatory      |                                  |

### 2.115 PSU Credentials

| Attribute         | Type   | Condition   | Description                                                |
|-------------------|--------|-------------|------------------------------------------------------------|
| password          | String | Conditional | Contains a password in plaintext.                          |
| encryptedPassword | String | Conditional | Is used when a password is encrypted on application level. |

This data type is used in [oFA-PFSM].

### 2.116 PSU Data

| Attribute  | Type           | Condition | Description |
|------------|----------------|-----------|-------------|
| psuld      | Max140Text     | Mandatory |             |
| psuldType  | PSU Id Type    | Optional  |             |
| psuName    | Max140Text     | Mandatory |             |
| psuAddress | Postal Address | Optional  |             |

| Attribute          | Type                         | Condition | Description               |
|--------------------|------------------------------|-----------|---------------------------|
| privateId          | Private Identification       | Optional  |                           |
| psuCorporateId     | Max140Text                   | Optional  |                           |
| psuCorporateIdType | PSU Corporate Id Type        | Optional  |                           |
| scaMethods         | Array of Authentication Type | Optional  |                           |
| limits             | Array of Limit               | Optional  | Global limits of the PSU. |

### 2.117 PSU Header Data

| Attribute         | Type    | Condition | Description                                                                                                                                                                                                                                                                                                                            |
|-------------------|---------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| psuIdRequired     | Boolean | Mandatory | <p>The ASPSP is providing whether the PSU-ID is mandatory for all of the following requests (if supported):</p> <ul style="list-style-type: none"> <li>• Payment Initiation Request</li> <li>• Consent Request</li> <li>• Payment Cancellation</li> <li>• Signing Basket</li> <li>• Subscription for Push AIS</li> </ul> <p>(IO15)</p> |
| psuIdTypeRequired | Boolean | Mandatory | <p>The PSU-ID-Type is necessary if different types of identifiers are allowed by the ASPSP to login and the ASPSP is either not able or does not want to identify the type from the content of the field PSU-ID. Usual examples for a PSU-ID-Types are:</p>                                                                            |



| Attribute                    | Type                 | Condition   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------|----------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              |                      |             | <ul style="list-style-type: none"> <li>• Username</li> <li>• Customer reference number</li> <li>• Account reference, e.g. an IBAN or MSISDN</li> <li>• email address</li> </ul> <p>The ASPSP has to decide whether the PSU-ID-TYPE (additional to the PSU-ID) is mandatory in all of the following requests where supported</p> <ul style="list-style-type: none"> <li>• Payment Initiation Request</li> <li>• Consent Request</li> <li>• Payment Cancellation</li> <li>• Signing Basket</li> <li>• Subscription Request</li> </ul> <p>(IO16)</p> |
| supportedPsuidTypes          | Array of PSU ID Type | Conditional | Shall be supported, if psu-id-typeRequired equals true.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| psuCorporate-idRequired      | Boolean              | Mandatory   | <p>PSU-Corporate-ID required in transaction initiation message, if a corporate account is affected</p> <p>(IO19)</p>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| psuCorporateIdTypeRequired   | Boolean              | Mandatory   | <p>PSU-Corporate-ID-TYPE required in message, if a corporate account is affected</p> <p>(IO20)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| supportedPsuCorporateIdTypes | Array of PSU         | Conditional | Shall be supported, if psu-corporate-id-typeRequired equals true.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Attribute | Type              | Condition | Description |
|-----------|-------------------|-----------|-------------|
|           | Corporate ID Type |           |             |

### 2.118 PSU Id Type

| Attribute   | Type       | Condition | Description                                          |
|-------------|------------|-----------|------------------------------------------------------|
| psuIdType   | Max35Text  | Mandatory | A supported PSU-Id-Type.                             |
| description | Max250Text | Optional  | Additional description of the addressed PSU-Id-Type. |

### 2.119 PSU Corporate Id Type

| Attribute          | Type       | Condition | Description                                                    |
|--------------------|------------|-----------|----------------------------------------------------------------|
| psuCorporateIdType | Max35Text  | Mandatory | A supported PSU-Corporate-Id-Type.                             |
| description        | Max250Text | Optional  | Additional description of the addressed PSU-Corporate-Id-Type. |

### 2.120 Push Account Entry Parameters

| Attribute            | Type                                | Condition | Description                                                                                                                                                                                                                                                                   |
|----------------------|-------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| accountEntryCriteria | Push Account Entry Trigger Criteria | Optional  | <p>This attribute is defining the account entries to be pushed to the API Client push URI: All entries which are fulfilling <b>all</b> the criteria listed in this attribute will be pushed.</p> <p>If the attribute is not provided, all account entries will be pushed.</p> |
| acceptedFormat       | Mime Type Code                      | Mandatory | This attribute describes the mime type which is supported for pushing the account entries by the API Client. If there is no match with the mime types offered by the ASPSP,                                                                                                   |

| Attribute           | Type               | Condition | Description                                                                                                                                                                                                             |
|---------------------|--------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |                    |           | the subscription is rejected with a dedicated error code.                                                                                                                                                               |
| preferredAttributes | Array of Max35Text | Optional  | These attributes are the preferred ones for every account entry to be pushed. The ASPSP might ignore this preference. Only attributes of the first level of Type Transactions as defined in [oFA-IG-Com] might be used. |
| documentsPreferred  | Boolean            | Optional  | <b>RFU:</b> This data element indicates that the API Client prefers to also receive documents attached to an account entry. This data attribute might be ignored by the ASPSP.                                          |

This data type is used in [oFA-IG-PAIS].

## 2.121 **Push** Account Entry Trigger Criteria

| Attribute                   | Type                                   | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|----------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bankTransactionCodePatterns | Array of Bank Transaction Code Pattern | optional  | <p>The criterion is fulfilled if the bankTransactionCode of the related account entry matches one of the bank transaction codes or code patterns (see the following paragraph) defined by this list.</p> <p>Any particle of the Bank Transaction Code triple might be replaced by the "???" character, indicating that all sub family codes apply. However, at least one particle of the triple must be stated explicitly (e.g. "???-CCRD-??").</p> |
| proprietaryBankCodes        | Array of Max35Text                     | Optional  | The criterion is fulfilled if the proprietaryBankTransactionCode of the related account entry equals one of the proprietary bank                                                                                                                                                                                                                                                                                                                    |

| Attribute          | Type                 | Condition | Description                                                                                                                                                                                                                                                                                    |
|--------------------|----------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |                      |           | transaction codes defined by this list.                                                                                                                                                                                                                                                        |
| accountEntryStatus | Account Entry Status | Optional  | <p>The criterion is fulfilled if the account entry has the related status booked or pending respectively.</p> <p><b>Note:</b> Entries might be pushed twice in case where this criterion is not provided if the transactionStatus is changed from pending to booked in the account system.</p> |
| maximumAmount      | Amount               | Optional  | The criterion is fulfilled if the (absolute value of the) transactionAmount of the related account entry is provided in the same currency and if the transactionAmount is less or equal to the amount provided in this attribute.                                                              |
| minimumAmount      | Amount               | Optional  | The criterion is fulfilled if the (absolute value of the) transactionAmount of the related account entry is provided in the same currency and if the transactionAmount is greater or equal to the amount provided in this attribute.                                                           |
| debtorAccount      | Account Reference    | Optional  | The criterion is fulfilled if the counterparty account of the related account entry is a debtor account and equals the account defined in this attribute.                                                                                                                                      |
| creditorAccount    | Account Reference    | Optional  | The criterion is fulfilled if the counterparty account of the related account entry is a creditor account                                                                                                                                                                                      |

| Attribute                          | Type              | Condition | Description                                                                                                                                                                                                                                              |
|------------------------------------|-------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    |                   |           | and equals the account defined in this attribute.                                                                                                                                                                                                        |
| creditDebitIndicator               | Credit Debit Code | Optional  | The criterion is fulfilled if the related account entry is a credit as defined in this attribute, or a debit respectively.                                                                                                                               |
| endToEndId                         | Max35Text         | Optional  | The criterion is fulfilled if the endToEndId of the related accountpayment entry equals the string provided in this attribute.                                                                                                                           |
| purpose                            | Purpose Code      | Optional  | The criterion is fulfilled if the related account entry has a purpose code equal to the code provided in this attribute.                                                                                                                                 |
| remittanceInformation Unstructured | Max140Text        | Optional  | The criterion is fulfilled if the related account entry has an unstructured remittance information where the string provided in this attribute is contained as a substring. When comparing substrings, the usage of small or capital letters is ignored. |

## 2.122 Push Account Statements Parameters

| Attribute                | Type                                    | Condition | Description                                                                                                                                                                                                                     |
|--------------------------|-----------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| accountStatementCriteria | Push Account Statement Trigger Criteria | Mandatory | This attribute is defining the account statements to be pushed to the API Client push URI: Every time, the criteria listed in this attribute are all fulfilled, the current account statement of the related account is pushed. |

This data type is used in [oFA -IG-PAIS].

**2.123 Push Account Statement Trigger Criteria**

| Attribute     | Type              | Condition | Description                                                                                                                                                                                                                                                                                                                          |
|---------------|-------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reports       | Max35Text         | {OR       | One of the values ""MT942" or one of the camt.052 versions offered in the Discovery API should be used, e.g. "camt.052.001.02", "BG.JSON2.0". " The criterion is fulfilled if a related report is available.If the related report format is not supported, then the ASPSP is rejecting the subscription with a dedicated error code. |
| notifications | Max35Text         | OR        | One of the values "MT941", "MT900", "MT910" or camt.054 versions.<br><br>If the related notification format is not supported, then the ASPSP is rejecting the subscription with a dedicated error code.                                                                                                                              |
| statements    | Max35Text         | OR}       | One of the values "MT940", and "camt.053", "BG.JSON2.0" shall be used. The criterion is fulfilled if a related statement is available.<br><br>If the related statement format is not supported, then the ASPSP is rejecting the subscription with a dedicated error code.                                                            |
| event         | Max35Text         | Optional  | Only the value "afterGeneration" may be used. The criterion is fulfilled if the related report is generated and made available to the PSU in the online channel.                                                                                                                                                                     |
| times         | Array of ISO Time | Optional  | The value is provided in format "hh:mm:ss". The criterion is fulfilled if the current local ASPSP time is                                                                                                                                                                                                                            |

| Attribute | Type | Condition | Description                                                                                        |
|-----------|------|-----------|----------------------------------------------------------------------------------------------------|
|           |      |           | <p>matching one of the defined times in the list of times.</p> <p>The list shall not be empty.</p> |

## 2.124 Push Balance Parameters

| Attribute       | Type                                   | Condition | Description                                                                                                                                                                             |
|-----------------|----------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| balanceCriteria | Push<br>Balance<br>Trigger<br>Criteria | Mandatory | This attribute is defining the balance information to be pushed to the API Client push URI: Every time, a balance is fulfilling all the criteria listed in this attribute it is pushed. |

This data type is used in [oFA -IG-PAIS].

## 2.125 Push Balance Trigger Criteria

| Attribute       | Type            | Condition | Description                                                                                                                                                                                                                                                     |
|-----------------|-----------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| balanceAmount   | Amount          | Optional  | The criterion is fulfilled if the current balance of type balanceType provided below fulfills the balanceOperator requirement provided below together with the provided balanceAmount in this attribute and where the last provided balance did not fulfill it. |
| balanceOperator | Max35Text       | Optional  | <p>Must be one of the following codes:</p> <ul style="list-style-type: none"> <li>• less,</li> <li>• lessOrEqual,</li> <li>• greaterOrEqual,</li> <li>• greater</li> </ul>                                                                                      |
| balanceType     | Balance<br>Type | Optional  | <p>This criterion is fulfilled if the balance is of this type.</p> <p>If the related balance type is not supported for the addressed account, then the ASPSP is</p>                                                                                             |

| Attribute           | Type              | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|-------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |                   |           | rejecting the subscription with a dedicated error code.                                                                                                                                                                                                                                                                                                                                          |
| creditLimitIncluded | Boolean           | Optional  | <p>This criterion is fulfilled if the related current balance is provided with the credit limit included (if this attribute is true) or not (if this attribute is false).</p> <p>If the attribute is true and if the related balance type defined above is not applicable to the condition whether a credit limit is included then the subscription is rejected with a dedicated error code.</p> |
| times               | Array of ISO Time | Optional  | <p>The value is provided in format "hh:mm:ss". The criterion is fulfilled if the current local ASPSP time is matching one of the defined times in the list of times.</p> <p>The list shall not be empty.</p>                                                                                                                                                                                     |

## 2.126 Push RTP Parameters

| Attribute      | Type                      | Condition | Description                                                                                                                                                                                                                                                                                                       |
|----------------|---------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rtpCriteria    | Push RTP Trigger Criteria | Optional  | <p>This attribute is defining the incoming RTP requests to be pushed to the API Client push URI: All RTPs which are fulfilling all the criteria listed in this attribute will be pushed.</p> <p>If the attribute is not provided, all RTPs will be pushed as soon as they are arriving at the ASPSP's system.</p> |
| acceptedFormat | Mime Type Code            | Mandatory | This is defining the mime type. If the requested mime type is not supported by the ASPSP, the                                                                                                                                                                                                                     |



| Attribute          | Type    | Condition | Description                                                                                                                                                          |
|--------------------|---------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |         |           | request is rejected. If not used, then the ASPSP is choosing the mime type.<br><br>At a minimum the ASPSP shall support a JSON mime type.                            |
| documentsPreferred | Boolean | Optional  | <b>RFU:</b> This data element indicates that the API Client prefers to also receive documents attached to an RTP. This data attribute might be ignored by the ASPSP. |

This data type is used in [oFA -IG-PAIS].

### 2.127 **Push** RTP Trigger Criteria

| Attribute      | Type      | Condition | Description                                                                                                                                                                                                         |
|----------------|-----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| event          | Max35Text | Optional  | "afterReception" is the only supported value.                                                                                                                                                                       |
| timeToExpire   | Integer   | Optional  | only forward incoming RTP with an expiry timestamp within that period, where the period is defined by the number of minutes as provided in this attribute. This criterion is meant to forward only short term RTPs. |
| expiryReminder | Boolean   | Optional  | This criterion is fulfilled if a long term RTP is still not confirmed 2 days before the RTP expires.                                                                                                                |

### 2.128 Remittance

The remittance type will be re-defined in version 2 to be able to support more complex requirements. This is not downwards compatible.

| Attribute                   | Type              | Condition | Description                                    |
|-----------------------------|-------------------|-----------|------------------------------------------------|
| referredDocumentInformation | Array of Referred | Optional  | Set of elements used to identify the documents |

| Attribute                       | Type                           | Condition | Description                                                                                                                                 |
|---------------------------------|--------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | Document Information           |           | referred to in the remittance information.                                                                                                  |
| creditorReferenceInformation    | Creditor Reference Information | Optional  | Reference information provided by the creditor to allow the identification of the underlying documents.                                     |
| additionalRemittanceInformation | Array of Max140Text            | Optional  | Additional information to be displayed to the PSU e.g. within authorisation processes. This will not be used in related interbank payments. |

This data type is used in [oFA-PFDom], [oFA-IG-XPIS].

## 2.129 References

| Attribute                | Type      | Condition | Description                                                                                                                                                                                                                                        |
|--------------------------|-----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| accountServicerReference | Max35Text | Optional  |                                                                                                                                                                                                                                                    |
| instructionId            | Max35Text | Optional  | Unique identification as assigned by an instructing party for an instructed party to unambiguously identify the instruction.                                                                                                                       |
| endToEndId               | Max35Text | Optional  |                                                                                                                                                                                                                                                    |
| uetr                     | UUID      | Optional  | Universally unique identifier to provide an end-to-end reference of a payment transaction.<br><br>It is recommended to use time-based version (version 1) and variant 8,9,a or b of the UUID as defined in [RFC4122] for UUID fields such as uetr. |
| mandateId                | Optional  | Optional  | direct debit related mandate.                                                                                                                                                                                                                      |

| Attribute   | Type      | Condition | Description |
|-------------|-----------|-----------|-------------|
| checkNumber | Max35Text | Optional  |             |

### 2.130 Referred Document Information

| Attribute       | Type                        | Condition      | Description                                                                |
|-----------------|-----------------------------|----------------|----------------------------------------------------------------------------|
| typeCode        | Referred Document Type Code | {Or - Optional | Specifies the type of referred document, provided as code.                 |
| typeProprietary | Max35Text                   | Or – Optional} | Specifies the type of referred document, provided in proprietary encoding. |
| typeIssuer      | Max35Text                   | Optional       | Issuer of the document type.                                               |
| number          | Max35Text                   | Optional       | Unique and unambiguous identification of the referred document.            |
| relatedDate     | ISO Date                    | Optional       | Date associated with the referred document.                                |

### 2.131 Related Document Information

| Attribute           | Type      | Condition | Description                                                                                                                                              |
|---------------------|-----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| resourceId          | UUID      | Mandatory |                                                                                                                                                          |
| relationCode        | Max35Text | {Or       | "parentDocument"<br><br><b>RFU:</b> In future, replacements might be supported, if underlying use cases have been clarified with the market demand side. |
| relationProprietary | Max35Text | Or}       |                                                                                                                                                          |

This code is used in [oFA-IG-Doc].

### 2.132 Referred Mandate Document Information

| Attribute         | Type                        | Condition      | Description                                                                                                                              |
|-------------------|-----------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| typeCode          | Referred Document Type Code | {Or – Optional | Specifies the type of referred document, provided as code.                                                                               |
| typeProprietary   | Max35Text                   | Or – Optional} | Specifies the type of referred document, provided in proprietary encoding.                                                               |
| typeIssuer        | Max35Text                   | Optional       | Issuer of the document type.                                                                                                             |
| number            | Max35Text                   | Optional       | Unique and unambiguous identification of the referred document.                                                                          |
| creditorReference | Max35Text                   | Optional       | Unique and unambiguous identification as assigned by the creditor to the referred document shared with the debtor for its own reference. |
| relatedDate       | ISO Date                    | Optional       | Date associated with the referred document.                                                                                              |

### 2.133 Regulatory Reporting

| Attribute                     | Type                           | Condition | Description                                                                                                                                                 |
|-------------------------------|--------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| debitCreditReportingIndicator | Regulatory Reporting Type Code | Optional  | Identifies whether the regulatory reporting information applies to the debit side, to the credit side or to both debit and credit sides of the transaction. |
| detailsCode                   | Max35Text                      | Optional  | Specifies the nature, purpose, and reason for the transaction to be reported for regulatory and statutory requirements in a coded form.                     |
| detailsInformation            | Array of Max35Text             | Optional  | Additional details that cater for specific domestic regulatory requirements.                                                                                |

This data type is used in [oFA-PFDom].

### 2.134 Report Exchange Rate

| Attribute              | Type          | Condition | Description                                                                                                                                      |
|------------------------|---------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| sourceCurrency         | Currency Code | Mandatory | Currency from which an amount is to be converted in a currency conversion.                                                                       |
| exchangeRate           | String        | Mandatory | Factor used to convert an amount from one currency into another. This reflects the price at which one currency was bought with another currency. |
| unitCurrency           | Currency Code | Mandatory | Currency in which the rate of exchange is expressed in a currency exchange. In the example 1EUR = xxxCUR, the unit currency is EUR.              |
| targetCurrency         | Currency Code | Mandatory | Currency into which an amount is to be converted in a currency conversion.                                                                       |
| quotationDate          | ISODate       | Mandatory | Date at which an exchange rate is quoted.                                                                                                        |
| contractIdentification | Max35Text     | Optional  | Unique identification to unambiguously identify the foreign exchange contract.                                                                   |

### 2.135 RTP Details

| Attribute               | Type                                 | Condition | Description                                                                                                                       |
|-------------------------|--------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
| paymentConditions       | Payment Conditions                   | optional  | Conditions for the execution of the payment.                                                                                      |
| creationDateTime        | ISODateTime                          | optional  | creationDateTime of the originator                                                                                                |
| requestedExpiryDateTime | ISODateTime                          | mandatory | Date and Time at which this request expires.                                                                                      |
| originatorPspld         | Financial Institution Identification | Optional  | Originator payment service provider in case of indirect participation regarding a related API Access Scheme or bilateral contract |

### 2.136 SCA Approach Methods Mapping

| Attribute   | Type                         | Condition | Description |
|-------------|------------------------------|-----------|-------------|
| scaApproach | SCA Approach Type            | Mandatory |             |
| scaMethods  | Array of Authentication Type | Mandatory |             |

### 2.137 Securities Account Fee

Represents a Fee rule that is applicable to a securities account:

| Attribute       | Type                                 | Condition | Description                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|--------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| typeCode        | Securities Fee Type Code             | {Or       | Type of the fee as a code.<br><br>Might be adjusted in the future based on market experience.                                                                                                                                                                                                                                                                         |
| typeProprietary | Max35Text                            | Or}       | Type of the fee as a proprietary string.                                                                                                                                                                                                                                                                                                                              |
| feeRules        | Array of Securities Account Fee Rule | Mandatory | Array of fee rules. Each rule represents how the amount of the fee is determined either explicitly by stating the amount or implicitly by providing a percentage, minimum and maximum amount.<br><br>This array will usually only contain one entry to represent the whole rule. Only in the case of the fee being calculated as a tiered percentage, several entries |

| Attribute             | Type       | Condition | Description                                                                                                                                                                          |
|-----------------------|------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |            |           | are included to represent the several tiers of the fee rule.                                                                                                                         |
| applicableFrom        | ISODate    | Optional  | First date, when this fee is applicable.<br><br>If not present, the fee is applicable starting with the creation of this account / the latest reporting date supported by the ASPSP. |
| applicableTo          | ISODate    | Optional  | Last date, when this fee is applicable.<br><br>If not present, the fee is applicable indefinitely.                                                                                   |
| additionalInformation | Max500Text | Optional  |                                                                                                                                                                                      |

### 2.138 Securities Account Fee Rule

| Attribute      | Type   | Condition | Description                                                                                                                                                    |
|----------------|--------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| amount         | Amount | {Or       | Amount of the fee.                                                                                                                                             |
| percentage     | String | Or}       | Percentage of the fee.<br><br>Up to 20 significant figures. The decimal separator is a dot.<br><br>Trailing zeroes must be truncated.Fee as a percentage.      |
| fromBaseAmount | Amount | Optional  | To be used for fees where a tiered percentage is used to calculate the fee relative to an implicitly defined base amount (such as a transaction's volume being |
| toBaseAmount   | Amount | Optional  |                                                                                                                                                                |

| Attribute     | Type   | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------|--------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |        |           | <p>the base amount for a courtage).</p> <p>The range of one tier is defined as all applicable entities with base amounts X and</p> $\text{fromBaseAmount} \leq X \leq \text{toBaseAmount}.$ <p>In any array of fee rules, only one of these entries may be missing the "fromBaseAmount" element (indicating that the rule represented by this entry is used without a lower bound to the base amount) and only one entry may be missing the "toBaseAmount" element (indicating that the rule represented by this entry is applied without an upper limit to the base amount).</p> <p>In cases, where the fee is not calculated as a tiered percentage, only one entry in array "feeRules" is included. This entry must be missing both subelements "fromBaseAmount" and "toBaseAmount".</p> |
| minimumAmount | Amount | Optional  | Minimum amount of the fee in each billing period / for each billed event if applicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| maximumAmount | Amount | Optional  | Maximum amount of the fee in each billing / for each billed event period if applicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |



## 2.139 Securities Order

| Attribute            | Type                  | Condition      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|-----------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| orderId              | String                | Optional       | <p>Can be used as access-ID in the API, where more details on an order is offered.</p> <p>If this data attribute is provided, this shows that the API Client can get access on more details about this transaction using the Read Securities Order Details Request.</p> <p><b>Remark:</b> Implementers might want to anticipate future services supporting placement or revocation of orders via the openFinance API. It is therefore recommended to define the orderId in a way that it will also identify an order in the case of a placement / revocation.</p> |
| side                 | Securities Order Side | Mandatory      | Type of transaction that is associated with this order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| financialInstrument  | Financial Instrument  | Mandatory      | Financial instrument the order refers to.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| unitsNumber Order    | Number                | {Or            | Nominal or numeric quantification of the financial instrument that is ordered.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| unitsNominal Order   | Amount                | Or}            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| unitsNumber Display  | Number                | {Or – Optional | Nominal or numeric quantification of the order that shall be displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| unitsNominal Display | Amount                | Or - Optional} |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Attribute               | Type                        | Condition      | Description                                                                                                                                                |
|-------------------------|-----------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| placeOfTrade            | Market Identification       | Optional       |                                                                                                                                                            |
| limitPrice Amount       | Amount                      | {Or- Optional  | Limit price displayed as an amount.                                                                                                                        |
| limitPrice Percent      | String                      | Or – Optional} | Limit price displayed as a percentage rate.<br><br>Up to 20 significant figures. The decimal separator is a dot.<br><br>Trailing zeroes must be truncated. |
| stopPrice Amount        | Amount                      | {Or- Optional  | Stop price displayed as an amount.                                                                                                                         |
| stopPrice Percent       | String                      | Or – Optional} | Stop price displayed as a percentage rate.<br><br>Up to 20 significant figures. The decimal separator is a dot.<br><br>Trailing zeroes must be truncated.  |
| tradingSessionIndicator | Trading Session Type Code   | Optional       |                                                                                                                                                            |
| typesOfOrder            | Array of Type of Order Code | Optional       | One or more codes to indicate characteristics of the order.                                                                                                |
| timeInForce             | Order Time Limit Code       | Optional       | If the order is restricted by a time limit, the type time limit is indicated by a value of the corresponding code set.                                     |
| expiryDate              | ISODate                     | {Or - Optional | Expiry date of the order if applicable.                                                                                                                    |

| Attribute          | Type              | Condition      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|-------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| expiryDateTime     | ISODateTime       | Or – Optional} | Expiry date and time of the order if applicable.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| relatedCashAccount | Account Reference | Optional       | If the execution of this order implies the transfer of money, cash account to be affected by this transfer from the PSU's side.                                                                                                                                                                                                                                                                                                                                      |
| orderSplit         | Boolean           | Optional       | Value true indicates that this order is the result of an order split. Missing values indicate "false".                                                                                                                                                                                                                                                                                                                                                               |
| orderModifyable    | Boolean           | Optional       | Value true indicates that this order can (still) be modified. Missing values indicate "false".                                                                                                                                                                                                                                                                                                                                                                       |
| orderStatus        | Order Status Code | Mandatory      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| details            | Max500Text        | Optional       | Additional details to the position                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| _links             | Links             | Optional       | <p>The following types of links are supported:</p> <p>"relatedOrders": An <b>array</b> of links to request order details of related orders (e.g. other orders that originated from the same order split as this order).</p> <p>"relatedTransactions": An <b>array</b> of links to request securities transaction details of transactions that resulted from this order.</p> <p>"orderDetails": <b>Single</b> link for retrieving details on this specific order.</p> |

## 2.140 Securities Position

This data structure represents one position within a securities account:

| Attribute           | Type                      | Condition | Description                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|---------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| financialInstrument | Financial Instrument      | Mandatory | Financial Instrument that is contained in this position.                                                                                                                                                                                                                                                                                                      |
| unitsNumber         | Number                    | {Or       | Nominal or numeric quantification of the financial instrument within this position, negative values for short positions.                                                                                                                                                                                                                                      |
| unitsNominal        | Amount                    | Or}       |                                                                                                                                                                                                                                                                                                                                                               |
| externalIdentifier  | Max35Text                 | Optional  | Name or identifier with an intrinsic meaning for the PSU to be displayed.                                                                                                                                                                                                                                                                                     |
| safekeepingPlace    | BICFI                     | Optional  | BIC of the place where the securities are safe-kept, physically or notionally.                                                                                                                                                                                                                                                                                |
| safekeepingCountry  | Country Code              | Optional  | Country where the securities are safe-kept.                                                                                                                                                                                                                                                                                                                   |
| balanceType         | SecuritiesBalanceTypeCode | Optional  | <p>Specifies the nature of the securities or investment fund balance.</p> <p>Type of balance in case of multiple positions for the same financial instrument for further distinction.</p> <p><b>Remark:</b> In consequence, several positions with the same financial instrument may occur in one position list, having different values of balanceType".</p> |
| averageBuyingPrice  | Amount                    | Optional  | Average buying price of the position excluding any fees or taxes.                                                                                                                                                                                                                                                                                             |

| Attribute             | Type                          | Condition | Description                                                                                                                                            |
|-----------------------|-------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| averageSellingPrice   | Amount                        | Optional  | Average selling price of the position, e.g. in case of a short position Price excluding any fees or taxes.                                             |
| totalBuyingPrice      | Amount                        | Optional  | In case of bonds, an average buying price does not make sense. Therefore, the total buying price excluding any fees or taxes can be displayed instead. |
| estimatedCurrentValue | Evaluated Amount              | Optional  | Estimated value of the position and timestamp of the estimation.                                                                                       |
| accruedInterest       | Accrued Interest              | Optional  | Accrued interest by the position.                                                                                                                      |
| currencyExchange      | Array of Report Exchange Rate | Optional  | The ASPSP might include exchange rates e.g. if the security is denoted in another currency than the account itself.                                    |
| details               | Max500Text                    | Optional  | Additional details describing the position.                                                                                                            |

## 2.141 Securities related Date or Time

| Attribute | Type   | Condition | Description                                                                                                                                                                                                                                                  |
|-----------|--------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type      | String | Mandatory | <p>Type of the date / date time. The following values are pre-defined:</p> <ul style="list-style-type: none"> <li>• effectiveSettlementDate,</li> <li>• settlementDate,</li> <li>• valueDate,</li> <li>• performanceDate,</li> <li>• bookingDate,</li> </ul> |

| Attribute   | Type        | Condition | Description                                                       |
|-------------|-------------|-----------|-------------------------------------------------------------------|
|             |             |           | <ul style="list-style-type: none"> <li>transactionDate</li> </ul> |
| date        | ISODate     | {Or       |                                                                   |
| dateAndTime | ISODateTime | Or}       |                                                                   |

## 2.142 Securities Transaction

| Attribute           | Type                                     | Condition | Description                                                                                                                                                                                                                                               |
|---------------------|------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| transactionId       | String                                   | Optional  | Can be used as access-ID in the API, where more details on a transaction is offered. If this data attribute is provided, this shows that the API Client can get access on more details about this transaction using the Read Transaction Details Request. |
| entryReference      | Max35Text                                | Optional  | Identification of the transaction as used e.g. for reference for deltafunction on application level.                                                                                                                                                      |
| relevantDates       | Array of Securities related Date or Time | Mandatory | At least one entry containing a pre-defined date type must be present.                                                                                                                                                                                    |
| financialInstrument | Financial Instrument                     | Mandatory | Financial instrument that was transferred within the transaction.                                                                                                                                                                                         |
| orderId             | String                                   | Optional  | Resource Id of the order resource that triggered this transaction, if applicable.                                                                                                                                                                         |
| unitsNumber         | Number                                   | {Or       | Nominal or numeric quantification of the financial instrument that has been transferred within this transaction.                                                                                                                                          |
| unitsNominal        | Amount                                   | Or}       |                                                                                                                                                                                                                                                           |

| Attribute                   | Type                      | Condition | Description                                                                                                                                                                                                                                                                                                           |
|-----------------------------|---------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |                           |           | Negative values indicate that the respective quantity of the financial instrument has been taken from the securities account, positive values indicate that the quantity has been added.                                                                                                                              |
| transactionType Code        | Transaction Activity Code | {Or       | Type of the transaction as code or as a proprietary string.                                                                                                                                                                                                                                                           |
| transactionType Proprietary | Max35Text                 | Or}       |                                                                                                                                                                                                                                                                                                                       |
| placeOfTrade                | Market Identification     | Optional  |                                                                                                                                                                                                                                                                                                                       |
| amountIncludes Fees         | Boolean                   | Optional  | Indicates whether the transactionAmount (see below) is including fees.<br><br>Default: false.                                                                                                                                                                                                                         |
| amountIncludes Taxes        | Boolean                   | Optional  | Indicates whether the transactionAmount (see below) is including taxes.<br><br>Default: false.                                                                                                                                                                                                                        |
| transactionAmount           | Amount                    | Optional  | Transaction amount for the transferred security for the PSU. Potential fees or taxes are reflected as indicated by elements "amountIncludesFees" and "amountIncludesTaxes".<br><br><b>Remark:</b> The price denoted here is the price for the whole lot of securities within this transaction, not for a single item. |

| Attribute             | Type                                | Condition       | Description                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------|-------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| relatedFees           | Array of Securities Transaction Fee | Optional        | <p>Only allowed if amountIncludesFees = true. In that case, the structure contains details of the fees that have been applied to this transaction (and therefore represent additional costs of the transaction beyond the transactionAmount).</p> <p>Fees with positive amount are debited by the PSU, only in the rare case of a fee credited to the PSU (e.g. in case of a reversal) negative amounts are used.</p> |
| currencyExchange      | Array of Report Exchange Rate       | Optional        | The ASPSP might include exchange rates e.g. if the transaction has been settled in another currency than the PSU's currency.                                                                                                                                                                                                                                                                                          |
| reversalIndicator     | Boolean                             | Optional        | <p>Indicates whether it is the reversal of a previously reported movement.</p> <p>Default: false</p>                                                                                                                                                                                                                                                                                                                  |
| reversedTransactionId | String                              | Optional        | "transactionId" of the reversed transaction, if applicable and supported by the ASPSP.                                                                                                                                                                                                                                                                                                                                |
| unitsNumberBeforeTx   | Number                              | {Or - Optional  | Nominal or numeric quantification of the financial instrument within the primary position before the transaction, negative values for short positions.                                                                                                                                                                                                                                                                |
| unitsNominalBeforeTx  | Amount                              | Or - Optional } |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| unitsNumberAfterTx    | Number                              | {Or - Optional  | Nominal or numeric quantification of the financial                                                                                                                                                                                                                                                                                                                                                                    |



| Attribute               | Type             | Condition       | Description                                                                                                                               |
|-------------------------|------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| unitsNominal<br>AfterTx | Amount           | Or - Optional } | instrument within the primary position after the transaction, negative values for short positions.                                        |
| accruedInterest         | Accrued Interest | Optional        | Accrued interest of the position to make transparent what the PSU really lost within this transaction.                                    |
| details                 | Max500Text       | Optional        | Additional details to the transaction.                                                                                                    |
| _links                  | Links            | Optional        | The following types of links are supported:<br><br>"transactionDetails": Single link for retrieving details on this specific transaction. |

### 2.143 Securities Transaction Fee

| Attribute       | Type                        | Condition | Description                                                                                                                                                                                                                             |
|-----------------|-----------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| typeCode        | Securities Fee<br>Type Code | {Or       |                                                                                                                                                                                                                                         |
| typeProprietary | Max35Text                   | Or}       |                                                                                                                                                                                                                                         |
| amount          | Amount                      | Mandatory |                                                                                                                                                                                                                                         |
| _links          | Links                       | Optional  | Links to resources, where the ASPSP can provide additional information on the fee, e.g. an overview on the general fee rules of the underlying securities account.<br><br>Currently no specific href types are defined in this context. |

**2.144 Source of Price**

| Attribute                     | Type             | Condition       | Description                                                                                                                                                                                             |
|-------------------------------|------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type                          | Market Type Code | Mandatory       |                                                                                                                                                                                                         |
| mic                           | Max4Text         | {Or - Optional  | If the type indicates local Market (Code: "LMAR"), the identification of the market should additionally be included here.<br><br>Technically, Source of Price is an extension of Market Identification. |
| market Identifier Proprietary | Max35Text        | Or - Optional } |                                                                                                                                                                                                         |

**2.145 Standing Order Details**

| Attribute     | Type        | Condition    | Description                                                                                                                                                                                                                                                                                                                    |
|---------------|-------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| startDate     | ISODate     | {Or          | The first applicable day of execution starting from this date the first payment was/will be executed.                                                                                                                                                                                                                          |
| startDateTime | ISODateTime | Or}          | The first applicable day and time of execution starting from this timestamp.<br><br>This attribute is only applicable to instant payments. Seconds and milli seconds might be ignored by the ASPSP.<br><br>The ASPSP might restrict the implementation to startDate. This restriction will be part of the ASPSP documentation. |
| endDate       | ISODate     | {Optional Or | The last applicable day of execution<br><br>If not given, it is an infinite standing order.                                                                                                                                                                                                                                    |
| endDateTime   | ISODateTime | Optional Or} | The last applicable day and time of requested execution.                                                                                                                                                                                                                                                                       |

| Attribute        | Type           | Condition | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|----------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |                |           | <p>This attribute is only applicable to instant payments.</p> <p>If none of the attributes is given, it is an infinite standing order.</p> <p>The ASPSP might restrict the implementation to endDate. This restriction will be part of the ASPSP documentation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                            |
| executionRule    | String         | Optional  | <p>"following" or "preceding" supported as values. This data attribute defines the behavior when a transaction date resulting from a standing order falls on a weekend or bank holiday. The payment is then executed either the "preceding" or "following" working day.</p>                                                                                                                                                                                                                                                                                                                                                                                                                    |
| withinAMonthFlag | Boolean        | Optional  | <p>This element is only used in case of frequency equals "Monthly".</p> <p>If this element equals false it has no effect. If this element equals true, then the execution rule is overruled if the day of execution would fall into a different month using the execution rule.</p> <p><b>Example:</b> executionRule equals "preceding", dayOfExecution equals "02" and the second of a month is a Sunday. In this case, the transaction date would be on the last day of the month before. This would be overruled if withinAMonthFlag equals true and the payment is processed on Monday the third of the Month.</p> <p><b>Remark:</b> This attribute is rarely supported in the market.</p> |
| frequency        | Frequency Code | Mandatory | <p>The frequency of the recurring payment resulting from this standing order.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Attribute         | Type              | Condition   | Description                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------|-------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| monthsOfExecution | Array of Max2Text | Conditional | <p>The format is following the regular expression <code>\d{1,2}</code>. The array is restricted to 11 entries. The values contained in the array entries shall all be different and the maximum value of one entry is 12.</p> <p>This attribute is contained if and only if the frequency equals "MonthlyVariable".</p> <p>Example: An execution on January, April and October each year is addressed by ["1", "4", "10"].</p> |
| multiplicator     | Numerical         | Optional    | <p>This is multiplying the given frequency resulting the exact frequency, e.g.</p> <p>Frequency=weekly and multiplicator=3 means every 3 weeks.</p> <p><b>Remark:</b> This attribute is rarely supported in the market.</p>                                                                                                                                                                                                    |
| dayOfExecution    | Max2Text          | Optional    | <p>"31" is ultimo.</p> <p>The format is following the regular expression <code>\d{1,2}</code>.</p> <p>Example: The first day is addressed by "1".</p> <p>The date is referring to the time zone of the ASPSP.</p>                                                                                                                                                                                                              |
| limitAmount       | Amount            | Conditional | <p>limitAmount</p> <p>Amount limit for fund skimming, e.g. skim all funds above this limit to savings account, i.e. typically a specific periodic payments with fixed remaining amount rather than fixed transaction amount. Amount may be zero as well as below zero, i.e. negative.</p> <p>Constraints: transactionAmount needs to be zero and bankTransactionCode needs</p>                                                 |

| Attribute | Type | Condition | Description                                 |
|-----------|------|-----------|---------------------------------------------|
|           |      |           | to specify PMNT-MCOP-OTHR for fund skimming |

## 2.146 Structured Additional Transaction Information

| Attribute              | Type                | Condition | Description |
|------------------------|---------------------|-----------|-------------|
| transactionStatus      | Transaction Status  | Mandatory |             |
| transactionCreated     | User Entry          | Optional  |             |
| transaction Authorised | Array of User Entry | Optional  |             |
| transaction Cancelled  | Array of User Entry | Optional  |             |

## 2.147 Structured Additional Information

| Attribute            | Type                   | Condition | Description                            |
|----------------------|------------------------|-----------|----------------------------------------|
| standingOrderDetails | Standing Order Details | {OR       | Details of underlying standing orders. |
| rtpDetails           | RTP Details            | OR}       | Details of underlying incoming RTPs    |

## 2.148 Subscription Entry

| Attribute  | Type              | Condition | Description                                                     |
|------------|-------------------|-----------|-----------------------------------------------------------------|
| accountId  | Account Reference | Mandatory | This is the account from which related entries shall be pushed. |
| validUntil | ISODate           | Optional  | Requested validity period of the service.                       |

| Attribute                       | Type       | Condition               | Description                                                                                                                                                                                                                                                                                     |
|---------------------------------|------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| subscriptionEntryName           | Max35Text  | Optional                | Name of the subscription entry                                                                                                                                                                                                                                                                  |
| subscriptionEntryId             | String     | Conditional             | Forbidden in HTTP Requests, e.g. <i>Initiate Subscription Request</i> and in an <i>Add a Subscription Entry Request</i> by the TPP (as this is assigned by the ASPSP). Mandatory in each entry of a response from an ASPSP.                                                                     |
| apiClientPrimaryPush URI        | Max256Text | Mandatory               | The URI where related account information shall be pushed to.                                                                                                                                                                                                                                   |
| apiClientSecondaryPush URI      | Max256Text | [Optional if supported] | <p>The URI where related account information shall be pushed to if the primary push URI is not reachable.</p> <p>If this attribute is used by the API Client and the ASPSP is not supporting this feature, then the request will be rejected with a dedicated additional error information.</p> |
| callbackWithLinkPreferred       | Boolean    | Optional                | API Client prefers to receive hyperlinks pointing to the related account information element if the related subservice criteria are met.                                                                                                                                                        |
| callbackWithStaticTextPreferred | Boolean    | Optional                | API Client prefers to get informed by static text if the related subservice criteria are met.                                                                                                                                                                                                   |
| staticCallbackText              | Max140Text | Optional                | The text to be provided by the ASPSP in case static text is sent if the related subservice criteria are met.                                                                                                                                                                                    |

| Attribute                      | Type                              | Condition | Description                                                                   |
|--------------------------------|-----------------------------------|-----------|-------------------------------------------------------------------------------|
| pushAccountEntryParameters     | Push Account Entry Parameters     | {Or       | Parameters for a subscription entry for the Push Account Entry Service.       |
| pushAccountStatementParameters | Push Account Statement Parameters | Or        | Parameters for a subscription entry for the Push Account Statement Service.   |
| pushBalanceParameters          | Push Balance Parameters           | Or        | Parameters for a subscription entry for the Push Balance Information Service. |
| pushRtpParameters              | Push RTP Parameters               | Or}       | Parameters for a subscription entry for the Push RTP Information Service.     |

This data type is used in [oFA -IG-PAIS].

## 2.149 Subscription Identification and Link

| Attribute          | Type      | Condition | Description                                   |
|--------------------|-----------|-----------|-----------------------------------------------|
| subscriptionId     | Max70Text | Mandatory | Identification of the generated subscription. |
| subscriptionStatus | Max35Text | Mandatory | Status of the related subscription.           |
| _links             | Links     | Mandatory | A hyperlink of type "subscription".           |

This data type is used in [oFA -IG-PAIS].

## 2.150 Transactions

| Attribute     | Type   | Condition | Description                                                                                                  |
|---------------|--------|-----------|--------------------------------------------------------------------------------------------------------------|
| transactionId | String | Optional  | Can be used as access-ID in the API, where more details on an transaction is offered. If this data attribute |

| Attribute                  | Type       | Condition   | Description                                                                                                                                                                                                                                                                                                                                             |
|----------------------------|------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |            |             | <p>is provided this shows that the AIS can get access on more details about this transaction using the GET Transaction Details Request.</p> <p><b>Remark:</b> ASPSPs should ensure that transactionID, if provided, is unique relative to the account that PSU is accessing, in order to ensure the quality of the data provided to the API Client.</p> |
| entryReference             | Max35Text  | Optional    | Is the identification of the transaction as used e.g. for reference for delta function on application level. The same identification as for example used within camt.05x messages.                                                                                                                                                                      |
| batchIndicator             | Boolean    | Optional    | If this indicator equals <code>true</code> , then the related entry is a batch entry.                                                                                                                                                                                                                                                                   |
| batchNumberOf Transactions | Integer    | Conditional | Shall be used if and only if the <code>batchIndicator</code> is contained and equals <code>true</code> .                                                                                                                                                                                                                                                |
| references                 | References | Optional    | <p>Might contain the payment identification attributes <code>endToEndId</code> as well as the new UETR field.</p> <p>In addition it is including <code>accountServicerReference</code>, <code>mandateId</code>, <code>checkId</code> (all level 1 fields in V1.3.x).</p>                                                                                |



| Attribute                  | Type                          | Condition    | Description                                                                                                                                                                                                                                                                                              |
|----------------------------|-------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| localInstrumentCode        | Local Instrument Code         | {Or Optional | User community specific instrument as a code provided by ISO20022.                                                                                                                                                                                                                                       |
| localInstrumentProprietary | Max35Text                     | Or Optional} | User community specific instrument as a proprietary attribute.                                                                                                                                                                                                                                           |
| bookingDate                | ISODate                       | Optional     | The Date when an entry is posted to an account on the ASPSPs books.                                                                                                                                                                                                                                      |
| valueDate                  | ISODate                       | Optional     | <p>Date at which assets become available to the account owner in case of a credit entry, or cease to be available to the account owner in case of a debit entry.</p> <p>Usage: If entry status is pending and value date is present, then the value date refers to an expected/requested value date.</p> |
| transactionAmount          | Amount                        | Mandatory    | The amount of the transaction or batch as billed to the account.                                                                                                                                                                                                                                         |
| currencyExchange           | Array of Report Exchange Rate | Optional     |                                                                                                                                                                                                                                                                                                          |
| amountDetails              | Amount Details                | Optional     | Additional information to the transaction amount.                                                                                                                                                                                                                                                        |
| interbankSettlementDate    | ISODate                       | Optional     | Date on which the amount of money ceases to be available to the agent that owes it and when the amount of money becomes                                                                                                                                                                                  |

| Attribute                           | Type                | Condition   | Description                                                                       |
|-------------------------------------|---------------------|-------------|-----------------------------------------------------------------------------------|
|                                     |                     |             | available to the agent to which it is due.                                        |
| cardTransaction                     | Card Transaction    | Optional    | Card transaction details in case of an underlying card transaction.               |
| creditor                            | Party Description   | Optional    | Name and potentially an identification of the creditor if a "Debited" transaction |
| creditor Account                    | Account Reference   | Conditional | Account of the payee.                                                             |
| creditorAgent                       | Agent Description1  | Optional    | Proprietary identifications might be provided under financialInstitutionId/other  |
| ultimate Creditor                   | Party Description4  | Optional    | Only name and identification is supported for now for this party.                 |
| debtor                              | Party Description   | Optional    | Name and potentially an identification of the debtor if a "Credited" transaction  |
| debtorAccount                       | Account Reference   | Conditional |                                                                                   |
| debtorAgent                         | Agent Description1  | Optional    | Proprietary identifications might be provided under financialInstitutionId/other  |
| ultimateDebtor                      | Party Description4  | Optional    |                                                                                   |
| remittance Information Unstructured | Array of Max140Text | Optional    |                                                                                   |
| remittance Information Structured   | Array of Remittance | Optional    |                                                                                   |

| Attribute                                      | Type                                          | Condition   | Description                                                                                                                                                                                                                                   |
|------------------------------------------------|-----------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| entryDetails                                   | Array of Entry Details                        | Optional    | Might be used by the ASPSP to transport details about transactions within a batch.                                                                                                                                                            |
| additionalTransactionInformation               | Max500Text                                    | Optional    | Might be used by the ASPSP to transport additional transaction related information to the PSU                                                                                                                                                 |
| additionalInformation<br>Structured            | Structured Additional Information             | Conditional | <p>Is used if and only if the bookingStatus entry equals "information".</p> <p>Every active standing order related to the dedicated payment account result into one entry.</p> <p>Each received RTP result into one entry.</p>                |
| additionalTransaction<br>informationStructured | Structured Additional Transaction Information | Optional    | Data about the transactional process of submission, authorisation and/or cancellation of the underlying payment.                                                                                                                              |
| purposeCode                                    | Purpose Code                                  | Optional    |                                                                                                                                                                                                                                               |
| bank<br>TransactionCode                        | Bank Transaction Code                         | Optional    | <p>Bank transaction code as used by the ASPSP and using the sub elements of this structured code defined by ISO20022.</p> <p>For standing order reports the following codes are applicable:</p> <p>"PMNT-ICDT-STDO" for credit transfers,</p> |

| Attribute                          | Type      | Condition | Description                                                                                                                                                                                                                                                                                                                        |
|------------------------------------|-----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    |           |           | <p>"PMNT-IRCT-STD0" for instant credit transfers</p> <p>"PMNT-ICDT-XBST" for cross-border credit transfers</p> <p>"PMNT-IRCT-XBST" for cross-border real time credit transfers and</p> <p>"PMNT-MCOP-OTHR" for specific standing orders which have a dynamical amount to move left funds e.g. on month end to a saving account</p> |
| bank<br>TransactionCodeProprietary | Max35Text | Optional  | proprietary bank transaction code as used within a community or within an ASPSP e.g. for MT94x based transaction reports. Renaming of the attribute to adapt to statements.                                                                                                                                                        |
| balanceAfter<br>Transaction        | Balance   | Optional  | This is the balance after this transaction. Recommended balance type is interimBooked.                                                                                                                                                                                                                                             |
| _links                             | Links     | Optional  | <p>The following links could be used here:</p> <p>transactionDetails for retrieving details of a transaction.</p>                                                                                                                                                                                                                  |

This data type is used in [oFA-IG-Com].

### 2.151 User Parameter

| Attribute           | Type                                             | Condition | Description |
|---------------------|--------------------------------------------------|-----------|-------------|
| userData            | PSU Data                                         | Mandatory |             |
| accountAccessRights | Array of Account Access and Authorisation Rights | Mandatory |             |

This data type is used in [oFA-Disc].

### 2.152 User Entry

| Attribute | Type        | Condition | Description |
|-----------|-------------|-----------|-------------|
| user      | Max70Text   | Mandatory |             |
| dateTime  | ISODateTime | Mandatory |             |

### 3 Code Lists

**NOTE:** It is indicated below each code, in which openFinance API Framework Implementation Guidelines the related data type is used in attributes of API requests and API responses on **first** level.

#### 3.1 Access Rights Codes

The current list of access right codes. This might be further extended.

| Codes                 | Description                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| ais                   | covers details, balances and transactions                                                                                                   |
| accountDetails        | covers generic account product information details                                                                                          |
| balances              | Allow to retrieve balances of the addressed account.                                                                                        |
| transactions          | Allow to retrieve transactions of the addressed account.                                                                                    |
| orders                | Allow to retrieve orders from an addressed order book related to a securities account.                                                      |
| ownerName             | Allow to retrieve the owner name/owner names related to this account.                                                                       |
| psuName               | Allow to retrieve the PSU name related to the consent authorisation.                                                                        |
| psuLeanIdentification | Allow to retrieve the PSU Name as well as PSU identification related data from the PSU parameters, e.g. birth date, address or passport ID. |
| trustedBeneficiaries  | Allow to retrieve the list of trusted beneficiaries related to this account.                                                                |
| initiatePayments      | Allow to initiate payments with the related token.                                                                                          |
| fundsConfirmations    | Allows to access an addressed account for a funds confirmation check.                                                                       |
| userParameters        | Allows to access the PSU's user parameters applicable to the account configuration.                                                         |
| ibanChecks            | Allows to access a restricted version of the user parameters related to signing rights.                                                     |

| Codes                  | Description                                                                                          |
|------------------------|------------------------------------------------------------------------------------------------------|
| corporateParameters    | RFU Will be supported in future.                                                                     |
| accountCheckParameters | Allows to access restricted user parameters e.g. role definitions of the PSU related to the account. |

### 3.2 Account Entry Status

| Code    | Description                                       |
|---------|---------------------------------------------------|
| booked  | Entry is booked on the account.                   |
| pending | Entry is pending, funds might be reserved or not. |

### 3.3 Account Reference Code

| Attribute         | Description                                              |
|-------------------|----------------------------------------------------------|
| iban              |                                                          |
| bban              |                                                          |
| msisdn            |                                                          |
| pan               |                                                          |
| maskedPan         |                                                          |
| proxy-<proxytype> | with proxytype being of Type Account Identification Code |

### 3.4 Account Status Code

| Code    | Description                               |
|---------|-------------------------------------------|
| enabled | account is available                      |
| deleted | account is terminated                     |
| blocked | account is blocked e.g. for legal reasons |

### 3.5 Balance Type

The following balance types are excluding credit limits unless the `creditLimitIncluded` element is present and equals true in the corresponding balance element.

**Remark:** This definition is following ISO20022 logic for defining balance types.

| Type             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| closingBooked    | <p>Balance of the account at the end of the pre-agreed account reporting period. It is the sum of the opening booked balance at the beginning of the period and all entries booked to the account during the pre-agreed account reporting period.</p> <p>For card-accounts, this is composed of</p> <ul style="list-style-type: none"> <li>• invoiced, but not yet paid entries</li> </ul>                                                                                                           |
| expected         | <p>Balance composed of booked entries and pending items known at the time of calculation, which projects the end of day balance if everything is booked on the account and no other entry is posted.</p> <p>For card accounts, this is composed of</p> <ul style="list-style-type: none"> <li>• invoiced, but not yet paid entries,</li> <li>• not yet invoiced but already booked entries and</li> <li>• pending items (not yet booked)</li> </ul>                                                  |
| openingBooked    | <p>Book balance of the account at the beginning of the account reporting period. It always equals the closing book balance from the previous report.</p>                                                                                                                                                                                                                                                                                                                                             |
| interimAvailable | <p>Available balance calculated in the course of the account 'servicer's business day, at the time specified, and subject to further changes during the business day. The interim balance is calculated on the basis of booked credit and debit items during the calculation time/period specified.</p> <p>For card-accounts, this is composed of</p> <ul style="list-style-type: none"> <li>• invoiced, but not yet paid entries,</li> <li>• not yet invoiced but already booked entries</li> </ul> |
| interimBooked    | <p>Balance calculated in the course of the account servicer's business day, at the time specified, and subject to further changes during the business</p>                                                                                                                                                                                                                                                                                                                                            |



| Type             | Description                                                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------------------------------------|
|                  | day. The interim balance is calculated on the basis of booked credit and debit items during the calculation time/period specified. |
| forwardAvailable | Forward available balance of money that is at the disposal of the account owner on the date specified.                             |
| nonInvoiced      | Only for card accounts, to be defined yet.                                                                                         |

### 3.6 Card Data Entry Mode

| Type | Description           |
|------|-----------------------|
| CICC | EMV based             |
| ECTL | EMV based contactless |
| MGST | Magnetic Stripe       |
| PHYS | Manual Entry          |
| BRCD | Barcode               |

This is following an ISO20022 Code: CardsDataReading1Code

### 3.7 Charge Bearer

| Type | Description                                                                                                                                                                                                                                                                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEBT | All transaction charges are to be borne by the debtor.                                                                                                                                                                                                                                                                                                                         |
| CRED | All transaction charges are to be borne by the creditor.                                                                                                                                                                                                                                                                                                                       |
| SHAR | In a credit transfer context, means that transaction charges on the sender side are to be borne by the debtor, transaction charges on the receiver side are to be borne by the creditor. In a direct debit context, means that transaction charges on the sender side are to be borne by the creditor, transaction charges on the receiver side are to be borne by the debtor. |
| SLEV | Charges are to be applied following the rules agreed in the service level and/or scheme.                                                                                                                                                                                                                                                                                       |

This is following ChargeBearerType1Code from ISO20022.

This code is used in [oFA-PFDom].

### 3.8 Classification Code

MandateClassification1Code from ISO20022.

### 3.9 Consent Status

| Code                | Description                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------|
| received            | The consent data have been received and are technically correct. The data is not authorised yet.                     |
| rejected            | The consent data have been rejected e.g. since no successful authorisation has taken place.                          |
| partiallyAuthorised | The consent is due to a multi-level authorisation, some but not all mandated authorisations have been performed yet. |
| valid               | The consent is accepted and valid for GET account data calls and others as specified in the consent object.          |
| revokedByPsu        | The consent has been revoked by the PSU towards the ASPSP.                                                           |
| expired             | The consent expired.                                                                                                 |
| terminatedByTpp     | The corresponding TPP has terminated the consent by applying the DELETE method to the consent resource.              |
| replacedByTpp       | The corresponding TPP has terminated the (recurring) consent implicitly by submitting a new (recurring) consent.     |

The ASPSP might add further codes. These codes then shall be contained in the ASPSP's documentation of the XS2A interface.

This code is used in [oFA-Con], [oFA-PFSM].

### 3.10 Consent Type Code

| Code   | Description                                                                                               |
|--------|-----------------------------------------------------------------------------------------------------------|
| global | This consent type asks certain rights for dedicated account types without further detailing the accounts. |

| Code         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| detailed     | This consent type asks for access for dedicated accounts and dedicated rights.                                                                                                                                                                                                                                                                                                                                                                                                               |
| aspspManaged | This consent type asks for access on certain account types for dedicated access rights. The accounts provisioned will be added during the authorisation process by the ASPSP.                                                                                                                                                                                                                                                                                                                |
| accountList  | <p>This consent type asks for the list of accounts for each addressed account type. In difference to other account information consents, this consent might be authorised by using one authentication factor only, if supported by the ASPSP.</p> <p>In addition to the "accountDetails" access rights, also other non-transactional access rights might be asked for. This might have an impact on the decision of the ASPSP to apply SCA or only one authentication factor of the PSU.</p> |

This code is used by [oFA-Con].

### 3.11 Contact Type

| Code        | Description                                                             |
|-------------|-------------------------------------------------------------------------|
| technical   | To be contacted in case of items related to the API implementation,     |
| operational | To be contacted in case of topics related to operational issues.        |
| business    | To be contacted in case of business related topics on premium services. |
| security    | To be contacted in case of IT security related topics.                  |

### 3.12 Credit Debit Code

| Code | Description               |
|------|---------------------------|
| CRDT | Operation is an increase. |
| DBIT | Operation is a decrease.  |

### 3.13 Credit Transfer Payment Method Code

This code set corresponds to ISO 20022 PaymentMethod3Code:

| Type | Description                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TRF  | Credit Transfer                                                                                                                                                               |
| CHK  | Cheque                                                                                                                                                                        |
| TRA  | Transfer Advice<br><br>Remark: Currently not supported by the openFinance payment data model. The code "TRA" is only mentioned for completeness as it is defined in ISO20022. |

This data type is used in [oFA-PFDom].

### 3.14 Creditor Reference Type Code

This code set is equivalent to the ISO 20022 definition DocumentType3Code:

| Type | Description                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------|
| RADM | Document is a remittance advice sent separately from the current transaction.                                                   |
| RPIN | Document is a linked payment instruction to which the current payment instruction is related, for example, in a cover scenario. |
| FXDR | Document is a pre-agreed or pre-arranged foreign exchange transaction to which the payment transaction refers.                  |
| DISP | Document is a dispatch advice.                                                                                                  |
| PUOR | Document is a purchase order.                                                                                                   |
| SCOR | Document is a structured communication reference provided by the creditor to identify the referred transaction.                 |

### 3.15 Document Status

The following status codes are supported for document resources.

| Code       | Description                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------|
| received   | Document received, request syntactically okay.                                                                |
| rejected   | Document rejected for several possible reasons.                                                               |
| accessible | Document has been made accessible for the addressed document receive.                                         |
| withdrawn  | Submitted document has been withdrawn successfully before made accessible to the addressed document receiver. |
| replaced   | <b>RFU:</b> Not yet fully defined.                                                                            |
| expired    | <b>RFU:</b> Not yet fully defined.                                                                            |

This code is used in [oFA-IG-Doc].

### 3.16 Exchange Rate Type Code

This code set is equivalent to the ISO 20022 definition ExchangeRateType1Code:

| Type | Description                                                       |
|------|-------------------------------------------------------------------|
| SPOT | Exchange rate applied is the spot rate.                           |
| SALE | Exchange rate applied is the market rate at the time of the sale. |
| AGRD | Exchange rate applied is the rate agreed between the parties.     |

### 3.17 Frequency Code

The following codes from the "EventFrequency7Code" of ISO 20022 are supported:

- Daily
- Weekly
- EveryTwoWeeks
- Monthly
- EveryTwoMonths
- Quarterly
- SemiAnnual

- Annual
- MonthlyVariable

This code is used in [oFA-IG-Com], [oFA-IG-XPIS].

### 3.18 Interest Type Code

The following ISO definitions apply for these codes (based on ISO InterestType3code as defined for message type reda.006.001.01, but here, only a restricted code list is supported):

| Code | Description                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIXD | <b>ISO 20022:</b> Indicates that the type of interest is fixed.<br><br><b>Remark:</b> This is a fixed interest for the period defined either in the interest entry or implicitly by the contract period. |
| INDE | <b>ISO 20022:</b> Indicates that the type of interest is index.<br><br><b>Remark:</b> This rate could relate e.g. to interest rates of central banks.                                                    |

### 3.19 Limit Type

| Attribute   | Description                                                                          |
|-------------|--------------------------------------------------------------------------------------|
| singleOrder | The limit applies to every transaction.                                              |
| daily       | It is the limit of transactions to be authorised aggregated during a day.            |
| weekly      | It is the limit of transactions to be authorised aggregated during a calendar week.  |
| monthly     | It is the limit of transactions to be authorised aggregated during a calendar month. |

### 3.20 Mandate Setup Reason Code

This code is covered by the ISO20022 ExternalMandateSetupReason1Code.

**NOTE:** Currently, there is no external mandate setup reason code defined yet in ISO20022.

### 3.21 Message Code

The permitted message error codes and related HTTP response codes are listed below. New services may introduce new Message Codes which are not yet reflected here.

#### 3.21.1 Service Unspecific HTTP Error Codes

| Message Code        | HTTP Response Code | Description                                                                                                                                                                                                       |
|---------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CERTIFICATE_INVALID | 401                | The contents of the signature/corporate seal certificate are not matching PSD2 general PSD2 or attribute requirements.                                                                                            |
| ROLE_INVALID        | 401                | The TPP does not have the correct PSD2 role to access this service.                                                                                                                                               |
| CERTIFICATE_EXPIRED | 401                | Signature/corporate seal certificate is expired.<br><br>Remark: If a resource has been created already, a related status reason code is used instead.                                                             |
| CERTIFICATE_BLOCKED | 401                | Signature/corporate seal certificate has been blocked by the ASPSP or the related NCA.<br><br>Remark: If a resource has been created already, a related status reason code is used instead.                       |
| CERTIFICATE_REVOKED | 401                | Signature/corporate seal certificate has been revoked by QSTP.<br><br>Remark: If a resource has been created already, a related status reason code is used instead.                                               |
| CERTIFICATE_MISSING | 401                | Signature/corporate seal certificate was not available in the request but is mandated for the corresponding.<br><br>Remark: If a resource has been created already, a related status reason code is used instead. |

| Message Code             | HTTP Response Code | Description                                                                                                                                                                                                                                                                                                                                       |
|--------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLIENT_INVALID           | 401                | Client certificate is valid but the related client is not contained in the related ASPSP or Scheme directory                                                                                                                                                                                                                                      |
| CLIENT_INCONSISTENT      | 401                | Client information is not consistent with related directory entries                                                                                                                                                                                                                                                                               |
| SIGNATURE_INVALID        | 401                | Application layer eIDAS Signature for API Client authentication is not correct.                                                                                                                                                                                                                                                                   |
| SIGNATURE_MISSING        | 401                | Application layer eIDAS Signature for API authentication is mandated by the ASPSP but is missing.                                                                                                                                                                                                                                                 |
| ROLE_INVALID             | 401                | The TPP does not have the correct PSD2 role to access this service                                                                                                                                                                                                                                                                                |
| FORMAT_ERROR             | 400                | <p>Format of certain request fields are not matching the XS2A requirements. An explicit path to the corresponding field might be added in the return message.</p> <p>This applies to headers and body entries. It also applies in cases where these entries are referring to erroneous or not existing data instances, e.g. a malformed IBAN.</p> |
| PARAMETER_NOT_CONSISTENT | 400                | Parameters submitted by API Client are not consistent. This applies only for query parameters.                                                                                                                                                                                                                                                    |
| PARAMETER_NOT_SUPPORTED  | 400                | The parameter is not supported by the API provider. This code should only be used for parameters that are described as "optional if supported by API provider."                                                                                                                                                                                   |
| PSU_CREDENTIALS_INVALID  | 401                | The PSU-ID cannot be matched by the addressed ASPSP or is blocked, or a password resp. OTP was not correct. Additional information might be added.                                                                                                                                                                                                |



| Message Code         | HTTP Response Code                                             | Description                                                                                                                                                                   |
|----------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SERVICE_INVALID      | 400 (if payload)<br>405 (if HTTP method)                       | The addressed service is not valid for the addressed resources or the submitted data.                                                                                         |
| SERVICE_BLOCKED      | 403                                                            | This service is not reachable for the addressed PSU due to a channel independent blocking by the ASPSP. Additional information might be given by the ASPSP.                   |
| CORPORATE_ID_INVALID | 401                                                            | The PSU-Corporate-ID cannot be matched by the addressed ASPSP.                                                                                                                |
| CONSENT_UNKNOWN      | 403 (if path)<br>400 (if header)                               | The Consent-ID cannot be matched by the ASPSP relative to the API Client.                                                                                                     |
| CONSENT_INVALID      | 401                                                            | The consent was created by this TPP but is not valid for the addressed service/resource.                                                                                      |
| CONSENT_EXPIRED      | 401                                                            | The consent was created by this TPP but has expired and needs to be renewed.                                                                                                  |
| TOKEN_UNKNOWN        | 401                                                            | The OAuth2 token cannot be matched by the ASPSP relative to the TPP.                                                                                                          |
| TOKEN_INVALID        | 401                                                            | The OAuth2 token is associated to the TPP but is not valid for the addressed service/resource.                                                                                |
| TOKEN_EXPIRED        | 401                                                            | The OAuth2 token is associated to the TPP but has expired and needs to be renewed.                                                                                            |
| RESOURCE_UNKNOWN     | 404 (if account-id in path)<br>403 (if other resource in path) | The addressed resource is unknown relative to the API Client.<br><br>An example for a payload reference is creating a signing basket with an unknown resource identification. |

| Message Code       | HTTP Response Code                | Description                                                                                                                        |
|--------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
|                    | 400 (if payload)                  |                                                                                                                                    |
| RESOURCE_EXPIRED   | 403 (if path)<br>400 (if payload) | The addressed resource is associated with the TPP but has expired, not addressable anymore.                                        |
| RESOURCE_BLOCKED   | 400                               | The addressed resource is not addressable by this request, since it is blocked e.g. by a grouping in a signing basket.             |
| TIMESTAMP_INVALID  | 400                               | Timestamp not in accepted time period.                                                                                             |
| PERIOD_INVALID     | 400                               | Requested time period out of bound.                                                                                                |
| SCA_METHOD_UNKNOWN | 400                               | Addressed SCA method in the Authentication Method Select Request is unknown or cannot be matched by the ASPSP with the PSU.        |
| SCA_INVALID        | 400                               | Method Application on authorisation resource (e.g. Confirmation Request) blocked since SCA status of the resource equals "failed". |
| STATUS_INVALID     | 409                               | The addressed resource does not allow additional authorisation.                                                                    |

### 3.21.2 PIS Specific HTTP Error Codes

| Message Code    | HTTP Response Code | Description                                                            |
|-----------------|--------------------|------------------------------------------------------------------------|
| PRODUCT_INVALID | 403                | The addressed payment product is not available for the PSU .           |
| PRODUCT_UNKNOWN | 404                | The addressed payment product is not supported by the ASPSP.           |
| PAYMENT_FAILED  | 400                | The payment initiation POST request failed during the initial process. |

| Message Code           | HTTP Response Code | Description                                                                                                                                                              |
|------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |                    | Additional information may be provided by the ASPSP.                                                                                                                     |
| KID_MISSING            | 401                | The payment initiation has failed due to a missing KID. This is a specific message code for the Norwegian market, where ASPSP can require the payer to transmit the KID. |
| EXECUTION_DATE_INVALID | 400                | The requested execution date is not a valid execution date for the ASPSP.                                                                                                |
| CANCELLATION_INVALID   | 405                | The addressed payment is not cancellable e.g. due to cut off time passed or legal constraints.                                                                           |

### 3.21.3 AIS Specific HTTP Error Codes

| Message Code                      | HTTP Response Code | Description                                                                                                                                                                                               |
|-----------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONSENT_INVALID                   | 401                | The consent definition is not complete or invalid. In case of being not complete, the bank is not supporting a completion of the consent towards the PSU.<br><br>Additional information will be provided. |
| CONSENT_TYPE_NOT_SUPPORTED        | 400                | The consentType chosen in the consent object is not supported by the ASPSP.                                                                                                                               |
| SESSIONS_NOT_SUPPORTED            | 400                | The combined service flag may not be used with this ASPSP.                                                                                                                                                |
| CONTENT_TEMPORARILY_NOT_AVAILABLE | 404                | The contents of the response can (temporarily) not be obtained because a service, the interface depends on is currently not available.                                                                    |
| ACCESS_EXCEEDED                   | 429                | The access on the account has been exceeding the consented multiplicity without PSU involvement per day.                                                                                                  |

| Message Code              | HTTP Response Code | Description                                                                                         |
|---------------------------|--------------------|-----------------------------------------------------------------------------------------------------|
| REQUESTED_FORMATS_INVALID | 406                | The requested formats in the Accept header entry are not matching the formats offered by the ASPSP. |

### 3.21.4 PIIS Specific Error Codes

| Message Code       | HTTP Response Code | Description                                                                                        |
|--------------------|--------------------|----------------------------------------------------------------------------------------------------|
| CARD_INVALID       | 400                | Addressed card number is unknown to the ASPSP or not associated to the PSU.                        |
| NO_PIIS_ACTIVATION | 400                | The PSU has not activated the addressed account for the usage of the PIIS associated with the TPP. |

### 3.21.5 Signing Basket Specific Error Codes

| Message Code             | HTTP Response Code | Description                                                                                        |
|--------------------------|--------------------|----------------------------------------------------------------------------------------------------|
| REFERENCE_MIX_INVALID    | 400                | The used combination of referenced objects is not supported in the ASPSPs signing basket function. |
| REFERENCE_STATUS_INVALID | 409                | At least one of the references is already fully authorised.                                        |

### 3.21.6 Push AIS specific Error Codes

Push Account Information service specific error codes are as follows:

| Message Code                    | HTTP Response Code | Description                                                                                                                                                                                                                                                                                   |
|---------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PRIOR_SUBSCRIPTION_AVAILABLE    | 409                | The PSU/Corporate initiating the Push Account Information Service subscription already holds a subscription in the quadruple API Client/PSU/ASPSP/subservice or API Client/Corporate/ASPSP/subservice respectively or in the triple API Client/ASPSP/subservice in the case of direct access. |
| SUBSCRIPTION_INVALID            | 400                | Return this code in case the API Client tries to add a subscription entry, delete a subscription entry or a subscription itself where it had been deactivated already before.                                                                                                                 |
| SECONDARY_URI_NOT_SUPPORTED     | 400                | An API Client is trying to register a secondary Push URI within a Push Account Information subscription, which is not supported by the ASPSP.                                                                                                                                                 |
| MIME_TYPE_NOT_SUPPORTED         | 400                | An API Client is trying to define a Push Account Information subservice with one or several mime types, where none of the mime types is supported by the ASPSP for this subservice.                                                                                                           |
| LIMIT_FOR_BALANCE_NOT_SUPPORTED | 400                | The subscription for a Push Balance Information Subservice is asking to involve a credit limit into the balance selection criteria, where the related balance does not support to include a related credit limit of the account.                                                              |
| ACCOUNT_CURRENCY_NOT_MATCHING   | 400                | The API Client initiated a subscription where an amount is involved which is not matching the currency of the addressed account.                                                                                                                                                              |
| BALANCE_TYPE_NOT_SUPPORTED      | 400                | The API Client initiated a subscription for a balance type that is not                                                                                                                                                                                                                        |

| Message Code | HTTP Response Code | Description                                       |
|--------------|--------------------|---------------------------------------------------|
|              |                    | supported for the addressed account by the ASPSP. |

This data type is used by [oFA-PFSM].

### 3.22 Mime Type Code

| Code             | Description                                 |
|------------------|---------------------------------------------|
| application/JSON | Mime type for body encoding in JSON syntax. |
| application/XML  | Mime type for body encoding in XML syntax.  |
| plain/text       | Mime type for body encoding in text.        |

### 3.23 Name Prefix Code

Specifies the terms used to formally address a person.

| Code | Description   |
|------|---------------|
| DOCT | Doctor        |
| MADM | Madam         |
| MISS | Miss          |
| MIST | Mister        |
| MIKS | GenderNeutral |

### 3.24 Onboarding Status

| Code     | Description                                                                    |
|----------|--------------------------------------------------------------------------------|
| received | The first POST has been received and the onboarding resource has been created. |

| Code                  | Description                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>The check of the input parameter was successful. The services identified by servicelds belong to the scheme identified by the schemelds. The API Client is known. The necessary service contract has been signed and correspond to the ID given by serviceContractId.</p> <p>The confirmation of the price lists is necessary and not done until now.</p> |
| rejected              | <p>The second step (PUT) with the confirmation of the price lists has been failed. This can be e.g. due to one of the following reasons:</p> <ul style="list-style-type: none"> <li>• apiClient not known or no match with QWAC.</li> <li>• Necessary confirmation of conditions failed.</li> </ul>                                                          |
| expired               | The status of the onboarding resource is set to expired on day validTo+1.                                                                                                                                                                                                                                                                                    |
| terminatedByApiClient | The status of the onboarding resource is set to terminatedByApiClient, if a DELETE request has been sent by the API Client.                                                                                                                                                                                                                                  |
| terminatedByAspsp     | The status of the onboarding resource is set to terminatedByAspsp, if the ASPSP has terminated the corresponding onboarding.                                                                                                                                                                                                                                 |
| valid                 | The onboarding of the API Client to the services identified by servicelds has been completed successfully. No changes of the validity since the onboarding has been completed.                                                                                                                                                                               |
| validTimeRestricted   | The onboarding of the API Client to the services identified by servicelds has been completed successfully. The validity has been changed since the onboarding has been completed. The validTo parameter has been set to a date when the onboarding will expire. Reasons for this will be defined by the business logic.                                      |

This data type is used in [oFA-IG-Adm].

### 3.25 Order Status Code

Code set to identify the status of an order:

| Code               | Description                                                                                                                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| unknown            | The status of this order can currently not be determined.                                                                                                                                              |
| new                | Outstanding order with no executions.                                                                                                                                                                  |
| partiallyFilled    | Outstanding order with executions and remaining quantity.                                                                                                                                              |
| filled             | Order completely filled, no remaining quantity.                                                                                                                                                        |
| doneForDay         | Order not, or partially, filled; no further executions forthcoming for the trading day.                                                                                                                |
| canceled           | Cancelled order with or without executions.                                                                                                                                                            |
| replaced           | Cancelled order due to a replacement with or without executions.                                                                                                                                       |
| pendingCancel      | Order with a request for cancellation pending. Does not indicate that the order has been cancelled.                                                                                                    |
| stopped            | Order has been stopped at the exchange. Used when guaranteeing or protecting a price and quantity.                                                                                                     |
| rejected           | Order has been rejected by sell-side (broker, exchange, ECN).<br><br><b>Note:</b> An order can be rejected subsequent to order acknowledgment, i.e. an order can pass from "new" to "rejected" status. |
| suspended          | Order has been placed in suspended state at the request of the client.                                                                                                                                 |
| pendingNew         | Order has been received by sell-side's (broker, exchange, ECN) system but not yet accepted for execution.                                                                                              |
| calculated         | Order has been completed for the day (either filled or done for day). Commission or currency settlement details have been calculated.                                                                  |
| expired            | Order has been cancelled in broker's system due to time in force instructions.                                                                                                                         |
| acceptedForBidding | Order has been received and is being evaluated for pricing.                                                                                                                                            |
| pendingReplace     | Order with an Order Cancel/Replace Request pending.                                                                                                                                                    |



### 3.26 Order Time Limit Code

Code set to identify the time an order shall be in force. Possible values:

| Code                | Description                                                                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| day                 | <b>Day Order:</b> Limit order that is only good for the remainder of the trading day on which it is placed.                                                                                                             |
| goodTillCancel      | <b>Good till cancelled (GTC) Order:</b> Order that is working regardless of the time frame, until the order is explicitly cancelled. (Might still be subject of time limits from the ASPSP / broker).                   |
| atTheOpening        | <b>At the Opening Order:</b> Order to be executed at the very beginning of the trading day.                                                                                                                             |
| immediateOrCancel   | <b>Immediate or cancel (IOC) Order:</b> Order to be executed immediately and any unfilled portion be cancelled.                                                                                                         |
| fillOrKill          | <b>Fill or kill (FOK) Order:</b> Order that is directed to be executed immediately at the market or a specified price or canceled if not filled.                                                                        |
| fillAndKill         | <b>Fill and kill Order:</b> Order that can be filled partially or cancelled. If the order cannot be filled in its entirety at the specified premium, as much as possible of the order is filled and the rest is killed. |
| goodTillCrossing    | <b>Good till Crossing (GTX) Order:</b> Order that is canceled prior to the market entering into an auction, or crossing phase.                                                                                          |
| goodTillDate        | <b>Good till Day or Date (GTD) Order:</b> Order which remains active till its specified date, unless it has already been fulfilled or cancelled.                                                                        |
| atTheClose          | <b>At the Close Order:</b> Order to be executed at the end of the trading day, at the price available.                                                                                                                  |
| goodThroughCrossing | <b>Good through Crossing Order:</b> Order that is valid up till and including a crossing phase.                                                                                                                         |
| atCrossing          | <b>At Crossing Order:</b> Order that is valid only during crossing (auction) phases.                                                                                                                                    |
| goodForTime         | <b>Good for Time (GFT) Order:</b> Order that is valid for a pre-defined time period.                                                                                                                                    |

| Code           | Description                                                                                                                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| goodForAuction | <b>Good for Auction (GFA) Order:</b> Order that is valid for an auction initiated by a trading firm.                                                                                               |
| goodForMonth   | <b>Good for this Month (GFM) Order:</b> Order that is valid until the end of the current month, i.e. from the time of order submission until the end of the last trading day of the current month. |

### 3.27 Party Name Match Code

| Code | Description                        |
|------|------------------------------------|
| MTCH | Match                              |
| NMTC | No match                           |
| CMTC | Close Match                        |
| NOAP | Validation check is not applicable |

This code is used in [oFA-IG-VOP].

### 3.28 Party Identification Match Code

| Code | Description                        |
|------|------------------------------------|
| MTCH | Match                              |
| NMTC | No match                           |
| NOAP | Validation check is not applicable |

This code is used in [oFA-IG-VOP].

### 3.29 Payment Product

The definition of payment products as used in URIs for payment initiation. The following table might be extended on domestic level. The encoding of the codes is Max35Text.

| Attribute             | Description |
|-----------------------|-------------|
| sepa-credit-transfers |             |

| Attribute                      | Description |
|--------------------------------|-------------|
| instant-sepa-credit-transfers  |             |
| cross-border-credit-transfers. |             |
| target-2-payments              |             |

### 3.30 Preferred Method

Preferred method used to reach the contact.

| Code | Description       |
|------|-------------------|
| LETT | Letter            |
| MAIL | Email             |
| PHON | Phone             |
| FAXX | Fax               |
| CELL | MobileOrCellPhone |

### 3.31 Priority Code

This code set is equivalent to the ISO 20022 definition Priority2Code:

| Type | Description               |
|------|---------------------------|
| HIGH | Priority level is high.   |
| NORM | Priority level is normal. |

This code is used in [oFA-PFDom].

### 3.32 Referred Document Type Code

This code set is equivalent to the ISO 20022 definition DocumentType6Code:

| Type | Description                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------|
| MSIN | Document is an invoice claiming payment for the supply of metered services, for example gas or electricity supplied to a fixed meter |
| CNFA | Document is a credit note for the final amount settled for a commercial transaction.                                                 |
| DNFA | Document is a debit note for the final amount settled for a commercial transaction.                                                  |
| CINV | Document is an invoice.                                                                                                              |
| CREN | Document is a credit note.                                                                                                           |
| DEBN | Document is a debit note.                                                                                                            |
| HIRI | Document is an invoice for the hiring of human resources or renting goods or equipment.                                              |
| SBIN | Document is an invoice issued by the debtor.                                                                                         |
| CMCN | Document is an agreement between the parties, stipulating the terms and conditions of the delivery of goods or services.             |
| SOAC | Document is a statement of the transactions posted to the debtor's account at the supplier.                                          |
| DISP | Document is a dispatch advice.                                                                                                       |
| BOLD | Document is a shipping notice.                                                                                                       |
| VCHR | Document is an electronic payment document.                                                                                          |
| AROI | Document is a payment that applies to a specific source document.                                                                    |
| TSUT | Document is a transaction identifier as assigned by the Trade Services Utility                                                       |
| PUOR | Document is a purchase order.                                                                                                        |

### 3.33 Regulatory Reporting Type Code

| Type | Description                                                    |
|------|----------------------------------------------------------------|
| CRED | Regulatory information applies to the credit side.             |
| DEBT | Regulatory information applies to the debit side.              |
| BOTH | Regulatory information applies to both credit and debit sides. |

### 3.34 Request Status

Syntactically, the same codes are used as in Transaction Status, but with a different semantic, since the codes are related to a Request To Pay transaction and **not** to the corresponding payment.

| Code | Description                                                                                                                                                                                                               |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACCP | The request to pay has been explicitly accepted by the PSU.                                                                                                                                                               |
| ACWC | The request to pay has been explicitly accepted by the PSU with changes related to the execution date or the accepted amount.                                                                                             |
| RCVD | The request to pay data can be mapped to a PSU, no restrictions of the PSU related to payment requests apply and the payment request can be further processed.                                                            |
| PDNG | <ul style="list-style-type: none"> <li>The request to pay transaction has not yet been mapped to a PSU, or</li> <li>PSU related restrictions regarding request to pay functionality could not yet been checked</li> </ul> |
| RJCT | The request to pay has been rejected by either the API Server, the ASPSP or the PSU.                                                                                                                                      |

This code is used in [oFA-IG-RTP].

### 3.35 SCA Approach Type

| Code      | Description            |
|-----------|------------------------|
| Decoupled | Decoupled SCA Approach |
| Embedded  | Embedded SCA Approach  |

| Code           | Description                                         |
|----------------|-----------------------------------------------------|
| OAuth2         | Redirect SCA Approach based on OAuth2 protocol      |
| Redirect       | Plain Redirect Approach                             |
| Online-Channel | Asynchronous SCA Approach via ASPSP online channels |

### 3.36 SCA Process Flow Requirement Code

| Attribute                  | Description                                                                                                                                                                                                                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| psuAuthenticationRequired  | The ASPSP will not select the SCA approach to be applied until the TPP has successfully updated the PSU Authentication or delivered an OAuth2 access token.                                                                                                                                                                                          |
| psuIdentificationRequired  | The ASPSP mandates the PSU-ID for services involving SCA (like PIS, AIS, ...) before offering a SCA Selection.                                                                                                                                                                                                                                       |
| scaMethodSelectionRequired | The ASPSP will not select the SCA approach to be applied until the PSU has selected the authentication method to be applied via the API Client.<br>Note: As the PSU is neither authenticated nor identified at this point, the ASPSP will offer the ASPSP-specific SCA methods. Accordingly, the PSU has to choose a method which is allowed to him. |
| none                       | The ASPSP is able to select the SCA approach to be applied without a preceding PSU Authentication, PSU Identification or Authentication Method Selection. This then could also be handled in the online channel during authorisation.                                                                                                                |

### 3.37 SCA Exemption Code

| Code                    | Description                                        |
|-------------------------|----------------------------------------------------|
| none                    | no SCA exemption supported for payment initiation. |
| lowValueTransactions    |                                                    |
| paymentsToSelf          |                                                    |
| transactionRiskAnalysis |                                                    |
| trustedBeneficiaries    |                                                    |

### 3.38 SCA Status

The following codes are defined for this data type.

| Code              | Description                                                                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| received          | An authorisation or cancellation-authorisation resource has been created successfully.                                                                                                               |
| psuIdentified     | The PSU related to the authorisation or cancellation-authorisation resource has been identified.                                                                                                     |
| psuAuthenticated  | The PSU related to the authorisation or cancellation-authorisation resource has been identified and authenticated e.g. by a password or by an access token.                                          |
| scaMethodSelected | The PSU/TPP has selected the related SCA routine. If the SCA method is chosen implicitly since only one SCA method is available, then this is the first status to be reported instead of "received". |
| started           | The addressed SCA routine has been started.                                                                                                                                                          |
| unconfirmed       | SCA is technically successfully finalised by the PSU, but the authorisation resource needs a confirmation command by the TPP yet.                                                                    |
| finalised         | The SCA routine has been finalised successfully (including a potential confirmation command). This is a final status of the authorisation resource.                                                  |
| failed            | The SCA routine failed. This is a final status of the authorisation resource.                                                                                                                        |
| exempted          | SCA was exempted for the related transaction, the related authorisation is successful. This is a final status of the authorisation resource.                                                         |

This data type is used in [oFA-PFSM].

### 3.39 Securing Method Code

This code is providing information about how an authorised amount is secured within the ASPSP.

| Code               | Description                                                                     |
|--------------------|---------------------------------------------------------------------------------|
| reservationOfFunds | The bank is reserving the funds via a disposition of the amount on the account. |
| creditLine         | The bank is reserving the funds via the usage of a credit line of the customer. |
| bankGuarantee      | The bank is offering a payment guarantee.                                       |
| consumerLoan       | The payment is secured by a consumer loan.                                      |

This code is used in [oFA-IG-XPIS].

### 3.40 Securities Fee Type Code

This code is providing information the type of a fee that is applicable to a securities account.

| Code           | Description                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------|
| transactionFee | Fee that applies to a transaction within this securities account.                                  |
| brokerageFee   | Fee a broker charges to execute transactions or provide specialized services on behalf of clients. |
| managementFee  | Fee levied by an investment manager for overseeing an investment fund.                             |
| courtage       | Courtage                                                                                           |
| custodyFee     | Fee charged by the ASPSP for safekeeping the securities within the securities account.             |
| exchangeFee    | Fee charged for transferring shares from one fund to another fund within the same group.           |
| thirdPartyFee  | Fees charged by a third party                                                                      |
| otherFee       | Fee that is not covered by any of the fee types above.                                             |



### 3.41 Securities Order Side

Code set to identify the "side" / intended action for the respective order. Possible values:

| Code         | Description               |
|--------------|---------------------------|
| buy          | Order to buy.             |
| sell         | Order to sell.            |
| subscription | Order for a subscription. |
| redemption   | Order for a redemption.   |

### 3.42 Service Type

| Code      | Description                        |
|-----------|------------------------------------|
| PIS       | single payments                    |
| RPIS      | periodic payments                  |
| BPIS      | bulk payments                      |
| XFPIIS    | secured-payments                   |
| XDPIS     | deferred-payments                  |
| XDFPIS    | secured-deferred-payments          |
| XMDPIS    | multiple-deferred-payments         |
| XMDFPIS   | secured-multiple-deferred-payments |
| XRPIIS    | recurring-payments                 |
| XRFPIS    | secured-recurring-payments         |
| XMRPIS    | multiple-recurring-payments        |
| COF       | funds confirmation                 |
| CONS-AIS  | consent on account information     |
| XCONS-DOC | consent for document services      |

| Code       | Description                                             |
|------------|---------------------------------------------------------|
| XCONS-PSUP | consent on psu parameter information                    |
| XCONS-COF  | consent for acceptance of COF requests                  |
| AIS        | Account information for payment accounts                |
| XAIS-LA    | Account information for loan accounts                   |
| XAIS-SA    | Account information for savings accounts                |
| XAIS-CA    | Account information for single cards                    |
| XAIS-SC    | Account information for securities accounts             |
| XPAIS-AE   | Push account information service for account entries    |
| XPAIS-AB   | Push account information service for account balances   |
| XPAIS-AS   | Push account information service for account statements |
| XPAIS-RTP  | Push account information service for incoming RTP       |
| RTP        | Request to Pay                                          |

This code is used in [oFA -IG-PAIS]

### 3.43 .Signing Class

The definition of authorisation rights of a user.

| Attribute           | Description                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| singleSigning       | User is entitled to sign related transactions on his/her own.                                                               |
| firstSigning        | User is entitled to sign related transactions as a first signer.                                                            |
| secondSigning       | User is only entitled to sign related transactions after a first signer has signed the transaction.                         |
| nonSpecifiedSigning | User is entitled to sign related transactions, without providing more detailed information about the related signing class. |
| transportSigning    | User is only entitled to sign the submission of a related transaction, but not to authorise the related transaction.        |
| none                | User has no rights related submission or authorisation of related transactions.                                             |

### 3.44 Status Reason Code

At a minimum, the following codes from the ISO20022 ExternalStatusReasonCode1 list need to be supported by the ASPSP. The codes should only used, if a related resource has been generated on API level, i.e. if a transaction status can be provided. Otherwise a related Message Code to be used in Error information would be sufficient.

| Code | Name                                                       | ISO 20022 Definition and openAPI Remarks                                                                                                                                                                                                                                                                                                                            |
|------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AM04 | InsufficientFunds                                          | Amount of funds available to cover specified message amount is insufficient.<br><b>Remark:</b> Indicates that the reason for rejecting the payment is that the required funds have been found to be not available for the specific (e.g due to missing funds or due to configured limits) during processing after the initial acceptance of the payment initiation. |
| AM21 | LimitExceeded                                              | Transaction amount exceeds limits agreed between bank and client.                                                                                                                                                                                                                                                                                                   |
| BEXX | There are several codes identifying specific invalid data. | Several circumstances.                                                                                                                                                                                                                                                                                                                                              |
| CN01 | AuthorisationCancelled                                     | Authorisation is cancelled.                                                                                                                                                                                                                                                                                                                                         |
| DS0C | SignerCertificateRevoked                                   | The signer certificate is revoked.<br><b>Remark:</b> Is used, if the signature/corporate seal certificate is revoked.                                                                                                                                                                                                                                               |
| DS0D | SignerCertificateNotValid                                  | The signer certificate is not valid (revoked or not active).<br><b>Remark:</b> Is used, if the signature/corporate seal certificate is blocked by the ASPSP or if the certificate is expired.                                                                                                                                                                       |
| DS0A | DataSignRequested                                          | Data signature is required.<br><b>Remark:</b> Is used, if a signing by a corporate seal/certificate is mandated by the ASPSP but is missing.                                                                                                                                                                                                                        |
| DS04 | OrderRejected                                              | The order was rejected by the bank side (for reasons concerning content)<br><b>Remark:</b> Indicates that the reason for rejecting the payment is that the content of the payment initiation has been found invalid during processing after the initial acceptance of the payment initiation.                                                                       |
| FOCR | FollowingCancellationRequest                               | Return following a cancellation request.                                                                                                                                                                                                                                                                                                                            |
| SL11 | CreditorNotOnWhitelistOfDebtor                             | Whitelisting service offered by the Debtor Agent; Debtor has not included the Creditor on its "Whitelist" (yet). In the Whitelist the Debtor may list all allowed Creditors to debit Debtor bank account.                                                                                                                                                           |
| TKSP | TokenSuspended.                                            | Token found with suspended status.                                                                                                                                                                                                                                                                                                                                  |

| Code | Name         | ISO 20022 Definition and openAPI Remarks |
|------|--------------|------------------------------------------|
| TKXP | TokenExpired | Token expired.                           |

This code is used in [oFA-IG-Com], [oFA-PFSM]..

### 3.45 Subscription Status

| Code                | Description                                                                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| received            | The subscription data have been received and are technically correct. The data is not authorised yet.                                                                                                |
| rejected            | The subscription data have been rejected e.g. since no successful authorisation has taken place.                                                                                                     |
| partiallyAuthorised | The subscription is due to a multi-level authorisation, some but not all mandated authorisations have been performed yet.                                                                            |
| valid               | The subscription is accepted and valid for the related subscribed services as specified in the subscription object.                                                                                  |
| validInChange       | The subscription is accepted and valid for the related subscribed services as specified in the subscription object. A new subscription entry has been submitted which is not yet (fully) authorised. |
| revokedByPsu        | The subscription has been revoked by the PSU towards the ASPSP.                                                                                                                                      |
| cancelledByAspsp    | The subscription has been cancelled by the ASPSP.                                                                                                                                                    |
| expired             | The subscription expired.                                                                                                                                                                            |
| terminatedByTpp     | The corresponding TPP has terminated the subscription by applying the DELETE method to the subscription resource.                                                                                    |

The ASPSP might add further codes. These codes then shall be contained in the ASPSP's documentation of the XS2A interface.

This code is used in [oFA-PFSM], [oFA -IG-PAIS].

### 3.46 Subscription Entry Status

| Code                | Description                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| received            | The subscription data have been received and are technically correct. The data is not authorised yet.                     |
| rejected            | The subscription data have been rejected e.g. since no successful authorisation has taken place.                          |
| partiallyAuthorised | The subscription is due to a multi-level authorisation, some but not all mandated authorisations have been performed yet. |
| valid               | The subscription is accepted and valid for the related subscribed service as specified in the subscription object.        |
| revokedByPsu        | The subscription has been revoked by the PSU towards the ASPSP.                                                           |
| expired             | The subscription expired.                                                                                                 |
| terminatedByTpp     | The corresponding TPP has terminated the subscription by applying the DELETE method to the subscription resource.         |

This code is used in [oFA-PFSM], [oFA-IG-PAIS].

### 3.47 Trading Session Type Code

Code set to identify the type of a trading session, in which an order shall be executed. Possible values:

| Code       | Description                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------------------|
| auctions   | Orders that are traded in regularly occurring auctions: orders are accumulated and executed on the basis of defined algorithms. |
| continuous | Orders that are executed directly against each other as they hit the marketplace.                                               |

### 3.48 Transaction Status

The transaction status is filled with codes of the ISO 20022 data table:

| Code | Name                               | ISO 20022 Definition                                                                                                                                                                                                                                                                                                                 |
|------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACCC | AcceptedSettlementCompleted        | Settlement on the creditor's account has been completed.                                                                                                                                                                                                                                                                             |
| ACCP | AcceptedCustomerProfile            | Preceding check of technical validation was successful. Customer profile check was also successful.                                                                                                                                                                                                                                  |
| ACSC | AcceptedSettlementCompleted        | Settlement on the debtor's account has been completed.<br><b>Usage:</b> this can be used by the first agent to report to the debtor that the transaction has been completed.<br><b>Warning:</b> this status is provided for transaction status reasons, not for financial information. It can only be used after bilateral agreement |
| ACSP | AcceptedSettlementInProgress       | All preceding checks such as technical validation and customer profile were successful and therefore the payment initiation has been accepted for execution.                                                                                                                                                                         |
| ACTC | AcceptedTechnicalValidation        | Authentication and syntactical and semantical validation are successful                                                                                                                                                                                                                                                              |
| ACWC | AcceptedWithChange                 | Instruction is accepted but a change will be made, such as date or remittance not sent.                                                                                                                                                                                                                                              |
| ACWP | AcceptedWithoutPosting             | Payment instruction included in the credit transfer is accepted without being posted to the creditor customer's account.                                                                                                                                                                                                             |
| RCVD | Received                           | Payment initiation has been received by the receiving agent.                                                                                                                                                                                                                                                                         |
| PDNG | Pending                            | Payment initiation or individual transaction included in the payment initiation is pending. Further checks and status update will be performed.                                                                                                                                                                                      |
| RJCT | Rejected                           | Payment initiation or individual transaction included in the payment initiation has been rejected.                                                                                                                                                                                                                                   |
| CANC | Cancelled                          | Payment initiation has been cancelled before execution                                                                                                                                                                                                                                                                               |
| ACFC | AcceptedFundsChecked               | Pre-ceeding check of technical validation and customer profile was successful and an automatic funds check was positive .                                                                                                                                                                                                            |
| PATC | PartiallyAcceptedTechnical Correct | The payment initiation needs multiple authentications, where some but not yet all have been performed. Syntactical and semantical validations are successful.                                                                                                                                                                        |
| PART | PartiallyAccepted                  | A number of transactions have been accepted, whereas another number of transactions have not yet achieved 'accepted' status.<br><b>Remark:</b> This code may be used only in case of bulk payments. It is only used in a situation where all mandated authorisations have been applied, but some payments have been rejected.        |

This code is used in [oFA-IG-Com], [oFA-PFSM], [oFA-IG-RTP].

### 3.49 Transaction Type

| Attribute                              | Description                                    |
|----------------------------------------|------------------------------------------------|
| payments                               | All payment types                              |
| payments.sepa-credit-transfers         | All sepa credit transfers.                     |
| payments.cross-border-credit-transfers | Cross border payments.                         |
| consents                               | Consents, e.g. on AIS                          |
| subscriptions                          | subscriptions for Push AIS Services            |
| all                                    | All transaction types that need authorisation. |

### 3.50 Type of Order Code

Code set to indicate characteristics of an order.

| Code                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allOrNone            | A round-lot market or limit-price order which must be executed in its entirety or not at all; unlike 'fill or kill', these orders are not cancelled if not executed as soon as received.                                                                                                                                                                                                                                                               |
| buyContraShortExempt | Order to buy contra short exempt.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| buyContraShort       | Order to buy contra short.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| buyMinus             | Order to buy at a price lower than the current market price. This is an order to buy a stated amount of a financial instrument provided that its price is not higher than the last sale if the last sale was a minus or zero minus tick, not higher than the last sale minus the minimum fractional change in the stock if the last sale was a plus or zero plus tick. The price limit indicates the highest price at which the order can be executed. |
| carefully            | Order that is not to be executed as a whole because it may disturb the price.                                                                                                                                                                                                                                                                                                                                                                          |

| Code               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| combinationOrder   | Order that is linked to another order to buy or sell and must be executed as a unit, both or none, or cancelled as a unit.                                                                                                                                                                                                                                                                                                                                                        |
| discretionary      | Order where the executing broker or investment manager decides on the quantity or price.                                                                                                                                                                                                                                                                                                                                                                                          |
| doNotIncrease      | Limit order to buy or stop order to sell or stop limit order that is not to be increased in shares on the ex-dividend date as a result of a stock dividend or distribution.                                                                                                                                                                                                                                                                                                       |
| doNotReduce        | Limit order to buy or stop order to sell, or stop-limit order to sell that is not to be reduced in price by the amount of an ordinary cash dividend. Applies only to ordinary stock dividends; should be reduced for other distributions.                                                                                                                                                                                                                                         |
| icebergOrder       | Type of limit order whose overall quantity is not transparent to the market. Rather, only a client-defined part of the order is shown to the market (the 'tip of the iceberg'). Upon execution of the first 'tip' (that is partial execution to the client while for the market it does not appear to be a 'partial') the system releases the next 'tip' of the same size to the market until the 'iceberg' has been melted down successfully. Different from a usual care order. |
| limitWith          | Order to be executed at a limit price, with a round-lot (or board-lot) sales; valid only for odd lot orders.                                                                                                                                                                                                                                                                                                                                                                      |
| limitWithout       | Order to be executed at a limit price, without a round-lot (or board-lot) sales; valid only for odd lot orders.                                                                                                                                                                                                                                                                                                                                                                   |
| limitOrder         | Order to buy at the indicated price limit or lower or an order to sell at the indicated limit price or higher.                                                                                                                                                                                                                                                                                                                                                                    |
| atMarket           | Order to buy or sell a specified amount of a financial instrument at the quoted market price or better.                                                                                                                                                                                                                                                                                                                                                                           |
| marketNotHeld      | Order to buy or sell a specified amount of a financial instrument at the quoted market price or better with some discretion on the price limit.                                                                                                                                                                                                                                                                                                                                   |
| marketToLimitOrder | Type of order that couples the high possibility of execution (Market Order) with a protection against unwanted price fluctuations (Limit Order).                                                                                                                                                                                                                                                                                                                                  |



| Code               | Description                                                                                                                                                                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| marketUntilTouched | Order to buy or sell a specified amount of a financial instrument at the quoted market price or better with some discretion on the price limit.                                                                                                                                             |
| notHeld            | Order that may be executed in partials or outside the hours of the exchange or other exchange rules.                                                                                                                                                                                        |
| orderLie           | Order that is related to another order where the second order may be cancelled without cancelling the first. Normally, the sell order must be executed before the buy order.                                                                                                                |
| stopLimit          | Stop order to buy (sell) that becomes a limit order at the limit price when the financial instrument trades at or above (below) the stop price after the order is submitted.                                                                                                                |
| stopOrder          | Order to buy that becomes a market order when the financial instrument trades at or above the stop price after the order is submitted or an order to sell which becomes a market order when the financial instrument trades at or below the stop price.                                     |
| stopLoss           | Order to sell that sets the sell price below the market price.                                                                                                                                                                                                                              |
| sellPlus           | Order to sell a stated amount provided that the price is not lower than the last sale price if the last sale was a plus or zero plus tick and not lower than the last sale minus the minimum fractional change in the financial instrument if the last sale was a minus or zero minus tick. |
| sellShortExempt    | Order to sell short which is exempt from short-sale rules.                                                                                                                                                                                                                                  |
| sellShort          | Order to sell a financial instrument that the seller does not own; a sale effected by delivering a financial instrument borrowed by or for the account of the seller.                                                                                                                       |

### 3.51 VOP Bulk Status

| Code     | Description                                            |
|----------|--------------------------------------------------------|
| received | Bulk technically received and bulk formatted correctly |
| pending  | processing ongoing                                     |

| Code      | Description                                                                       |
|-----------|-----------------------------------------------------------------------------------|
| rejected  | Some processing errors have occurred which did not allow to process the requests. |
| completed | all verification results, requested in the vop bulk, are completed                |

This code is used in [oFA-IG-VOP].

### 3.52 Other ISO-related basic Types

The following codes and definitions are used from ISO 20022

- **Account Identification Code:** ExternalAccountIdentification1Code
- **Address Type Code:** AddressType2Code
- **Bank Transaction Code:** ExternalBankTransactionDomain1Code
- **BBAN:** BBANIdentifier
- **BICFI:** BICFIIdentifier
- **Category Purpose Code:** ExternalCategoryPurpose1Code
- **Cash Account Type:** ExternalCashAccountType1Code
- **Clearing System Identification Code:** ExternalClearingSystemIdentification1Code
- **Document Format Code:** ExternalDocumentFormat1Code
- **Document Type Code:** ExternalDocumentType1Code
- **Exchange Rate Type Code:** ExchangeRateType1Code
- **External Financial Instrument Identification Type Code:** ExternalFinancialInstrumentIdentificationType1Code (see [ISO20022\_SR])
- **External Local Instrument Code:** ExternalLocalInstrument1Code
- **Financial Institution Code:** ExternalFinancialInstitution1Code
- **IBAN:** IBAN2007Identifier

Pattern: [A-Z]{2}[0-9]{2}[A-Z0-9]{1,30}

- **Instruction Code:** Instruction3Code
- **ISIN:** ISINOct2015Identifier (see [ISO20022\_SR])  
Pattern: [A-Z]{2,2}[A-Z0-9]{9,9}[0-9]{1,1}
- **Language Code:** Code following ISO – 639-3 (three letter code)
- **LEI:** LEIIdentifier as defined in ISO 17442 "Financial Services - Legal Entity Identifier (LEI)".
- **Mandate Reason Code:** ExternalMandateReason1Code
- **Market Type Code:** MarketType4Code (see [ISO20022\_SR])
- **Merchant Category Code:** Category code conform to ISO 18245
- **Organisation Identification Code:** ExternalOrganisationIdentification1Code
- **PersonIdentificationCode:** ExternPersonIdentification1Code
- **Phone Number:** PhoneNumber
- **Purpose Code:** ExternalPurpose1Code
- **Proxy Account Type Code:** ExternalProxyAccountType1Code
- **Sequence Type Code:** SequenceType3Code
- **Service Level Code:** ExternalServiceLevel1Code
- **Status Reason Code:** ExternalStatusReason1Code
- **Transaction Activity Code:** TransactionActivity1Code (see [ISO20022\_SR])
- **Type of Price Code:** TypeOfPrice17Code (see [ISO20022\_SR])

The following code is a concatenated code from ISO20022

- **BankTransactionCode:** This code type is concatenating the three ISO20022 Codes Domain Code, Family Code and SubFamily Code by hyphens, resulting in "DomainCode"- "FamilyCode"- "SubFamilyCode".

**Example:** PMNT-RCDT-ESCT defining a transaction assigned to the PayMeNT Domain (PMNT), belonging to the family of ReceivedCreditTransfer (RCDT) that facilitated the EuropeanSEPACreditTransfer (ESCT)

To reflect a securities position's balance type, **SecuritiesBalanceTypeCode** is defined as the code list, containing all codes from ISO 20022 code lists

- SecuritiesBalanceType1Code (see [ISO20022\_IF])
- SecuritiesBalanceType2Code (see [ISO20022\_IF])
- SecuritiesBalanceType11Code (see [ISO20022\_SR])
- SecuritiesBalanceType12Code (see [ISO20022\_SR])

For all codes used in JSON structures, not the abbreviation defined for XML encoding, but the name of the code is used as value.

The following Codes are used from other ISO standards:

- **Currency Code:** Codes following ISO 4217 Alpha 3
- **Country Code:** Two characters as defined by ISO 3166

Further basic ISO data types:

- **ISODateTime:** A particular point in the progression of time defined by a mandatory date and a mandatory time component, expressed in either UTC time format (YYYY-MM-DDThh:mm:ss.sssZ) or local time with UTC offset format (YYYY-MM-DDThh:mm:ss.sss+/-hh:mm),. These representations are defined in "XML Schema Part 2: Datatypes Second Edition - W3C Recommendation 28 October 2004" which is aligned with ISO 8601.
- **ISODate:** A particular point in the progression of time in a calendar year expressed in the YYYY-MM-DD format.

### 3.53 Additional Cryptographic Code Lists

Codes to identify algorithms for signature generation and encryption shall be used as defined by [RFC7518]. At least the following codes have to be supported:

#### Signature Algorithm Code

##### PS256

for RSASSA-PSS using SHA-256 and MGF1 with SHA-256 (section 3.5 of [RFC7518])

#### Key Encryption Algorithm Code

##### RSA-OAEP-256

for RSAES OAEP using SHA-256 and MGF1 with SHA-256 (section 4.3 of [RFC7518])

## **Content Encryption Algorithm Code**

### **A256GCM**

for AEC in Galois/Counter Mode using 256-bit key (section 5.3 of [RFC7518])

## 4 References

### 4.1 Normative References

- [EBA-RTS] Commission Delegated Regulation (EU) 2018/389 of 27 November 2017 supplementing Directive 2015/2366 of the European Parliament and of the Council with regard to Regulatory Technical Standards for Strong Customer Authentication and Common and Secure Open Standards of Communication, C(2017) 7782 final, published 13 March 2018
- [EBA-FR] Final Report, Draft Regulatory Technical Standards, amending Commission Delegated Regulation (EU) 2018/389 supplementing Directive (EU) 2015/2366 of the European Parliament and of the Council with regard to regulatory technical standards for strong customer authentication and common and secure open standards of communication, published 5 April 2022
- [eIDAS] Regulation (EU) No 910/2014 of the European Parliament and of the Council on Electronic Identification and Trust Services for Electronic Transactions in the Internal Market, 23 July 2014, published 28 August 2014
- [ETSI PSD2] ETSI TS 119 495 V1.1.2; Electronic Signatures and Infrastructures (ESI); Sector Specific Requirements; Qualified Certificate Profiles and TSP Policy Requirements under the payment services Directive (EU) 2015/2366
- [PSD2] Directive (EU) 2015/2366 of the European Parliament and of the Council on payment services in the internal market, published 23 December 2015
- [HAL] Kelley, M., "HAL - Hypertext Application Language", 2013-09-18, [http://stateless.co/hal\\_specification.html](http://stateless.co/hal_specification.html)
- [ISO20022\_IF] ISO 20022, Message Definition Report - Part 2 Investment Funds, February 2020
- [ISO20022\_SR] ISO 20022, Message Definition Report - Part 2 Settlement And Reconciliation - ISO – Maintenance 2019 – 2020, February 2020
- [RFC2426] Dawson, F. and T. Howes, T., "vCard MIME Directory Profile", September 1998, <https://tools.ietf.org/html/rfc2426>
- [RFC3230] Mogul, J. and A. Van Hoff, "Instance Digests in HTTP", RFC 3230, DOI 10.17487/RFC3230, January 2002, <https://www.rfc-editor.org/info/rfc3230>
- [RFC3986] [T. Berners-Lee, R. Fielding and L. Masinter, "Uniform Resource Identifier \(URI\): Generic Syntax", RFC 3986, January 2005, https://tools.ietf.org/html/rfc3986](#)
- [RFC4122] P. Leach, M. Mealling, R. Salz, "A Universally Unique IDentifier (UUID) URN Namespace", July 2005, <https://tools.ietf.org/html/rfc4122>

- [RFC4648] Josefsson, S., "The Base16, Base32, and Base64 Data Encodings", October 2006, <https://tools.ietf.org/html/rfc4648>
- [RFC5843] Bryan, A, "Additional Hash Algorithms for HTTP Instance Digests", RFC 5843, DOI 10.17487/RFC5843, April 2010, <https://www.rfc-editor.org/info/rfc5843>
- [RFC6749] Hardt, D., "The OAuth 2.0 Authorization Framework", October 2012, <https://tools.ietf.org/html/rfc6749>
- [\[RFC7231\] R. Fielding, J. Reschke, Hypertext Transfer Protocol \(HTTP/1.1\): Semantics and Content](#)
- [RFC7518] JSON Web Algorithms (JWA), Request for Comments 7518, Internet Engineering Task Force (IETF), M. Jones (Microsoft), May 2015
- [RFC7807] M. Nottingham, Akamai, E. Wilde, „Problem Details for HTTP APIs“, March 2016, <https://tools.ietf.org/html/rfc7807>
- [\[RFC 8414\] M. Jones, N. Sakimura, J. Bradley; "OAuth 2.0 Authorization Server Metadata";](#)

## 4.2 Informative References

- [oFA-IG-Com] openFinance API Framework, Implementation Guidelines, Compliance Services, Version 2.2, 24 January 2025.
- [oFA-IG-Doc] openFinance API Framework, Implementation Guidelines, (Standalone) Document Services, Version 1.0, 16 December 2024
- [oFA-IG-Man] openFinance API Framework, Implementation Guidelines, Mandate Services, Version 1.0, 21 October 2024
- [oFA-IG-VOP] openFinance API Framework, Implementation Guidelines, VOP Services, Version 1.0, Version 1.0, 24 January 2025
- [oFA-IG-XPIS] openFinance API Framework, Implementation Guidelines, Extended Payment Initiation Services, Version 2.1, 31 July 2024.
- [oFA-IG-XAIS] openFinance API Framework, Implementation Guidelines, Extended Account Information Services, Version 2.1, 31 July 2024
- [oFA-IG-RTP] openFinance API Framework, Implementation Guidelines, Request to Pay Services, version 2.1, 18 October 2024
- [oFA-IG-PAIS] openFinance API Framework, Implementation Guideleines, Push Account Information Services, 31 July 2024
- [oFA-PFSM] openFinance API Framework, Protocol Functions and Security Measures, Version 2.1, 31 July 2024

- [oFA-PFDom] openFinance API Framework, Payment Formats for Version 2.x and Domestic AIS/PIS formats, current version
- [oFA-Con] openFinance API Framework, Consent API, Version 2.1, 31 July 2024
- [oFA-IG-Adm] openFinance API Framework, Implementation Guidelines, Administrative Services, Version 1.0, 21 February 2024
- [oFA-Disc] openFinance API Framework, Implementation Guidelines, Discovery Services, Version 1.1, 24 January 2025
- [oFA-TDM2] openFinance API Framework, Transactions Data Model and Account Statements for Version 2.0 of the openFinance API Framework, Version 1.0, 16 November 2022



## 5 Annex A: Change Log

### 5.1 Changes from Version 2.1 to Version 2.2

The following changes have been applied in version 2.2 relative to version 2.1:

| Section  | Change                                                                                    | Reason                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preamble | The licence conditions of Berlin Group have been removed.                                 | To support the European community in re-using the openFinance data model, when not directly using openFinance APIs.                                                                  |
| 2.1      | The attribute cardAccounts have been added.                                               | single card accounts and reconciliation card accounts need now to be addressed explicitly in the consent. The former solution did not allow a proper differentiation.<br><br>Erratum |
| 2.7      | The attribute resourceId has been restricted from string to max70Text.                    | Clarification, as applied everywhere in the API Framework.                                                                                                                           |
| 2.7      | The description of the attribute ownerNames was extended.                                 | Clarification                                                                                                                                                                        |
| 2.16     | New data type to support the features of adding a debtor account to a resource.           | CR 113                                                                                                                                                                               |
| 2.43     | The spelling of the attribute clearingSystemIdentificationCode was corrected.             | Erratum                                                                                                                                                                              |
| 2.68     | Added the status reason to the initiation data attributes                                 | Clarification.                                                                                                                                                                       |
| 2.72     | Added an attribute to "Links" for indicating an update of a resource by a debtor account. | CR113                                                                                                                                                                                |
| 2.81     | The attributes oauth2PrestepMetadataUri and oauth2OfflineScopeRequired have               | Better support of documentation.                                                                                                                                                     |

| Section | Change                                                                    | Reason |
|---------|---------------------------------------------------------------------------|--------|
|         | been added to the API Security parameters.                                |        |
| 2.83    | Added the value "pageSize" for supported query parameters in transactions | CR115  |
| 2.83    | Added the attribute maxPageSize                                           | CR115  |
| 2.112   | Some attributes adapted to lowerCamelCase                                 | Errata |
| 3.52    | Do not allow pure local time zone anymore (only with UTC offset).         | CR111  |

## 5.2 Changes from Version 2.2 to Version 2.2.1

The following changes have been applied in version 2.2.1 relative to version 2.2:

| Section | Change                                                     | Reason                 |
|---------|------------------------------------------------------------|------------------------|
| 2.50    | New attribute "Duration" for Mandate API                   | Support of new service |
| 0       | New attribute "Extended Party Description" for Mandate API | Support of new service |
| 2.74    | New attribute "Mandate Type" for Mandate API               | Support of new service |
| 2.75    | New attribute "Mandate Status" for Mandate API             | Support of new service |
| 2.77    | New attribute "Occurrences" for Mandate API                | Support of new service |
| 2.78    | Condition of attributes changed                            | Errata                 |
| 2.132   | New attribute "Referred Mandate Document information"      | Support of new service |

| Section | Change                                                             | Reason                                                 |
|---------|--------------------------------------------------------------------|--------------------------------------------------------|
| 3.8     | New "Classification code" for Mandate API                          | Support of new service                                 |
| 3.20    | New "Mandate Setup Reason Code" for Mandate API                    | Support of new service                                 |
| 3.21.1  | New codes "CLIENT_INVALID" and "CLIENT_INCONSISTENT"               | Support error information for directory related issues |
| 3.52    | Added "Organisation Identification Code" and "Mandate Reason Code" | Errata and support of new service                      |

### 5.3 Changes from Version 2.2.1 to Version 2.2.2

The following changes have been applied in version 2.2.2 relative to version 2.2.1:

| Section | Change                                                                                                  | Reason                                                                          |
|---------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 2.77    | Corrected Occurences to Occurrences in the title<br><br>Changed the code OOF to OOFF in the description | Errata                                                                          |
| 2.82    | New attribute<br>psuAccountSelectionSupported                                                           | New feature resulting from CR113                                                |
| 2.86    | new data Type Parameters<br>Mandate Services                                                            | The new Mandate Services imply additions to the discovery services data model.  |
| 2.87    | new data Type Parameters<br>Document Services                                                           | The new Document Services imply additions to the discovery services data model. |
| 2.90    | corrected the description for attribute serviceType.<br><br>Added attributes for character set items    | Errata                                                                          |

| Section | Change                                                                  | Reason                                                                                                                                                           |
|---------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.91    | new data type Parameters VOP Services                                   | The new VOP Services imply additions to the discovery services data model.                                                                                       |
| 2.131   | new data type Related Document Information                              | Needed for the new Document Services                                                                                                                             |
| 2.132   | data type of attribute type code was corrected                          | Erratum                                                                                                                                                          |
| 3.15    | new status code Document Status                                         | Needed for the new Document Services                                                                                                                             |
| 3.33    | added the new Approach Type code "Online-Channel"                       | Needed as new SCA approach for asynchronous authorisations in online channels, introduced in the openFinance API Framework Release 2.1 e.g. for Mandate Services |
| 3.44    | added several Status Reason Codes                                       | Needed for several services and related updates.                                                                                                                 |
| 4.2     | implementation guidelines for document, mandate and VOP services added. | New services                                                                                                                                                     |
| several | Replaced the notion of "Remark for Future" by "RFU"                     | harmonising both expressions used in the document before to one expression.                                                                                      |

#### 5.4 Changes from Version 2.2.2 to Version 2.2.3

| Section | Change                                                                                  | Reason                                               |
|---------|-----------------------------------------------------------------------------------------|------------------------------------------------------|
| 2.60    | New data type From To Date                                                              | Needed for the Discovery API.                        |
| 2.79    | New data type for Compliance Specs for Extended Bulk Payment Status                     | New Function in Compliance Specs.                    |
| 2.80    | New data type for Compliance Specs for Extended Bulk Payment to add also the VOP Status | Integration of VOP Services into Payment Initiation. |

| Section                | Change                                                                                                                                                                                         | Reason                                                                                           |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 2.81                   | Adding the value "ASPSP-Channel" to SCA approaches in Discovery APIs                                                                                                                           | Recognising the new SCA approach for asynchronous SCA in ASPSP online channels.                  |
| 2.81                   | renaming the attribute client-explicit-authorisation-preferredSupported into clientExplicitAuthorisationPreferenceSupported and client-sca-preferenceSupported to clientScaPreferenceSupported | lowerCamelCase is used for header parameters if they appear in the Discovery API in the payload. |
| 2.82                   | The value "JSON-Extended" has been added to the values for the attribute supportedStatusFormats.                                                                                               | Support of the new Extended Status endpoint needs to be defined.                                 |
| 2.82                   | Three attributes with prefix "clientVop" has been added.                                                                                                                                       | Potential support of the VOP related header parameters needs to be defined.                      |
| 2.99<br>2.100<br>2.101 | Three new data types have been added to support bulk VOP requests and responses.                                                                                                               | New VOP service.                                                                                 |
| 2.116 to<br>2.119      | Several header parameters which come as part of attributes in the Discovery API payload have been transformed from "-" syntax to lowerCamelCase                                                | Enhance consistency and readability.                                                             |
| 2.145                  | Timestamps have been added to Standing Order Details for startpoint and endpoint of a standing order for instant payments.                                                                     | New function                                                                                     |
| 3.27, 3.28             | Two new code lists have been added for VOP Results.                                                                                                                                            | New VOP Service / VOP service integration                                                        |
| 3.51                   | New code list for a VOP bulk status.                                                                                                                                                           | New VOP Service                                                                                  |
| 3.52                   | Status Reason Code was added as ISO20022 External Code List, since it was missing.                                                                                                             | Erratum                                                                                          |